Chapter 1. Review of Thermodynamics and Heat Transfer

1.1 Introduction
1.2 Basic concepts of Heat Transfer
1.3 First Law of Thermodynamics
1.4 Second Law of Thermodynamics
1.5 Ideal Gas

Readings:

- M.J. Moran and H.N. Shapiro, Fundamentals of Engineering Thermodynamics,3rd ed., John Wiley & Sons, Inc., or Other thermodynamics texts

1.1 Introduction

1.1.1 Thermodynamics

Thermodynamics is the science devoted to the study of energy, its transformations, and its relation to the status of matter.

energy — the first law (conservation of energy)
entropy — the second law (quality of energy)
every naturally occurring transformation of energy is accompanied somewhere by a loss in the availability of energy for future performance of work

1.1.2 System and Surroundings

System: an object, any quantity of matter, any region of space, etc. selected for study
Surroundings: the rest
Basic system types: *Closed system (control mass)* and *Open system (control volume)*

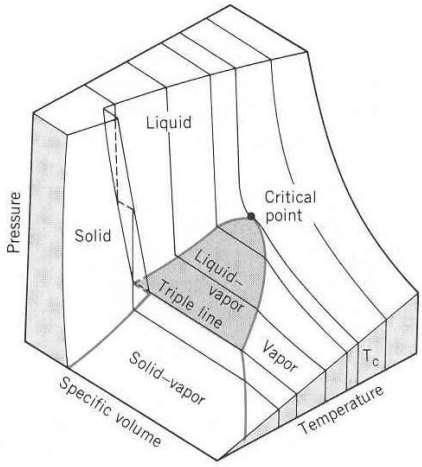
1.1.3 Property, State, Process and Equilibrium

A *property* is the observable characteristic of a system such as temperature, pressure and density.

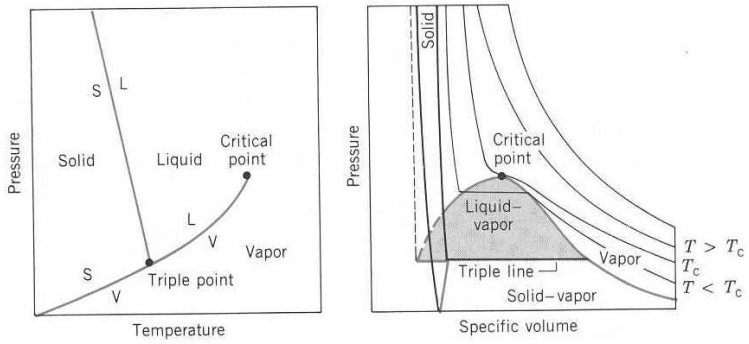
A *state* is the condition of the system defined by properties.

A *process* is transformation of the system from one state to another.

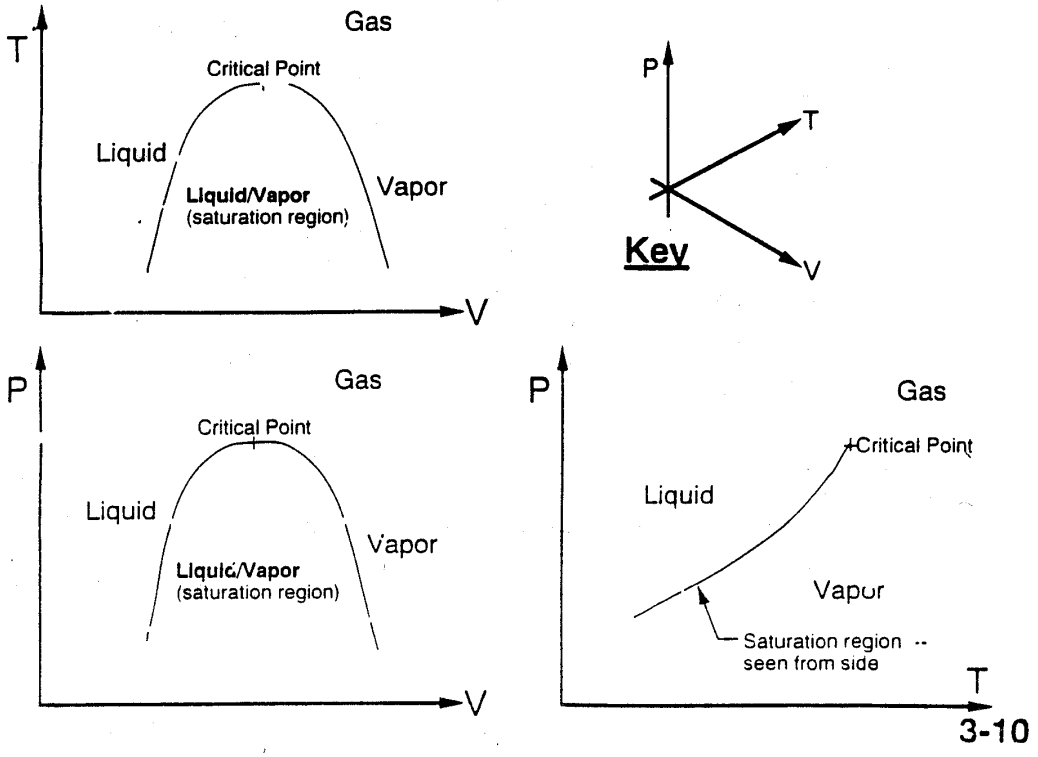
A thermodynamic *cycle* is a process that begins and ends at the same state.



(a)



Pressure-volume-temperature surface for a substance that expands on freezing



1.1.4 p-v-T Relationships

Thermodynamic properties: tables, graphs, equations and programs.

1.2 Basic Concepts

The engineering discipline of heat transfer is concerned with methods of calculating rates of heat transfer. These methods are used by engineers to design components and systems in which heat transfer occurs.

(1) Temperature: Degree of molecular movement

S.I. ^{o}C, K
I-P: ^{o}F, ^{o}R

(2) Energy: E (Btu, J)

Capacity to do work. Examples:

- Thermal
- Light
- Mechanical
- Electrical
- Chemical

1 Btu=1055 J

Work (W) is an action of a force on a moving system.

(3) Heat (Thermal energy): Q (Btu, J)

Heat is energy transferred across the system boundary by temperature difference (ΔT).

(4) Specific heat: C_p (Btu/lb^{o}F, J/kg K)

Heat needed to raise temperature of 1 lb (1 kg) material for 1 ^{o}F (1 ^{o}C).

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(5) Heat capacity: (Btu/ft^3 ^{o}F) and **thermal mass** (Btu/^{o}F)

Ability to store energy
Heat capacity = density x specific heat $= \rho C_p$
Thermal mass = density x volume x specific heat $= \rho V C_p$

(6) Energy change (heat transfer): Q (Btu)

$Q=\rho V C_p \Delta T$

where ρ - density, V - volume, ΔT - temperature difference.

(7) Heat flow (heat transfer rate or energy change rate): $\dot{Q}$ (Btu/h)

Energy transferred per unit time.

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1.3 First Law of Thermodynamics

The *first law of thermodynamics* expresses principle of conservation of energy.

Forms of energy:

$KE = \frac{1}{2}mv^2$	Kinetic (motion of the system)
$PE = mgz$	Potential (position of the system)
$U = U(T)$	Internal (stored in matter)

System total energy (E) = sum of all forms of energy

Energy transfer mechanisms are work (W) and heat (Q), which are not properties of the system.

Conventions: Work done by a system is positive.
Heat transfer to a system is positive.

1.3.1 First Law for the Closed System

Heat (Q) – Work (W) = Change in Total Energy (ΔE)

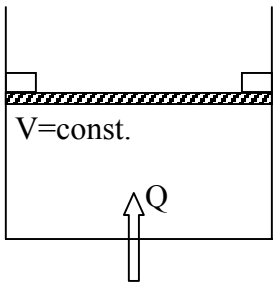
Closed system with process between states "1" and "2":

$$Q_{1-2} - W_{1-2} = m\left[(u_2 - u_1) + \frac{v_2^2 - v_1^2}{2g_C} + \frac{g(z_2 - z_1)}{g_C}\right]$$

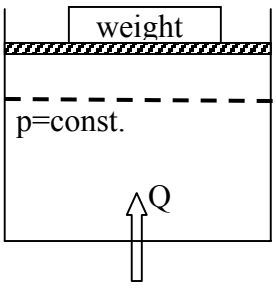
Example 1

Constant volume heating of a cylinder filled with a gas.

1st Law: $Q_{1\text{-}2} = m\,\Delta u = m\,c_V\,\Delta T$



Example 1: Constant volume heating



Example 2: Constant pressure heating

Example 2

Constant pressure heating of a cylinder filled with a gas.

1st Law:

where i = u + pv is enthalpy.

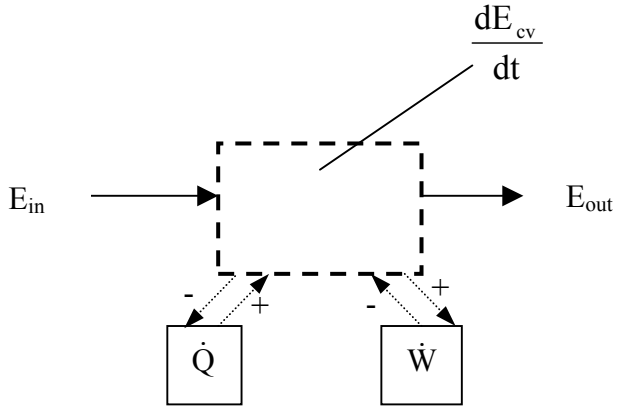
Enthalpy is a property that combines Δu and pv work that are only forms of energy change in many processes.

$$c_P = c_V + R$$

where R - specific ideal gas constant, and $c_P > c_V$ because constant input does work.

1.3.2 First Law for the Open System (Control Volume Formulation)

Heat Transfer ($\dot{Q}$) – Net Rate of Work ($\dot{W}$) = Net Rate of Energy Flow
(across the system boundary)



$$\dot{Q}_{CV} - \dot{W}_{CV} = \sum_{out}\left[\dot{m}\left(i + \frac{v^2}{2g_C} + \frac{gz}{g_C}\right)\right] - \sum_{in}\left[\dot{m}\left(i + \frac{v^2}{2g_C} + \frac{gz}{g_C}\right)\right] + \frac{dE_{cv}}{dt}$$

Special case of the 1st law is for steady-state flow that has constant flow across the boundary and no mass or energy change in CV.

For a typical HVAC system two assumptions are valid:

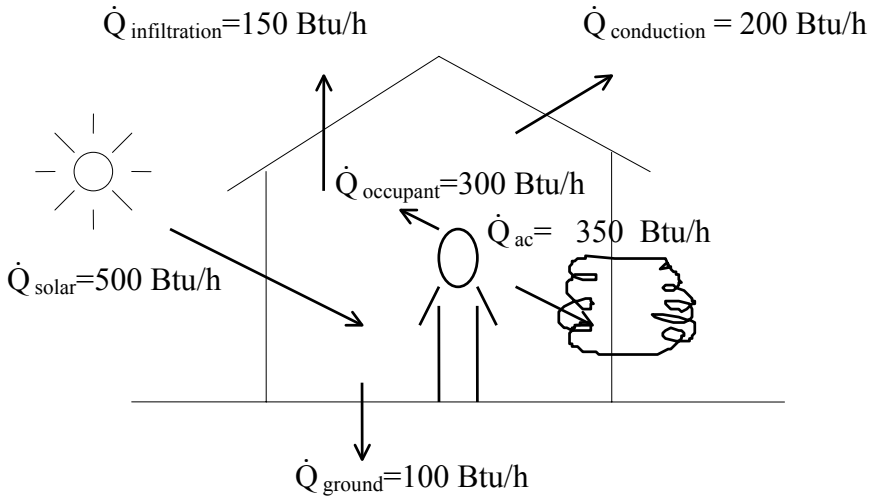
1) Steady-state ($\frac{dE_{cv}}{dt}=0$, $\frac{d\dot{m}_{cv}}{dt}=0$) flow across the system boundary
2) KE and PE terms usually small

Therefore, for the typical HVAC system:

$$\dot{Q}_{CV} - \dot{W}_{CV} = \sum_{out} \dot{m} \cdot i - \sum_{in} \dot{m} \cdot i$$

Example 3

A house/building is a thermal system and its envelope is the boundary. Let us consider some energy transfer in a single family house.



The thermal mass of the house is assumed to be 700 Btu/^{o}F.

(a) Is the system in equilibrium?

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Since $\dot{Q}_{in} = \dot{Q}_{out}$, the system is in equilibrium. The heat flows are steady state. The temperature will not change.

(b) What will happen if the A/C is shut off?

1.4 Second Law of Thermodynamics

All processes obey the 1st Law of Thermodynamics. However, some 1st law processes never occur. For example, heat transfer from cold reservoir to hot reservoir or flow from low pressure to high pressure.

The 2nd Law of Thermodynamics defines:

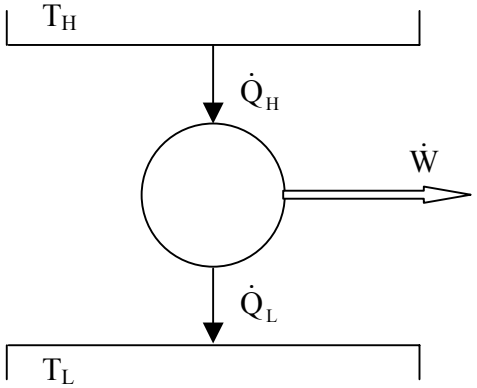
- direction of change for processes
- final equilibrium for spontaneous processes
- criterion for theoretical performance limits of cycles
- quality of energy

Energy changes and transfer involves both conservation principle and degradation in quality. Therefore, the thermal efficiency of all heat engines must be less than 100% due to *dissipative effects*. Processes occurring in a system such as heat engine are *irreversible* since either the system or its surroundings cannot be returned to their initial states. A *reversible* process is an idealization.

Heat engines (heat pumps) are closed systems, which operates continuously, or cyclically, and produce (use) work while exchanging heat across its boundaries.

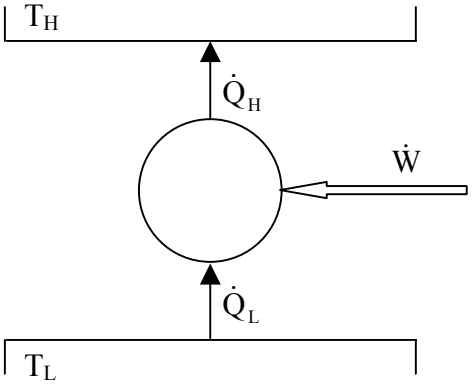
1.4.1 Heat Engine

Work produced while heat extracted from high temperature (T_H) reservoir and rejected to low temperature reservoir (T_L).



1.4.2 Heat Pump

Work used to extract heat from low temperature reservoir (T_L) and reject to high temperature (T_H) reservoir.



1.4.3 Performance evaluation of cycles

Performance evaluation of cycles: comparisons with the ideal *Carnot heat engine* that is a totally reversible heat engine or pump.

$$\dot{Q} \propto T \text{ of reservoir (absolute scale)}$$

(1) "Efficiency" (η) of a Heat Engine

$$\eta = \frac{W}{\dot{Q}_H} = \frac{\dot{Q}_H - \dot{Q}_L}{\dot{Q}_H} = 1 - \frac{\dot{Q}_L}{\dot{Q}_H}, \qquad 0 < \eta < 1$$

Carnot cycle: $\eta_{Carnot} = 1 - \frac{T_L}{T_H}$, $\eta_{Real\,Process} < \eta_{Carnot}$

$\eta_{Real\,Process} < 1$

Real irreversibilities:

- Friction, electrical resistance, heat transfer across finite temperature difference, …

(2) "Coefficient of Performance" (COP) of a Heat Pump

Heating conditions (heat pump):

Carnot cycle: $COP_{Heat,\,Carnot} = \frac{T_H}{T_H - T_L}$

Cooling conditions (refrigeration):

Carnot cycle: $COP_{Cool,\,Carnot} = \frac{T_L}{T_H - T_L}$

For energy exchange with the same two thermal reservoirs:

Carnot efficiency of a heat pump: $\eta_{pump} = \frac{COP_{Actual}}{\dot{Q}_{Carnot}}$

1.5 Ideal Gas

$$pv = \frac{R_u}{M}T$$

where

p — gas pressure, psi, Pa
v — specific volume, ft^3/lbm, m^3/kg, $v=1/\rho$
M — molecular mass, lbm/mol, kg/mol
R_u — universal gas constant, R_u=1545.32 ft-lbf/(mol-^{o}R) = 8314 J/(mol-K)
T — temperature, ^{o}R, K

Define:

$$R = \frac{R_u}{M}$$

where

R — specific gas constant
air: R_a=53.34 ft-lbf/(lbm-^{o}R) = 287 J/(kg-K)
water: R_v = 85.76 ft-lbf/(lbm-^{o}R) = 462 J/(kg-K)

Then:

$$pv=RT$$

Chapter 2.
Moist Air Properties and Air-Conditioning Processes

2.1 Moist Air and Its Properties
2.2 Methods of Measurements and Analysis
2.3 Typical Air Conditioning Processes
2.4 Characteristics of "Real" Systems
2.5 Psychrometric Analysis of Complete Systems

Readings:
- McQuiston & Parker (M&P) Ch 3
- Texts on moist air in most thermodynamics books

2.1 Moist Air and Its Properties

2.1.1 Air Composition (two components)

Dry air composition (volume fraction):

Nitrogen	78.084%	Minor gases	0.003%
Oxygen	20.948%	Water Vapor	
Argon	0.934%	Dust, Fog, Microbe	
Carbon dioxide	0.031%		

2.1.2 Ideal Gas Law

For dry air

$$R_a = \frac{R_u}{M_a} = \frac{1545.32}{28.965} = 53.352(\text{ft} - \text{lbf})/(\text{lbm} - \text{R}) =$$

For water vapor

$$R_v = \frac{R_u}{M_v} = \frac{1545.32}{18.02} = 85.78\text{ft} - \text{lbf}/(\text{lbm} - \text{R}) =$$

The following data for *US STANDARD ATSMOPHERE* are from ASHARE Handbook of Fundamentals (Chapter6). Several assumptions are introduced such as:

The atmosphere consists of dry air that behaves as an ideal gas.(see pg.50 in the textbook)

Table 1 Standard Atmospheric Data for Altitudes to 10 000 m

Altitude, m	Temperature, °C	Pressure, kPa
−500	18.2	107.478
0	15.0	101.325
500	11.8	95.461
1000	8.5	89.875
1500	5.2	84.556
2000	2.0	79.495
2500	−1.2	74.682
3000	−4.5	70.108
4000	−11.0	61.640
5000	−17.5	54.020
6000	−24.0	47.181
7000	−30.5	41.061
8000	−37.0	35.600
9000	−43.5	30.742
10000	−50	26.436
12000	−63	19.284
14000	−76	13.786
16000	−89	9.632
18000	−102	6.556
20000	−115	4.328

Data adapted from NASA (1976).

Table 1 Standard Atmospheric Data for Altitudes to 60,000 ft

		Pressure	
Altitude, ft	Temperature, °F	in. Hg	psia
−1000	62.6	31.02	15.236
−500	60.8	30.47	14.966
0	59.0	29.921	14.696
500	57.2	29.38	14.430
1000	55.4	28.86	14.175
2000	51.9	27.82	13.664
3000	48.3	26.82	13.173
4000	44.7	25.82	12.682
5000	41.2	24.90	12.230
6000	37.6	23.98	11.778
7000	34.0	23.09	11.341
8000	30.5	22.22	10.914
9000	26.9	21.39	10.506
10,000	23.4	20.58	10.108
15,000	5.5	16.89	8.296
20,000	−12.3	13.76	6.758
30,000	−47.8	8.90	4.371
40,000	−69.7	5.56	2.731
50,000	−69.7	3.44	1.690
60,000	−69.7	2.14	1.051

Data adapted from NASA (1976).

Reference p and T are functions of altitude. Formula for p as function of elevation Z:

$$p[psi] = 14.696\ (1-6.8753\times10^{-6}\ Z[ft])^{5.2559}$$

$$p[bar] = 1.013\ (1-2.256\times10^{-6}\ Z[m])^{5.2559}$$

Example

Sea level: p=14.696 psi, T=59°F

3.1.3 Fundamental Parameters

(1) *Pressure*

The air layer above the earth forms atmospheric pressure. Atmospheric pressure:

- sea level 14.692 psi
- elevation of 6600 ft, 11.513 psi

○

(note: pressure inch mercury)

a,b ---Table 3-2 in M&P

Partial pressure (Dalton's Law)

○

Component/partial pressure: $p = p_1 + p_2 + p_3 + \ldots$

Gibbs Dalton's Law for Moist air:

○

p_a—dry air component (constant)
p_v—vapor component (change with moisture content)

Note: When applying ideal gas law to each component of a mixture (e.g., moist air), should use partial pressure for the component.

For component i: $P_i v_i = R_i T_i$

Where P_i is partial pressure for component i.

Example

One lbm H_2O vapor in 100 lbm dry air at standard pressure.

(a) What is p_v vapor pressure? (b) What is saturation T at this p_v?

(a) ,

○

(b) Saturation: State of maximum concentration for mixture components.

○

(2) *Temperature*

Temperature is the macro results of molecular kinetics.

0th law of thermodynamics

$$T_a = T_b \qquad T_b = T_c \quad \Rightarrow \quad T_a \equiv T_c$$

Table 2.1 Relationship between Different Temperature Scales

Relationship between	Kelvin (K)	Celsius (°C)	Fahrenheit (°F)	Rankine (R)
Kelvin (K)	--	$K = {}^{\circ}C + 273.15$		$K = \frac{5}{9}R$
Celsius (°C)	${}^{\circ}C = K - 273.15$	--	${}^{\circ}C = \frac{5}{9}({}^{\circ}F - 32)$	${}^{\circ}C = \frac{5}{9}R - 273.15$
Fahrenheit (°F)		${}^{\circ}F = \frac{9}{5}{}^{\circ}C + 32$	--	^{o}F = R-459.67
Rankine (R)	$R = \frac{9}{5}K$			--

Example

Temperature in	Celsius	Kelvin	Fahrenheit	Rankine
Water Boiling	100°C	373.15K	212°F	617.67°R
Ice Point	0°C	273.15K	32°F	491.67°R
Absolute Zero	-273.15°C	0 K	-459.67°F	0°R

(3) *Humidity Ratio, W*

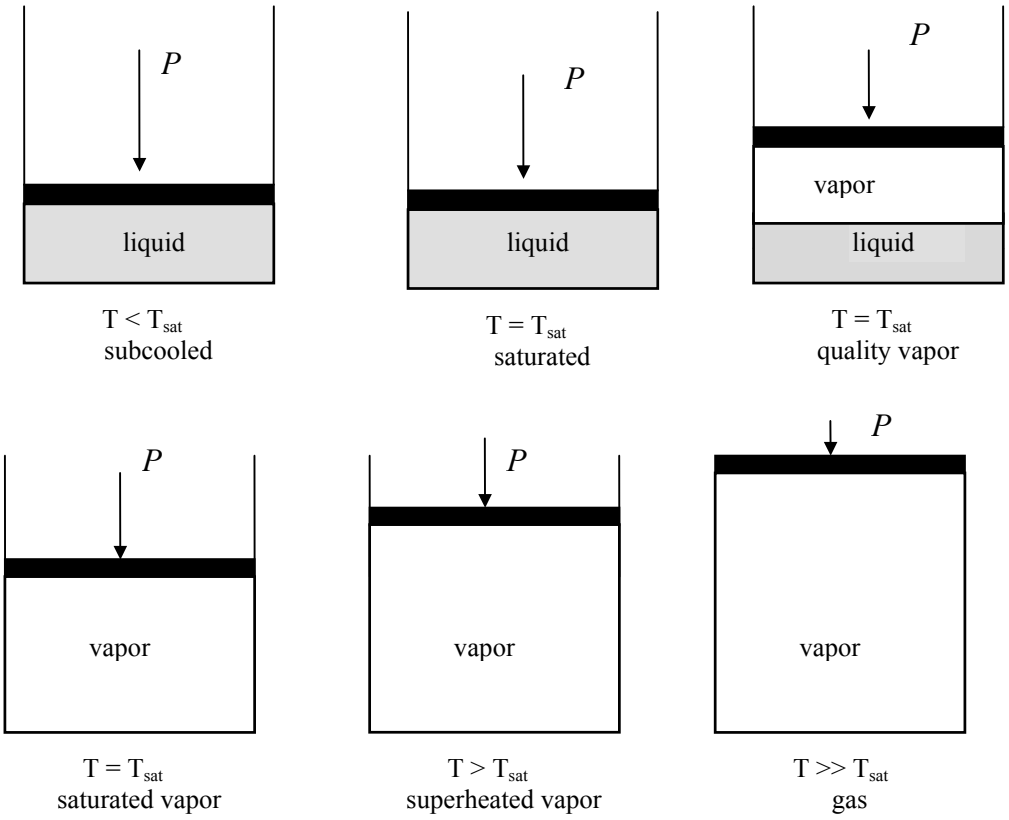
Definition: $W = \frac{m_v}{m_a} \quad (Kg / Kg_{drry\ air})$

i.e. 1 kg dry air + w kg water vapor = (1+W) kg moist air

where P is the atmospheric pressure. Because $p_v << P$, thus $W \propto p_v$

(4) *Relative Humidity ϕ*

Thermodynamic fluid states



$$\phi = \frac{p_v}{p_{v,s}} \times 100\% = \frac{p_v}{p_s} \times 100\% \qquad p_{v,s} = p_s$$

p_v = partial pressure of the water vapor in the air
p_s = partial pressure of the water vapor in a saturated mixture under the same temperature

Dry air: ϕ=0% Saturated air: ϕ=100%

- Difference between W and ϕ:

Moist air: $W = 0.622 \frac{p_v}{P - p_v}$

Saturated air: $W_s = 0.622 \frac{p_s}{P - p_s}$

$$\therefore \quad \frac{W}{W_s} = \frac{p_v}{p_s} \cdot \frac{P - p_s}{P - p_v} = \phi \cdot \frac{P - p_s}{P - p_v}$$

$$\phi = \frac{W}{W_s} \frac{P - p_v}{P - p_s} \times 100\%$$

Since P>> p_v and P >>p_s

-

Further

-

Example

Determine the humidity ratio of moist air at a temperature of 24°C and a relative humidity of 50% at a standard pressure 1atm

Given: T, ϕ
Find: W

Solution:

-

(5) *Dewpoint Temperature, T_d*

T_d — the saturated temperature of a given mixture at the same pressure and humidity ratio.

Example

Find T_d of the air in the above example.

Solution:

At the dew point temperature: air mixture ϕ=100%

○

(6) *Enthalpy*

Enthalpy of the moist air = enthalpy of the dry air + enthalpy of the water vapor
Enthalpy is energy per unit mass.

$$i = i_a + W\, i_v$$
$$i_a = C_{p,a}\, T$$
$$i_v = i_g + C_{p,v}\, T$$

where

$C_{p,a}$ = specific heat of dry air kJ/(kg^{o}C), Btu/(lb^{o}F)
$C_{p,v}$ = specific heat of water vapor kJ/(kg^{o}C), Btu/(lb^{o}F)
i_g = the enthalpy of saturated water vapor at 0°C or 0°F.
i_g = 2501.3 kJ/kg at 0°C; i_g = 1061.2 Btu/lb at 0°F.

Therefore, we have

$$i = C_{p,a}T + W(i_g + C_{p,v}\, T)$$

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Example

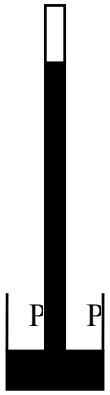
Find the enthalpy of the air mixture in the above example.

Solution:

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2.2 Methods of Measurement and Analysis

2.2.1 Measurement of air temperature



- Liquid-in-glass thermometers
- Thermocouples

2.2.2 Measurement of pressure

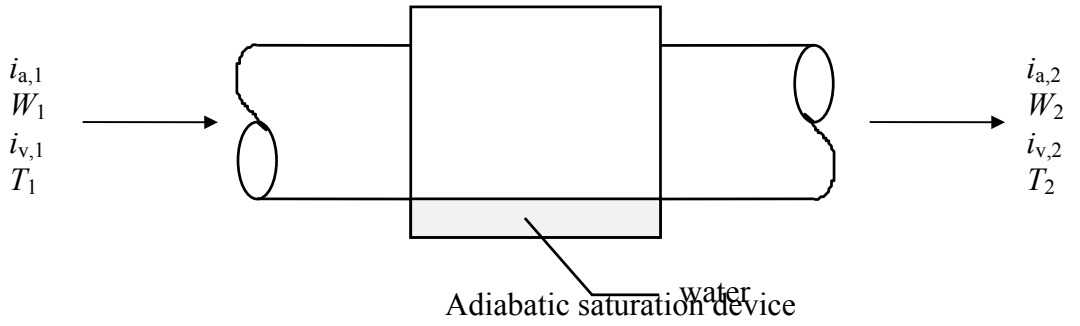
- Absolute pressure (vacuum tube with mercury)
- Differential pressure
  - Manometers
  - Pressure transducer

2.2.3 Measurement of other parameters of moist air

To determine state of moist air, one property in addition to the pressure and temperature must be known. It can be v, I, ϕ, or W. However, none of them can be directly measured. As an alternative, we seek an indirect measuring technique. In this section, the method used to determine air humidity will be introduced.

- **Adiabatic saturation device**

  First we look at a special process:



The equation that describes above process is:

- 1st Law
- The process is adiabatic, and only flow work is present.:
- where $\dot{m}_1 = \dot{m}_a + W_1 \dot{m}_a$, $\dot{m}_2 = \dot{m}_a + W_2 \dot{m}_a$, and form mass balance

The equation that describe above process becomes:

○

$$\underset{(in)}{dry\ air} + \underset{(in)}{water\ vapor} + \underset{(added)}{water\ vapor\ added} = \underset{(out)}{dry\ air} + \underset{(out)}{water\ vapor}$$

so we have

○

where i_{fg} is enthalpy difference between liquid water and saturated vapor at the temperature T_2

$$W_2 = 0.622 \frac{p_{v,2}}{P - p_{v,2}}$$

Then the state of moist are can be determined.

T_2 is called ***wet-bulb temperature***.

Example

Find W and ϕ of the above adiabatic saturation device.

Given: P=1.01325×10^{5}Pa, T_1=30°C, T_2=26°C

Find: W_1 , ϕ_1

Solution:

Since the state 2 is in saturation, from the Table A-1b (McQuiston & Paker, p587), we can find:

for T_2 = 26°C, $p_{v,2} = p_s = 0.03363\times10^5$ (ϕ=100%), i_{fg} = 2439.1 *kJ/kg*, i_w=109.07 *kJ/kg*
for T_1 = 30°C, $i_{v,1}$ = 2555.3 *kJ/kg*

1) Find W_1

○

2) Find ϕ_1

$$\phi = \frac{p_{v,1}}{p_{s1}} \times 100\%$$

$p_{v,1}$ can be found via

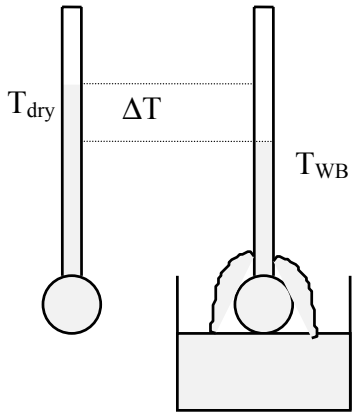
From Table A-1b

- **The Psychrometer**

Psychrometer T_{WB} used in place of T_2 for practical humidity measurement.

In wet-bulb, heat transfer from air →bulb

p+T_{DB}+T_{WB} =>Moist air state



Key issues to measure T_{WB}

- Wet bulb unshielded
- Wet bulb well ventilated (V>100 fpm)

For thermocouples V could be lower.
Then the accuracy is in order of 0.27°C (0.5°F).

2.2.4 The psychrometric chart

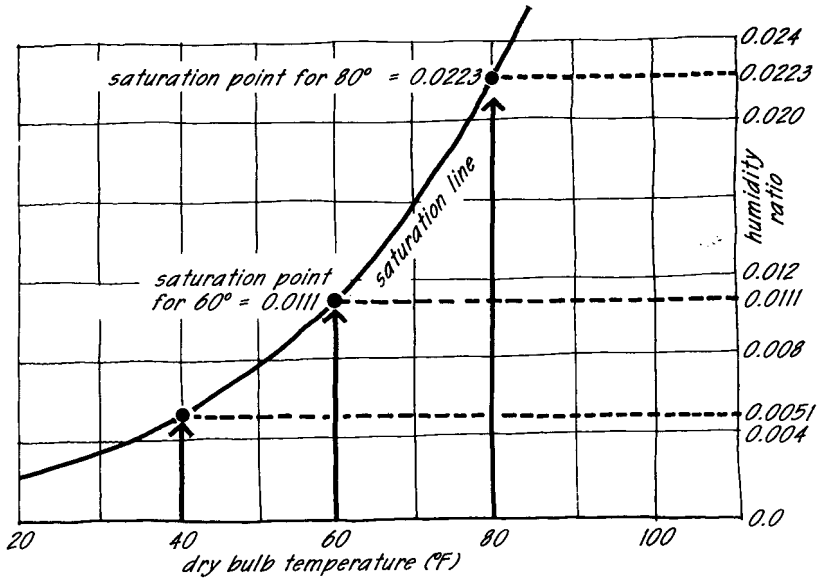
(1) *Selection of the Coordinates*

Horizontal coordinate — Enthalpy (155°)
Vertical coordinate — Humidity Ratio

(2) *Dry Bulb Temperature*

$$i = 1.01T + W(2501.3 + 1.86T) \quad \text{(SI Unit)}$$
$$T = const \Rightarrow i \propto W$$

Isothermal lines are not parallel.

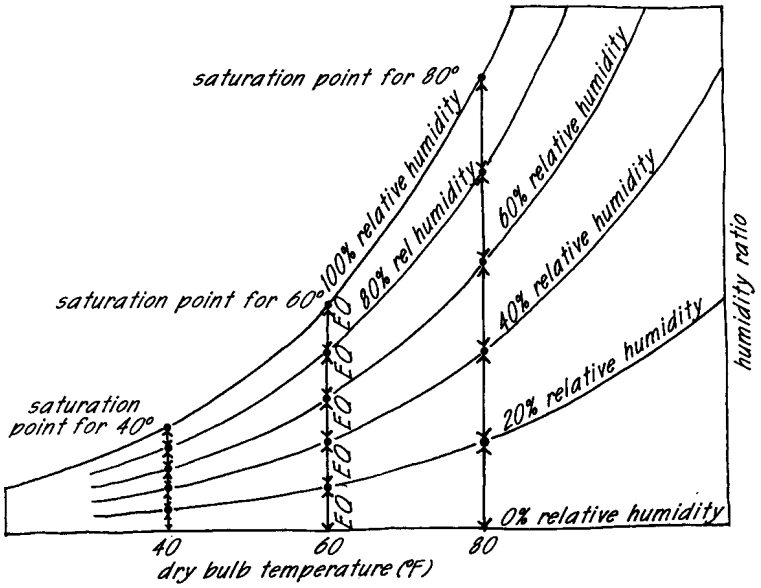


(3) *Relative Humidity*

$$W = 0.622\frac{\phi\, p_{v,s}}{P - \phi\, p_{v,s}} \qquad \phi = \frac{W}{W_s}\frac{P - p_v}{P - p_s} \times 100\%$$

Under a certain P, $W = f(p_{v,s})$ $p_{v,s} = f(T)$

From P587 Table A-1b. find W-T relationship
Approximately $\phi \approx \frac{W}{W_s} \times 100\%$ (equal division)



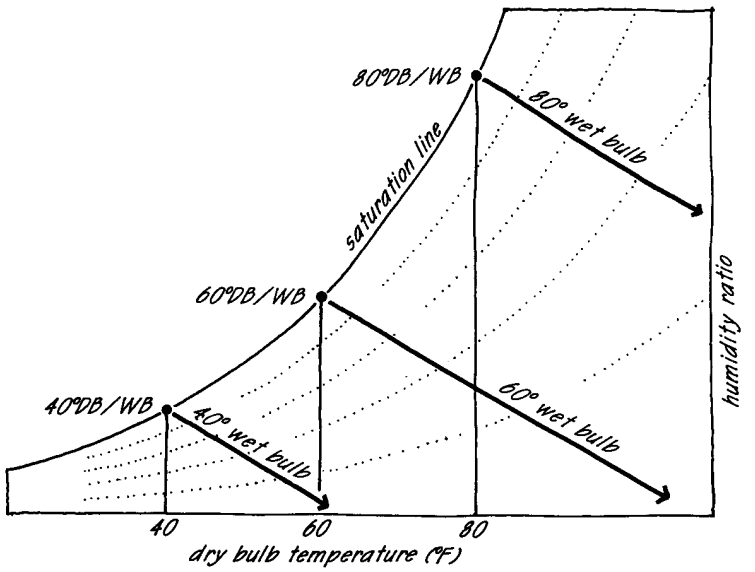
(4) *Specific Volume*

$$v = \frac{1}{\rho} = \frac{1}{\rho_a + \rho_v}$$

$$\rho_a + \rho_v = \frac{p_a}{R_a T} + \frac{p_v}{R_v T}$$

$$v = \frac{T}{\frac{p_a}{R_a} + \frac{p_v}{R_v}}$$

(5) *Wet-Bulb Temperature*

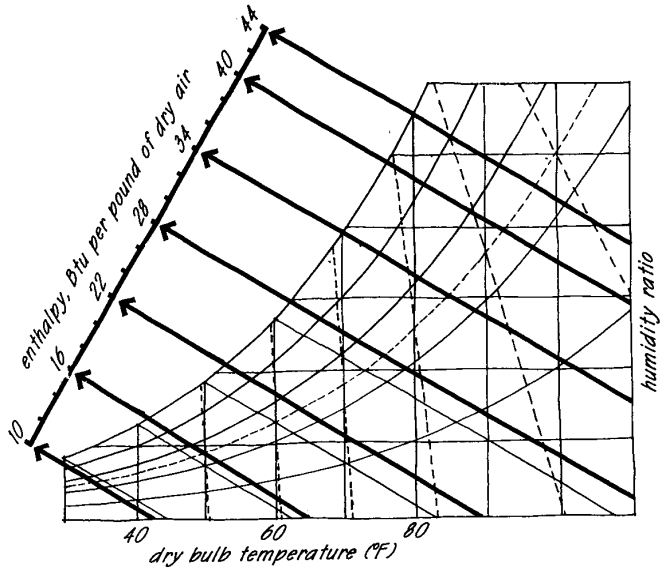


$i = (C_{p,a} + W C_{p,v}) T_{wet} + W \times 2501$ when T_{wet} = const, linear relationship

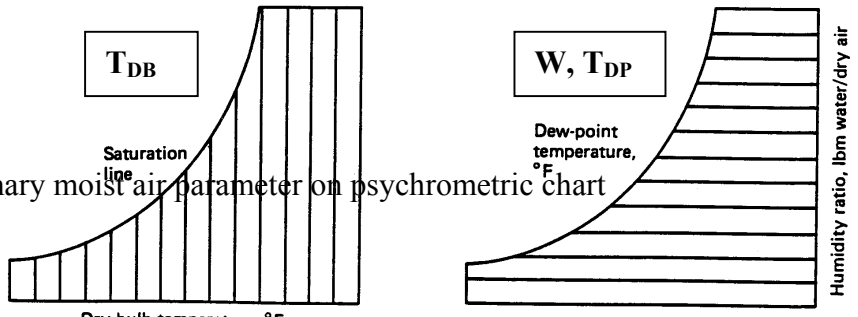
$i = (C_{p,a} + W C_{p,v}) T_{wet} + W \times 2501 + W i_w$ when $\phi \neq 100\%$

(6) *Sensible Heat /Total Heat*

Enthalpy/Humidity ratio
See figure: Primary moist air parameter on the psychrometric chart.



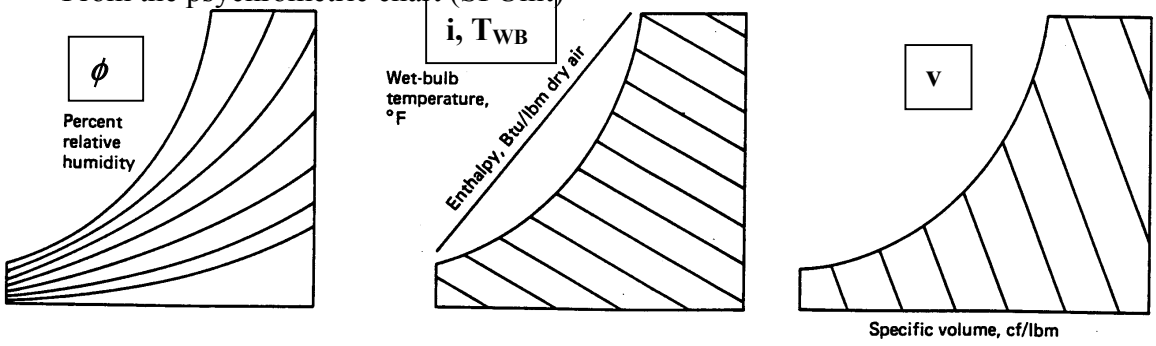
For specific pressure:



Find: Primary moist air parameter on psychrometric chart

Solution:

From the psychrometric chart (SI Unit)



Example

Repeat the previous problem by using the psychrometric chart.

Given: $T_{dry} = 30^{\circ}C, \quad T_{wet} = 26^{\circ}C$

Find: Primary moist air parameter on psychrometric chart

Solution:

2.3 Typical Air-Conditioning Process

Typical Air-Conditioning processes are:

- Sensible Heating / Cooling
- Cooling and dehumidification
- Heating and humidification
- Adiabatic Humidification
- Adiabatic Mixing of Air

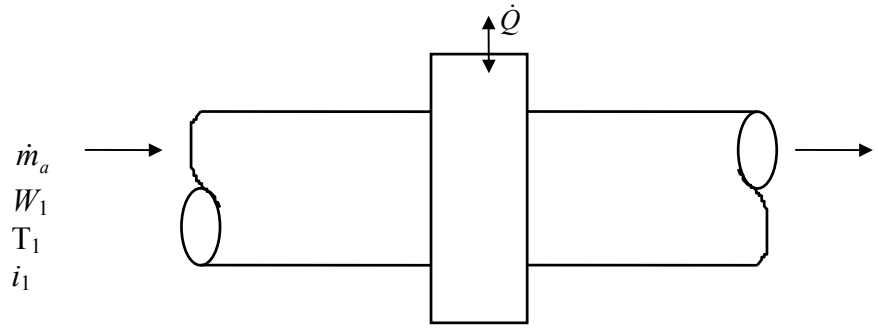
Governing equations:

STATE is a point, and PROCESS (sequence of states) is a line on the Chart.

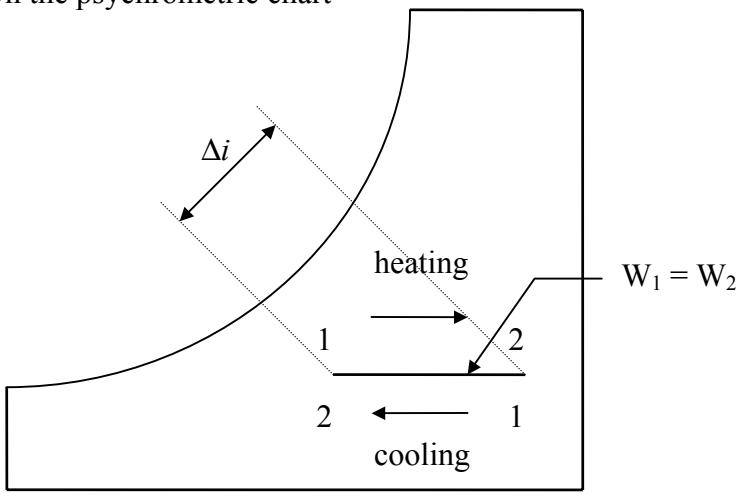
Process may involve:

- Sensible Heat (change T_{DB}, constant W)
- Latent Heat (constant T_{DB}, change W)
- Both

2.3.1 Sensible heating and cooling



On the psychrometric chart



Energy conservation (1st Law)

$$q = i_2 - i_1 = C_p(T_2 - T_1)$$.

Example

Determine the energy (heat flux) required for sensible heating of air at 15°C and 50% RH to 32°C. Also find ϕ_2.

Given: State 1: 15°C, RH=50%, State 2: 32°C
Find: $\dot{q}$, ϕ_2

Solution:

Heat Flux:

Relative humidity:

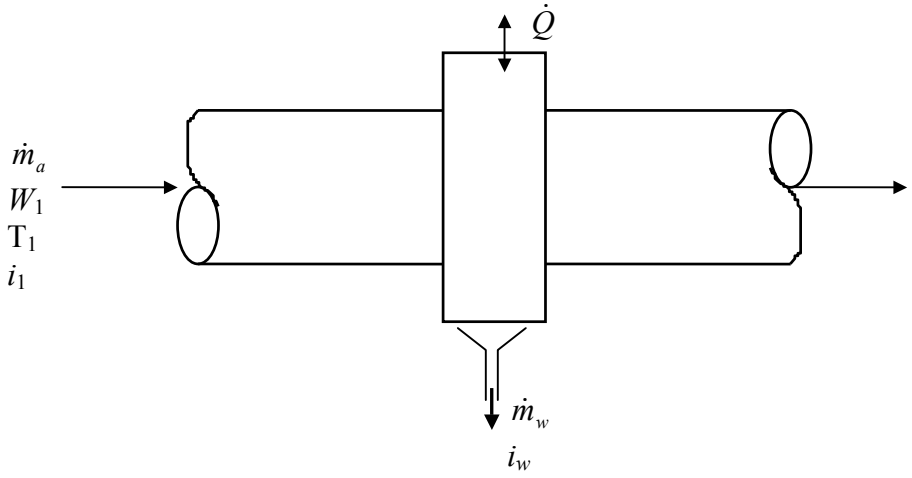
From the psychrometric chart (the process is a horizontal line):

Heat flux is

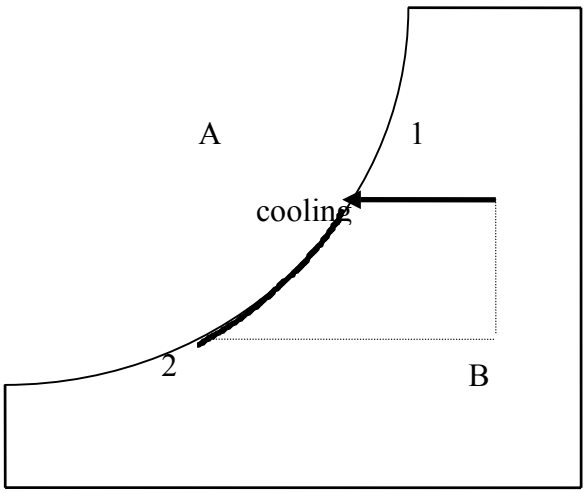
Relative humidity:

How to check if W is constant, i.e. no latent heat?

2.3.2 Cooling and dehumidification



Moisture is removed as saturated liquid.



where i_w — water enthalpy

$\dot{m}_a(W_1 - W_2)i_W$ is normally small

Sensible heat:

$$\dot{q}_{sensible} = C_p(T_2 - T_1)$$

Latent heat

$$\dot{q}_{latent} = (W_2 - W_1)i_{fg}$$

Example

Air at 60% RH, T_{dry} =30°C, Cooled to 18°C. Determine ϕ_{new}, $q_{sensible}$, q_{latent}

Solution:

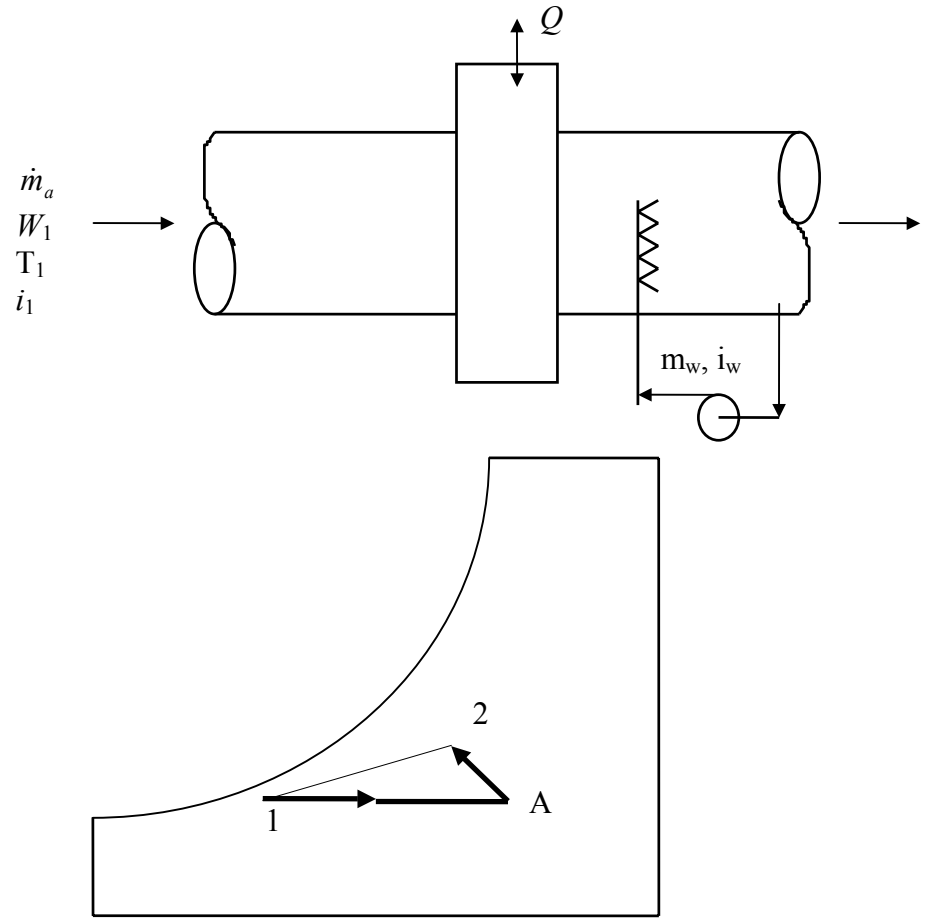
ϕ_{new}=100%, from the psychrometric chart, we can find

By using the formula

Sensible Heat Factor **(SHF)** is $\frac{\dot{Q}_S}{\dot{Q}}$

Defines process slope on the chart. Use protractor (semicircular scale) in the upper left hand corner to read the sensible heat factor.

2.3.3 Heating and humidification



$\frac{enthalpy}{humidity\ ratio}$ -> Look at the semicircular scale in the psychrometric chart.

Defines the process slope.

For adiabatic humidification, then

$$\dot{Q} = 0$$

$$i_2 - i_1 = \Delta W\, i_w$$

$$W_2 - W_1 = \Delta W$$

Since i_w of the water is rather small, $i_2 \approx i_1$

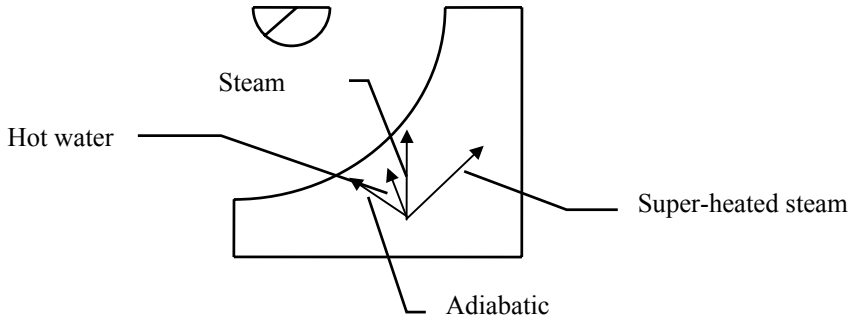
Example

In Phoenix, it is possible to use evaporative cooling in summer. In a room of 50 m^3, the air temperature is 45°C and relative humidity is 20%. Comfort standard allows the relative humidity to be increased to 60% by evaporative cooling. Determine the dry bulb temperature and water needed if there is no internal heat source and no air infiltration. Assume local pressure is 101325 Pa.

Solution:

This is an adiabatic humidification process. The air process in a psychrometric chart is iso-enthalpy. For the psychrometric chart, we can determine the starting humidity ratio and ending humidity ratio as

2.3.4 Other Humidity Process



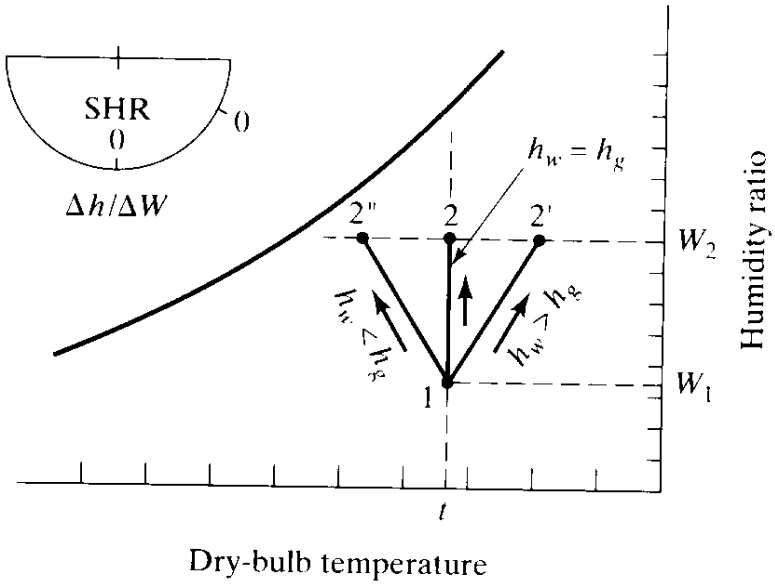


Figure 8.7 Schematic illustration of humidification processes.

For humidification: $i_w = \dfrac{\Delta i}{\Delta W}$

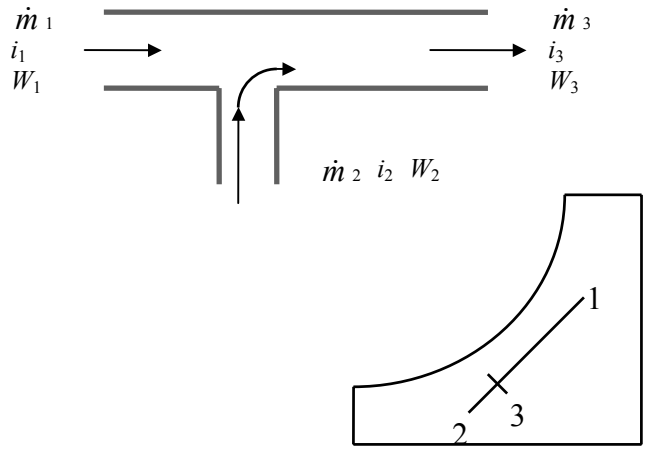
Adiabatic humidification: $i_w = i_f$ at T_{WB}, process $i = const.$

Hot water: $i_w < i_g$ at T_{DB}, process $T_{DB} \downarrow$

Saturated steam: $i_w = i_g$ at T_{DB}, process $T_{DB} = const.$

Super-heated steam: $i_w > i_g$ at T_{DB}, process $T_{DB} \uparrow$

2.3.5 Adiabatic Mixing of Air



Eliminate $\dot{m}_3$, We obtain

$$\dot{m}_1(i_1 - i_3) = \dot{m}_2(i_3 - i_2)$$
$$\dot{m}_1(W_1 - W_3) = \dot{m}_2(W_3 - W_2)$$

so

$$\frac{\dot{m}_1}{\dot{m}_2} = \frac{i_3 - i_2}{i_1 - i_3} = \frac{W_3 - W_2}{W_1 - W_3}$$

Example

Return air at 25°C, 50% relative humidity and flowing at a rate of 5 m^3/s is mixed with outside air at 35°C and 60% relative humidity and flowing at a rate of 1.25 m^3/s. Determine the mixed air condition and flow rate.

Given: $T_1, T_2, \phi_1, \phi_2, Q_1(V_1), Q_2(V_2)$
Find: T_3, ϕ_3, W_3, m_3

Solution:

From the psychrometric chart (ASHRAE PSYCHROMETRIC CHART NO.1[sea level], chart 1 *b*), we can determine the point 1 and 2

		Return Air	Outdoor Air
i_1	(*kJ/kg*)	50.8	90.5
W_1	(*kg water/kg dry air*)	0.010	0.0215
v_1	(m^3/*kg dry air*)	0.858	0.902

then we find the mass flow rates:

○

and the enthalpy, etc.:

○

2.4 Characteristics of "Real" Systems

2.4.1 Design conditions

The processes described in the previous chapter are used to condition moist air in a real air-conditioning systems. The AC system is used to remove both **sensible** and **latent** heat from a space. Relationship between sensible and latent heat is defined as **SHF** (sensible heat factor).

Sensible loads:

Latent loads:

Sensible and Latent loads:

Example

Dishwasher (100 dishes/h): sensible 167 W (570 Btu/h), and latent 65 W (220 Btu/h)
Person (male, moderate office work): sensible 70 W (250 Btu/h), and latent 30 W (105 Btu/h)
Light bulb: sensible 100 W = 341 Btu/h

AC system provides airflow:

- at certain **T** to meet **sensible** loads
- at certain **W** to meet **latent** loads

where T_{supply} and W_{supply} must give sensible and latent conditioning proportional to the loads.

Condition line represents line in the psychrometric chart through space conditions with the slope defined by SHF. This line contains all feasible supply air states.

Supply farther from space condition => Smaller mass flow required

Design condition is defined by:

- dry bulb temperature,
- humidity and
- pressure.

Design condition + SHF + T_{supply} => Fix mass flow and supply air condition

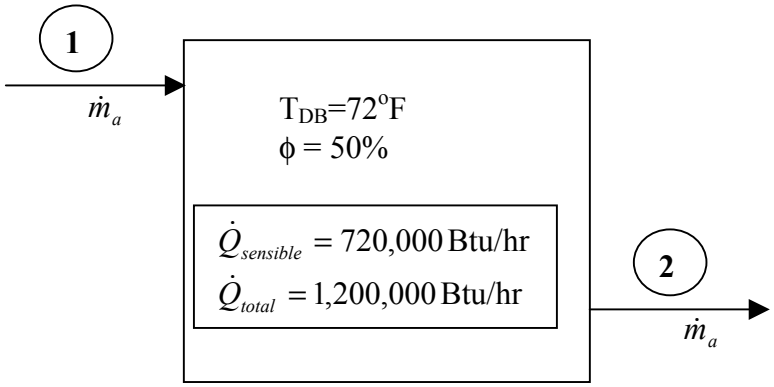
Example

Space design condition is 72°F and 50% RH. The total cooling load is 1,200,000 Btu/hr (100ton), and the sensible cooling load is 720,000 Btu/hr (60 ton). Compare flow rates for (a) ΔT = 10°F and (b) ΔT = 20°F. $\Delta T = (T_{space} - T_{supply})$

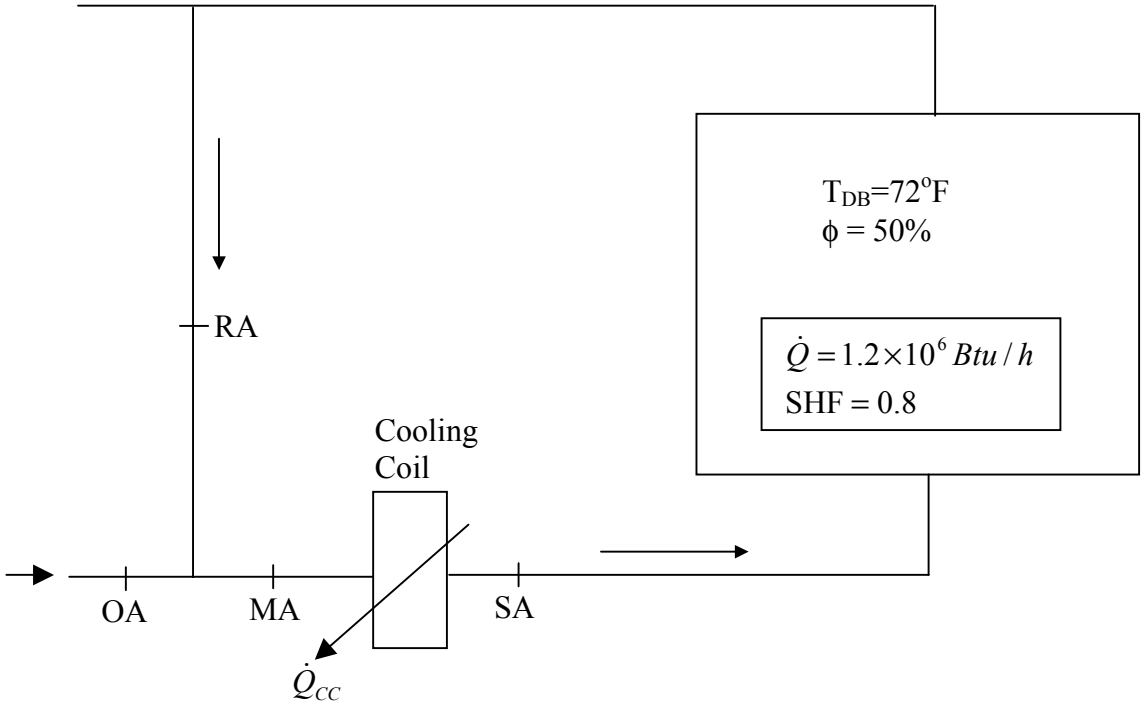
$$1 ton = 12{,}000\ Btu/h$$

Solution:

Construct condition line by using the protractor.



2.4.2 Analysis of the single zone system



Example

Cooling design conditions for OA are 16,262 cfm and 90°F db/72°F wb.
Design space cooling load is 1,200,000 Btu/h (80% sensible). Supply air temperature is 55°F.
Determine (a) supply airflow rate and (b) cooling coil load.

Solution:

Plot known state on psych chart: OA, SA, and RA.

Assumption of perfect mixing => RA is equal to room air conditions => i_{RA}=26.4 Btu/lbm
OA: i_A=35.6 Btu/lbm, v_{OA}=14.1 ft^3/lbm
SA: i_{SA}=21.2 Btu/lbm, v_{SA}=13.1 ft^3/lbm

(a) Flow rate

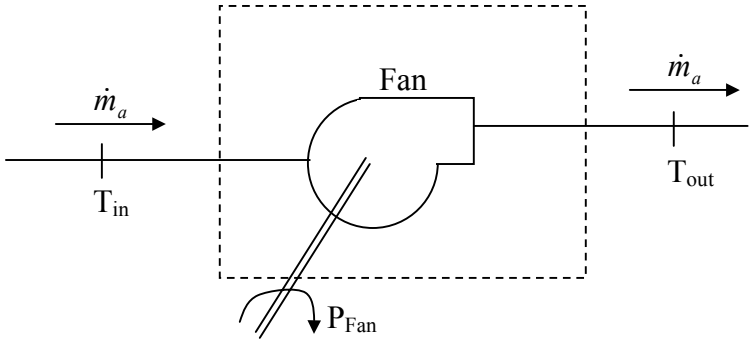
(b) Cooling coil load

○

2.4.3 Component characteristics

Fan Heat Gain

Due to irreversibility, i.e. friction, fan contributes to sensible heat gain, and increases temperature of moist air.



○

where $-\dot{W} \equiv P_{Fan}$, therefore $$T_{out} = T_{in} + \frac{P_{Fan}}{\dot{m}_a c_p}$$

Example

A 1.5 kW fan moves 1m^3/s of dry air entering at 15°C. What is T_{out}?

Solution:

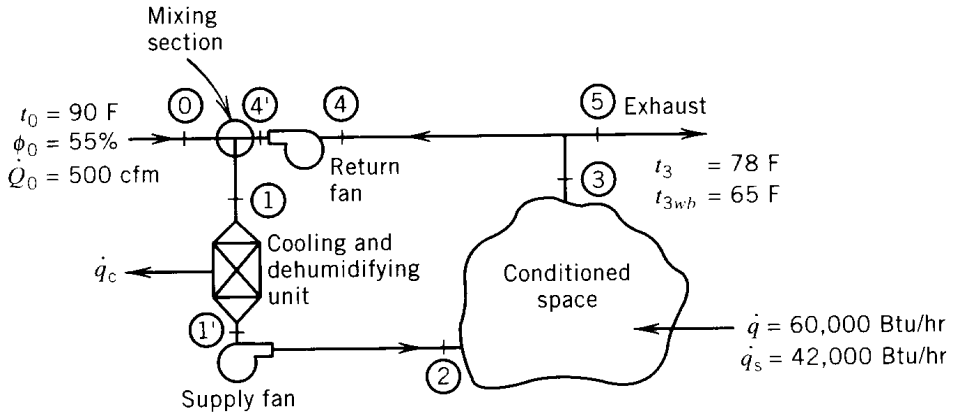
○

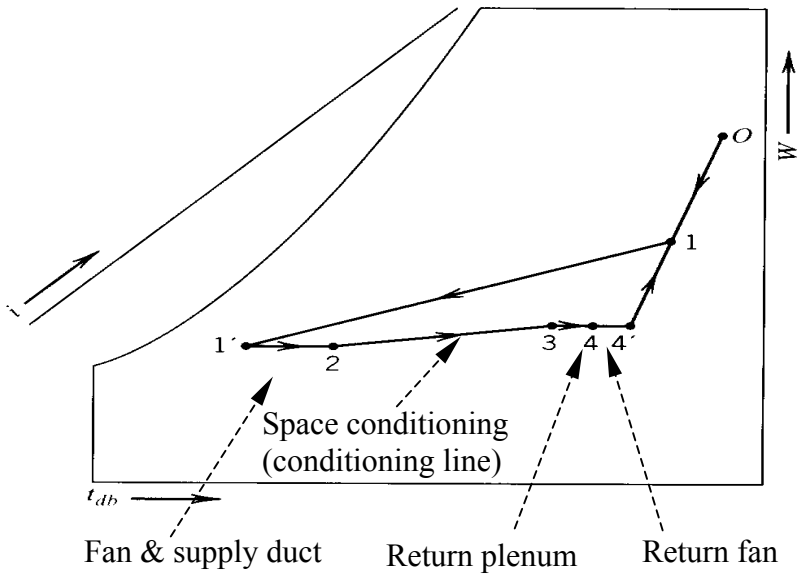
Duct and Plenum Heat Gain

Plenum heat gains: lights; hot pipes.

Ducted supply/return: convection in hot spaces; "sweating" on cool duct surfaces.

Example





Cooling Coils

Cooling coil are indirect contact heat exchangers. Different types:

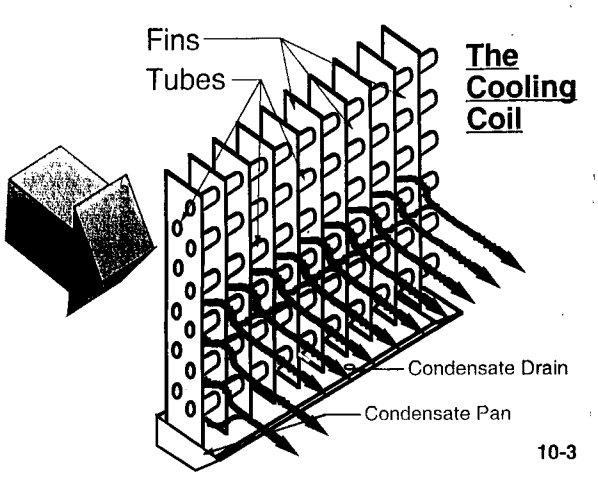
- Air-to-Water
- Air-to-Refrigerants

Air is outside, and liquid is inside the tubes.

Air-side --- fins promote heat transfer (larger area)
Water-side --- shape promotes heat transfer (higher turbulence)

For an **ideal coil**: $T_{DB} = T_{coil,\ mean}$
$\phi = 100\%$

For a **real coil**: $T_{DB} > T_{coil,\ mean}$
$\phi < 100\%$



Approximate methods for rough estimate of cooling coil condition:

- fixed relative humidity (assume coil leaving RH, say 90%)
- bypass factor "**b"** (assume fraction of flow bypasses the coil, reminder in perfect contact with coil)

○

where LA-air leaving the coil, EA - air entering coil, ADP - coil (apparatus) dew point

Example:

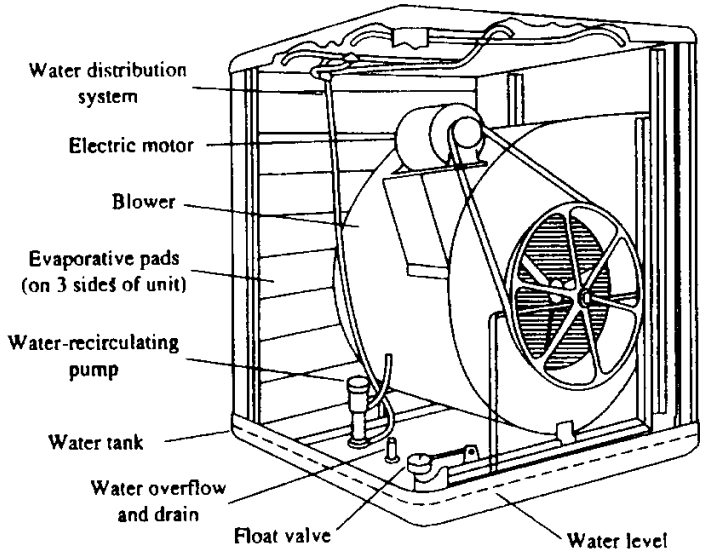
A chilled water coil with 8°C entering water conditions air from 26°C db/ 19°C wb to 15°C db/ 14°C wb. What are T_{ADP}, b, and leaving relative humidity?

Solution:

○

Evaporative Cooling

If sufficiently dry air is available, an evaporative process can be used to cool the air stream. Direct evaporative coolers (see Figure) add moisture to air adiabatically. The evaporation uses air sensible heat => T_{air} drops.



Direct Evaporative Cooler

Direct evaporative cooling effectives:

○

where ε_e varies with air flow rate and media thickness. Range is 60-95%. Typical value is 80%.

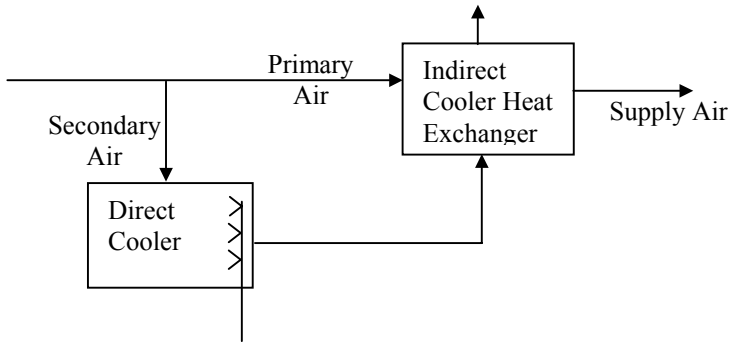
Unassisted Direct Evaporative Cooling

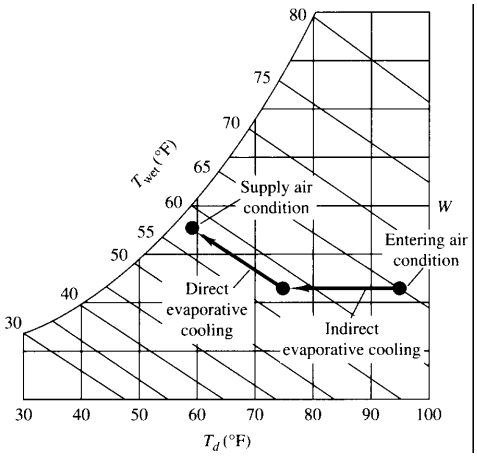
Applicable if the wet bulb temperature is less than 24°C (75°F). Regional applicability is limited in U.S.:

- the western U.S.
- the north central states
- the northeastern U.S.

May require large supply airflow rates. May give high space humidity at times.

Indirect evaporative cooling is sensible process of cooling an outdoor air stream without humidifying (sensible cooling).





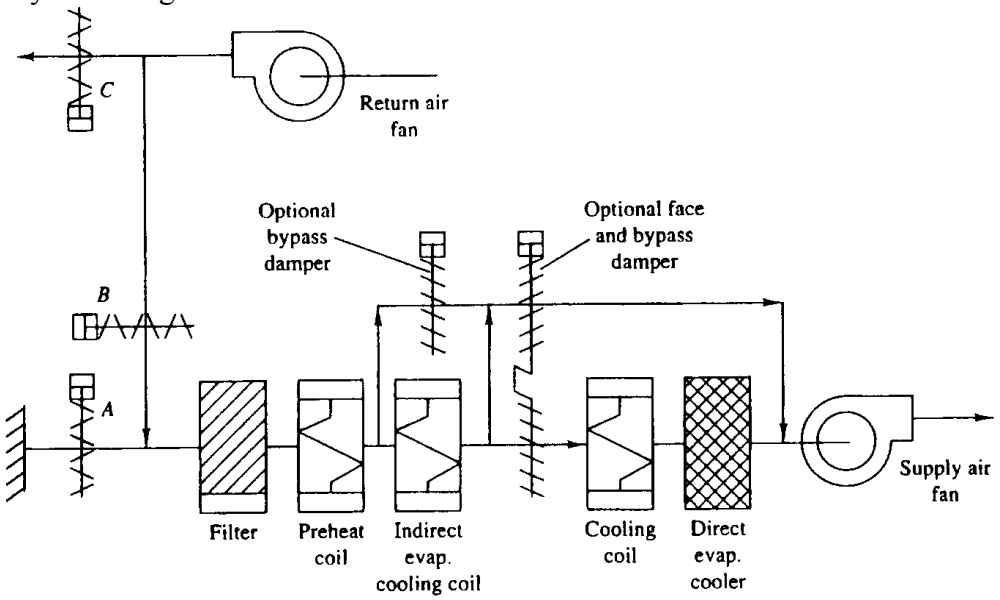
Indirect evaporative cooling effectives:

○

Range is 40-80%.

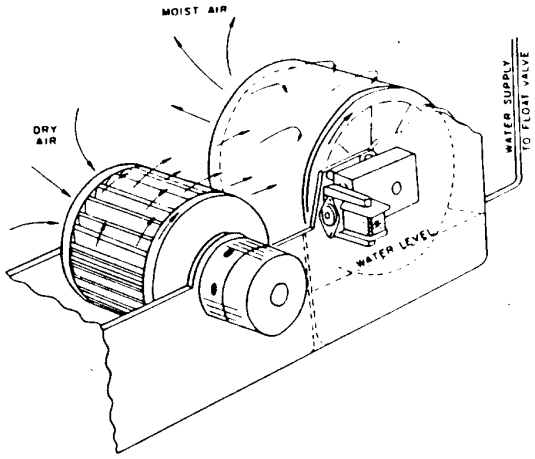
Benefits of evaporative cooling:
large savings possible, "environmentally friendly", inexpensive for direct cooling.
Problems of evaporative cooling:
may not meat peak loads, may increase duct size, may allow wider variation of humidity, potential microbial growth problems.

Example of a system that extends application of direct and indirect coolers. The system better control humidity by combining evaporative and mechanical cooling, and increase energy efficiency by including an economizer.

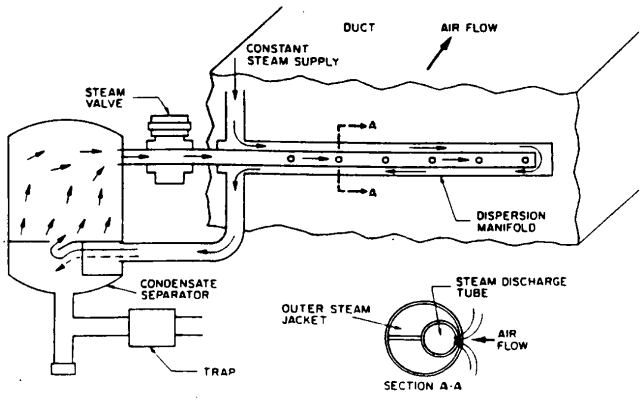


Heating and Humidification Systems

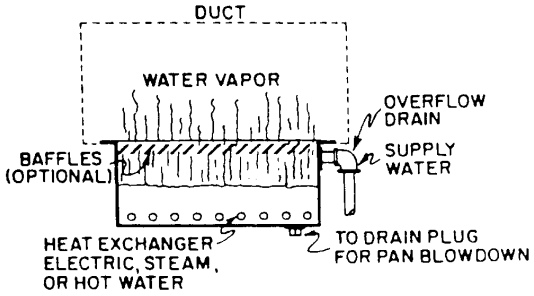
Humidification types: wetted media, heated pan, atomizer and steam.



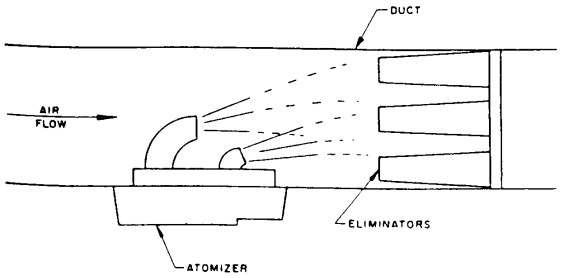
Wetted Drum Humidifier



Jacketed Steam Humidifier



Heated Pan Humidifier



Atomizing Humidifier with Optional Filter Eliminator

2.4.3 Off-Design Conditions

In operation, the cooling or heating loads are only a part of the design loads. HVAC system needs to respond to this lower demand, and some of the strategies are:

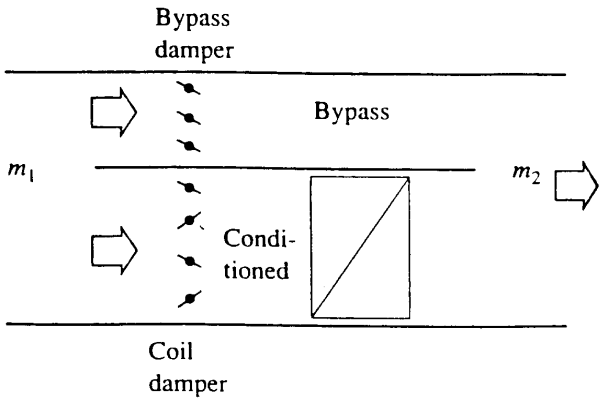
- CAV-RH (Constant Air Volume)
- VAV-RH (Variable Air Volume)
- Face and bypass coil
- Economizer
- Variable T for the heat exchangers

Analyze these processes by the same methods as design conditions.
Problems for cooling systems: thermostat controls T_{DB}, and therefore humidity correct only at design (in general). As a result, space humidity varies with loads, and may need to iterate to space conditions. This is not a problem for heating conditions.
CAV-RH

- same flow rate and W as design conditions, variable T_{DB}
- good control at part loads

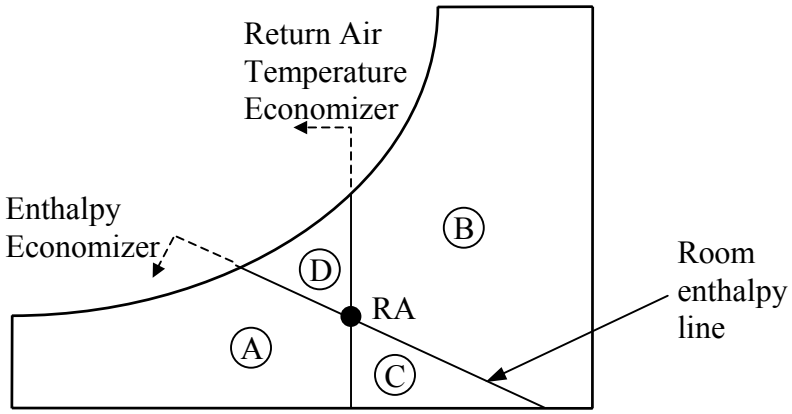
VAV-RH
- lower flow rate in proportion to the sensible loads
- same coil dewpoint temperature => less dehumidification

Face and bypass coil
- bypass fraction is proportional to sensible loads
- no dehumidification of bypassed air => supply humidity is proportional to bypass fract.



Face and Bypass Coil

Economizer
- used in spring or fall
- supplies outdoor air without operating a cooling coil; potential humidity problems
- limit is 100% outdoor air; control humidity rise with reheat



C: RA Economizer is off => lost cooling opportunity
D: RA Economizer is on => energy penalty results
No reliable, durable and inexpensive enthalpy sensor for the enthalpy economizer.

2.5 Psychrometric Analysis of Complete Systems

Space Heating and Cooling Loads

Heating Load — the maximum probable net rate of *heat loss* from a conditioned space which would have to be made up by addition of heat from the heating system to maintain some desired temperature and humidity conditions in the space
Cooling Load — for cooling

Example:

Cooling and heating load of a classroom at PSU with 10 occupants are estimated as follows:

Sensible	Cooling (W)	Heating (W)
Walls	1000	2000
Window (conduction)	1000	2000
Window (radiation)	1000	-
People: 70 W/person	700	-
Lighting	300	-

Latent	Cooling (W)	Heating (W)
People 30 W/person	300	-
Plants	700	

Outdoor Design Conditions		
T	31°C	-14°C
T_{wet}	23°C	-

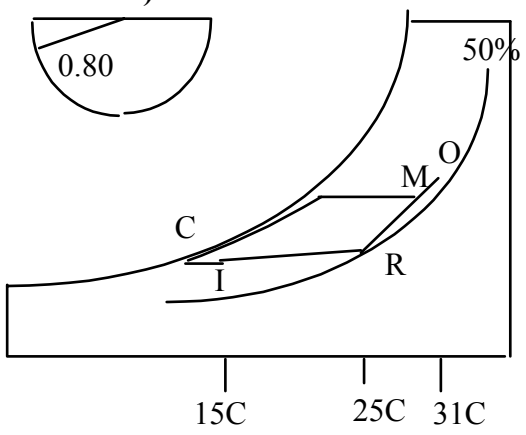
Indoor Design Conditions		
T	25°C	22°C
ϕ	50%	50%

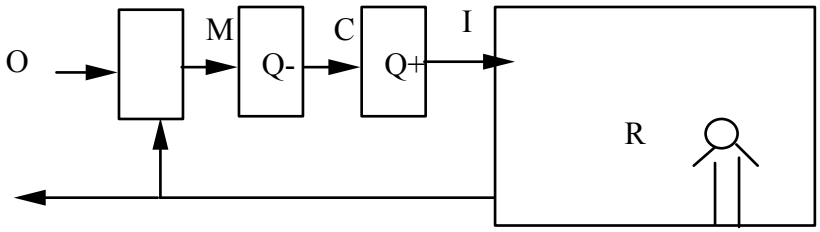
Minimum Outdoor Air		
Fresh air: 8 L/s person	80 L/s	80 L/s

Air Supply Temperatures		
Maximum		60°C
Minimum	15°C	

Design the air-conditioning system.

Solution (Summer cooling conditions):





Determine the enthalpy at all the status. We use the psych chart in sea level, p = 101325 Pa.

Outdoor (O): $T_o = 31^oC$, $T_{o,wet} = 23^oC$

From Table A-1b, $p_{s,wet} = 2815$ Pa

$W_{o,wet} = 0.622\ p_{s,wet}/(p - p_{s,wet})$

$= 0.622 \times 2815/(101325 - 2815) = 0.0178\ kg_v/kg_a$

$$W_o = \frac{C_p(T_{o,wet} - T_o) + W_{o,wet} i_{fg}}{i_o - i_w} = \frac{1.01(23-31) + 0.0178 \times 2447}{2558 - 96} = 0.0144\ kg_v/kg_a$$

$i_o = 1.01T_o + W_o(2501 + 1.86T_o)$

$= 1.01\ kJ/(kg_a\ ^oC) \times 31^oC$

$+ 0.0144\ kg_v/kg_a(2501\ kJ/kg_v + 1.86\ kJ/(kg_v\ ^oC) \times 31^oC)$

$= 68.15\ kJ/kg_a$

Room (R): $T_R = 25\ ^oC$, $\phi = 50\%$

From Table A-1B, $p_{s,R} = 3174$ Pa

$W_R = 0.622\ \phi\ p_{s,R}/(p - \phi\ p_{s,R})$

$= 0.622 \times 0.5 \times 3174\ /(101325 - 3174)$

$= 0.01\ kg_v/kg_a$

$i_R = 1.01T_R + W_R(2501 + 1.86T_R)$

$= 1.01\ kJ/(kg_a\ ^oC) \times 25^oC$

$+ 0.01\ kg_v/kg_a(2501\ kJ/kg_v + 1.86\ kJ/(kg_v\ ^oC) \times 25^oC)$

$= 50.72\ kJ/kg_a$

Mixture (M): $m_a = \frac{Q_{sensible}}{C_p(T_R - T_I)} = \frac{4000\,W}{1.01x1000^3\,J/kg_a(25^oC - 15^oC)} = 0.396kg/s$

$m_o = 80$ L/s = $80x10^{-3}$ m^3/s x 1.2 kg_a/m^3 = 0.096 kg_a/s

$m_R = m_a - m_o = 0.396 - 0.096 = 0.3$ kg_a/s

$i_M = \frac{m_o i_o + m_R i_R}{m_o + m_R} = \frac{0.096 \times 68.15 + 0.3 \times 50.72}{0.096 + 0.3} = 54.95kJ/kg$

Supply air at the inlet (I): $T_I = 15\ ^oC$

$\Delta W = m_a(W_R - W_I) = Q_{latent}/i_{fg}$

$W_I = W_R - Q_{latent}/(i_{fg}\ m_a)$=

= 0.01kg_v/kg_a - 1 kW /(2454 kJ/kg_v x 0.396 kg_a/s) = 0.009 kg_v/kg_a

$i_I = 1.01T_I + W_I(2501 + 1.86T_I)$

= 1.01 x 15 + 0.009(2501 + 1.86 x 15)

= 37.91 kJ/kg_a

Cooling coil (C): $\phi = 90\%$

$W_c = W_I = 0.009$ kg_v/kg_a

$W_C = 0.622\frac{p_C}{p - p_C}$

$0.009 = 0.622\frac{p_C}{101325 - p_C}$

p_C= 1447.5 Pa

$p_{s,C} = p_C / \phi$ = 1447.5 / 0.9= 1608 Pa

From Table A1-b, $T_C = 14\ ^oC$

$i_C = 1.01T_C + W_C(2501 + 1.86T_C)$

= 1.01 x 14 + 0.009 (2501 + 1.86 x 14)

= 36.88 kJ/kg_a

Fan: $\dot{V} = \dot{m}_a\ v$ = 0.396 kg/s x 0.84 m^3/kg = 0.332 m^3/s = 1200 m^3/hr

Cooling coil: $Q_{cooling} = \dot{m}_a\ (i_M - i_C)$ = 0.396 kg_a/s (54.95 kJ/kg_a - 36.88 kJ/kg_a)

= 7.156 kW

Heating coil: $Q_{heating} = \dot{m}_a\ (i_I - i_C)$ = 0.396 kg_a/s (37.91 kJ/kg_a - 36.88 kJ/kg_a)

= 0.4 kW

The capacity of the heating coil will be larger in winter. Therefore, the final size of the equipment should be the greater of the summer and winter capacities. In many cases, economizers are used to recover energy. Then re-heat in the present design becomes unnecessary.

Chapter 3 Design Conditions

3.1 Outdoor design conditions
3.2 Indoor design conditions

Readings: Chapter 4 and pp. 191-192 of the text
Chapters 7, 8, 9, 12, 13 & 26 of ASHRAE Handbook-Fundamentals, 1997
ASHRAE 62-1999: IAQ Standard
ASHRAE 55-1992: Thermal Comfort Standard

3.1 Outdoor Design Conditions

3.1.1 Winter design conditions

Use Table B-1 in the textbook or Table 1A of ASHRAE Fundamental Handbook Chapter26.

The 99.6% and 99% indicate the risk level desired. When 99% is selected, it means the outdoor temperatures have been equaled or exceeded by 99% of the total number of hours in a year (8760 hrs).

99.6% (0.4%) ~ 35 hrs
99.0% (1.0%) ~ 88 hrs
98.0 (2.0%) ~ 175 hrs
95.0 (5.0%) ~ 438 hrs

Design conditions - Column 2
Wind - Column 3
Peak load for infiltration - Columns 4 and 5

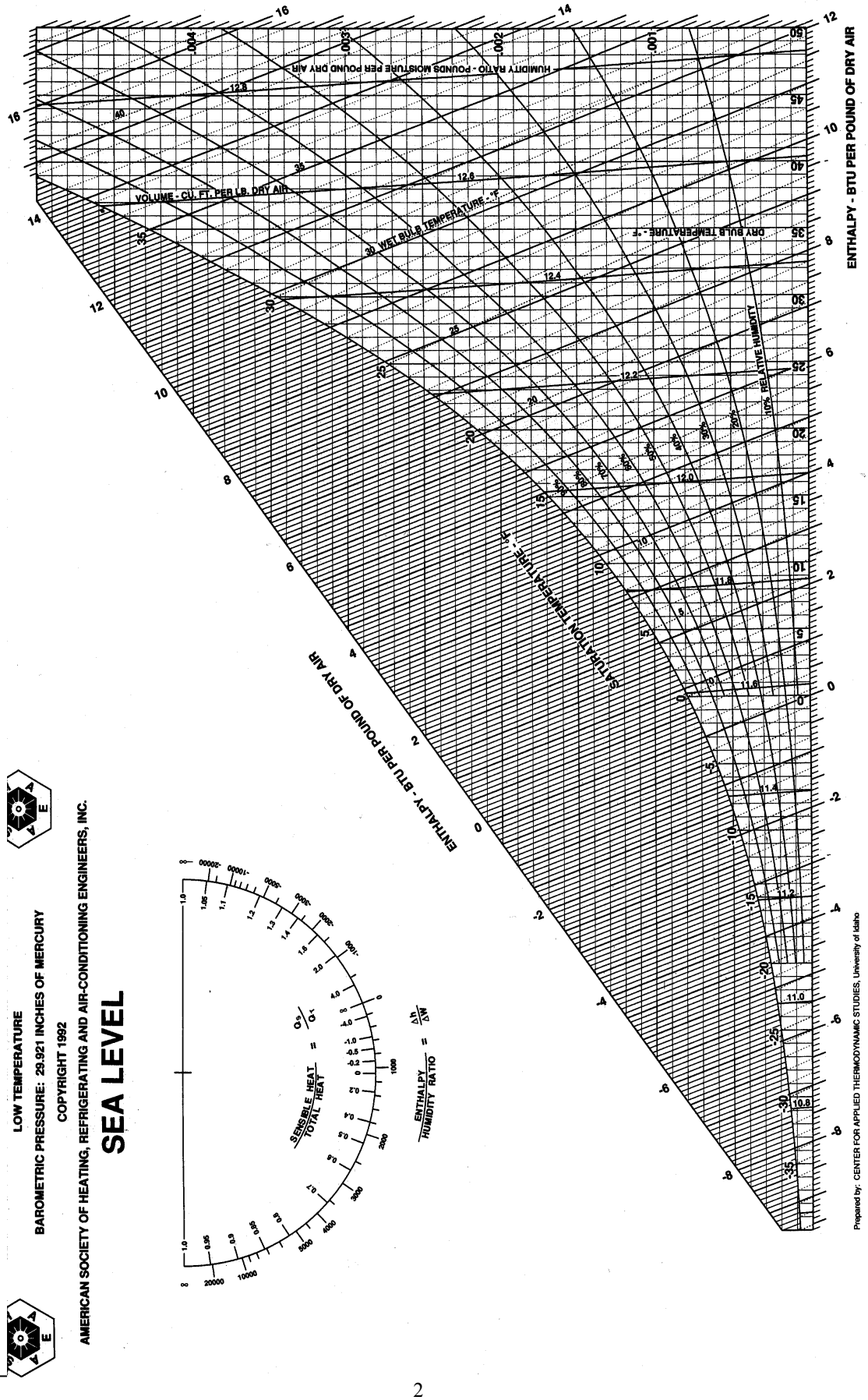
ASHRAE Handbook 1997 and the textbook use annual percentiles of 99.6% and 99%.
Design rule of thumb: design outdoor relative humidity in winter 60%

3.1.2 Summer design conditions

Use Table B-1 in the textbook or Table 1B in ASHRAE Fundamental Handbook Chapter26.

DB is dry-bulb temperature and MWB is the mean-coincident-wet-bulb temperature. The 0.4%, 1% and 2% mean the percentile of the total hours may not meet indoor design conditions.

Design conditions - Column 2
Cooling towers - Column 3
Peak moisture load - Column 4


LOW TEMPERATURE
BAROMETRIC PRESSURE: 29.921 INCHES OF MERCURY
COPYRIGHT 1992
AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS, INC.
SEA LEVEL
ENTHALPY - BTU PER POUND OF DRY AIR
DRY BULB TEMPERATURE - °F
HUMIDITY RATIO - POUNDS MOISTURE PER POUND DRY AIR
VOLUME - CU. FT. PER LB. DRY AIR
WET BULB TEMPERATURE - °F
SATURATION TEMPERATURE - °F
RELATIVE HUMIDITY
SENSIBLE HEAT / TOTAL HEAT
ENTHALPY / HUMIDITY RATIO
Prepared by: CENTER FOR APPLIED THERMODYNAMIC STUDIES, University of Idaho

3.2 Indoor Design Conditions

Indoor air quality (IAQ):

defines safety and comfort effect of contaminants on indoor environment.

Thermal Comfort:

defines effects of temperature, humidity, air motion on perceived quality of surrounding

Noise level:

Space type	Noise Criteria (NC)
Radio and TV broadcasting rooms	NC 20-30
Theaters, concert halls, meeting rooms	NC 20-30
Private offices	NC 35
Gymnasiums	NC 40-50
Workshops	NC 45-70

Pressure:

$p_{room} - p_{environment} = 5\text{-}10$ Pa

3.2.1 Indoor air quality

Good **IAQ**: a clean, healthy, and odor-free indoor environment.

"Sick Building Syndrome" (SBS) is discomfort/illness caused by indoor air.
SBS is often comparable to a cold or influenza

○

A few facts about poor IAQ:

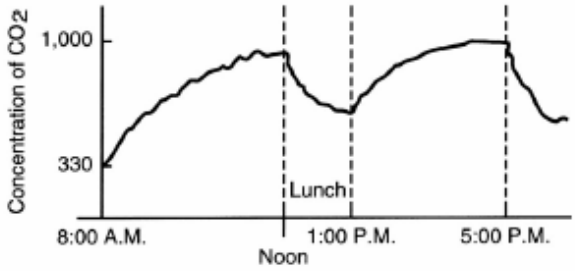
Health problems: 30 – 70 million people (U.S. EPA)
Health complaints: 30% new and remodeled buildings (U.S. EPA)
Occupational illness: 439,000 cases in 1996 (NIOSH)
Economic loss: $40 – 120 billion/year ($200-600/person.year)

Causes of sickness:

- COx, NOx and other common gases (human, combustion)
- Environmental tobacco smoke ETS (smoking)
- Volatile organic compounds VOC
  (building materials: construction materials, furnishings, finishes)
- Particulate matters (outdoor air, activities, ETS)
- Biohazards (molds, bacteria, etc.)
- Radon (soil)

Contaminants	Sources	Permitted level	Health effects
CO_2 CO SO_x NO_x Ra	Human, combustion Combustion, ETS Combustion Combustion Soil	1000 ppm 15 ppm 100 $\mu g/m^3$ 4 picocuries/l	Stuffing Body chemistry Irritation, asthma Not very clear Lung cancer
VOCs (Formaldehyde)	Combustion, pesticides, building materials, etc.	0.1 ppm	Eyes and mucous membrane irritation
Particulate (0.01 micro-insects)	Outdoor air, activities, ETS, furnishings, pets, etc		Lung diseases Cancer (ETS)

The following diagram illustrates the buildup of indoor carbon dioxide (due to occupant exhalation) throughout a normal day:



Carbon Dioxide Concentration (ppm)	Results
250-350	normal outdoor ambient air
600	minimal air quality complaints
600-1000	effect on indoor air quality is less clearly interpreted
1000	indicates inadequate ventilation; indoor air quality complaints are more widespread

Common Causes of Poor Indoor Air Quality (IAQ)

Tight buildings (Energy conservation)=>
=>Less outdoor air (Infiltration) =>
=>High contaminant concentrations (New building materials release VOCs)

Numerous indoor air quality investigations over the last decade by the National Institute for Occupational Safety & Health (NIOSH) have found the primary source of indoor air quality problems are:

- Inadequate ventilation 52%
- Contaminant from inside the building 16%
- Contaminant from outside the building 10%
- Microbial contamination 5%
- Contamination from building fabric 4%
- Unknown Sources 13%

Control of contaminants

○

New parameters

- Supply air - Total air supplied to a space
- Fresh air - Outdoor air
- Infiltration - Uncontrolled air entered to a space
- Exfiltration - Uncontrolled air left a space

For mechanical ventilated space, infiltration should be zero because of positive pressure in the conditioned space.

Ventilation

Most common method for contaminant control is ventilation. Ventilation can be either natural or forced. Ventilation dilutes contaminants with outdoor air, and requirements are defined by state/local codes and referenced standards.

ANSI/ASHRAE Standard 62-1999:
Ventilation for Acceptable Indoor Air Quality

Defines:
- Acceptable outdoor air quality
- Procedure for acceptable ventilation

1. Ventilation Rate Procedure

Fixed OA requirements:
- Per person/per m^2
- Varies with occupancy type

The standard prescribes the rate at which outdoor air must be delivered to a space.
Example: Office space 8 l/(s person) (15 cfm/person).

Basis:
- Presumed CO_2 concentration < 1000 ppm
- Sufficient outdoor air to remove odors

The standard uses CO_2 as indicator of IAQ since CO_2 is "marker" for human contaminants.
Typical quantities 8-10 l/(s person) (15-20 cfm/person)

2. Indoor Air Quality Procedure

This procedure limits measured contaminant concentrations.
Must sense specific contaminants.

Ventilation rate procedure predominates.

Pros:
- Prescriptive
- Low capital cost

Cons:
- May fail off-design with VAV
- Energy cost

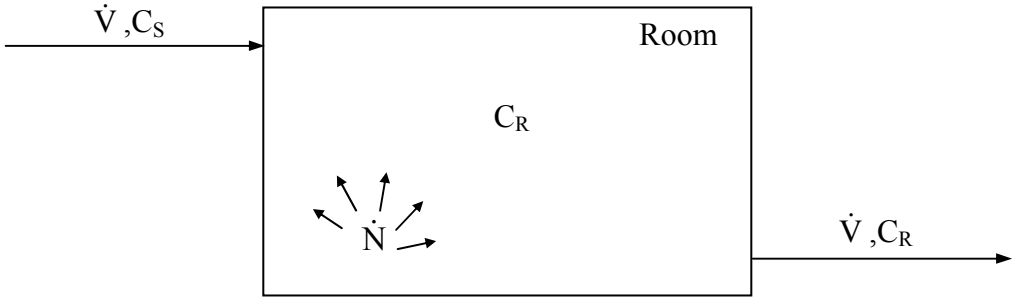
Perfect dilution (Steady State Ventilation Conditions)

"Well-Mixed Spaces":

- Perfect mixing of supply/space
- Exhaust at mixed condition
- Constant contaminant generation rate
- Fixed supply flow rate and conditions

Mass balance for contaminants: Mass IN + Generation = Mass OUT

$\dot{V}$ - Supply air flow rate (m_a^3/s)
C_S - Contaminant concentration of the supplied air (m_c^3/m_a^3)
$\dot{N}$ - Contaminant sources generated within the space (m_c^3/s)
C_R - Average contaminant concentration in the room (m_c^3/m_a^3), (ppm = $10^{-6} \times m_c^3/m_a^3$)



Example

In a French home, the CO_2 concentration in a bedroom was 4000 ppm. The bedroom size is 12 m^2 and room height 2.5 m. Find the air change rate if two occupants were in the bedroom. Suppose that the outdoor CO_2 level was 300 ppm and a person breathes out 0.30 L/min. CO_2.

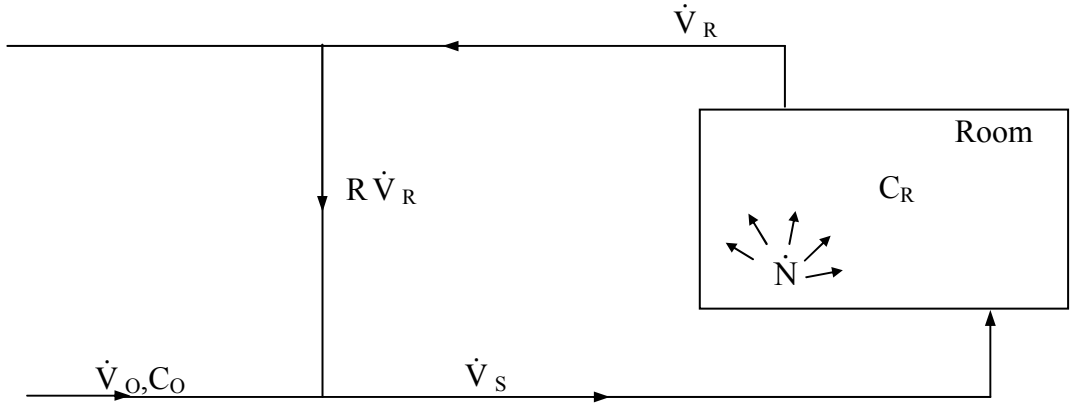
Solution:

Air change rate - ACH (Air Change rate per Hour):

ACH = $\dot{V}$ / (Room volume) = 9.72 m^3/hr / (12 m^2 x 2.5 m) = 0.324 ACH

Ventilation with Recalculation (Single Space)

If outdoor air is mixed with the return air, the supply contaminant concentration has to be determined form the contaminant balance for the mixing process.



Steady state: $\dot{V}_S = \dot{V}_R$, $R\dot{V}_S = R\dot{V}_R$, $\dot{V}_O = (1\text{-}R)\ \dot{V}_S$

Mass balance: $\dot{V}_S C_R = \dot{V}_O C_O + R\dot{V}_R C_R + \dot{N} = (1\text{-}R)\ \dot{V}_S C_O + R\dot{V}_S C_R + \dot{N}$

Space concentration:

For a single space: R=>0 minimum concentration
R=>1 concentration => ∞

System Serving Multiple Spaces

HVAC systems are usually serving multiple spaces that have different requirements for airflow rate of fresh air. The airflow rate requirements for the fresh (outdoor) air $\dot{V}_{O,i}$ are determined from Table 2 (Standard 62) that is presented as Table 4-5 in the textbook (M&P).

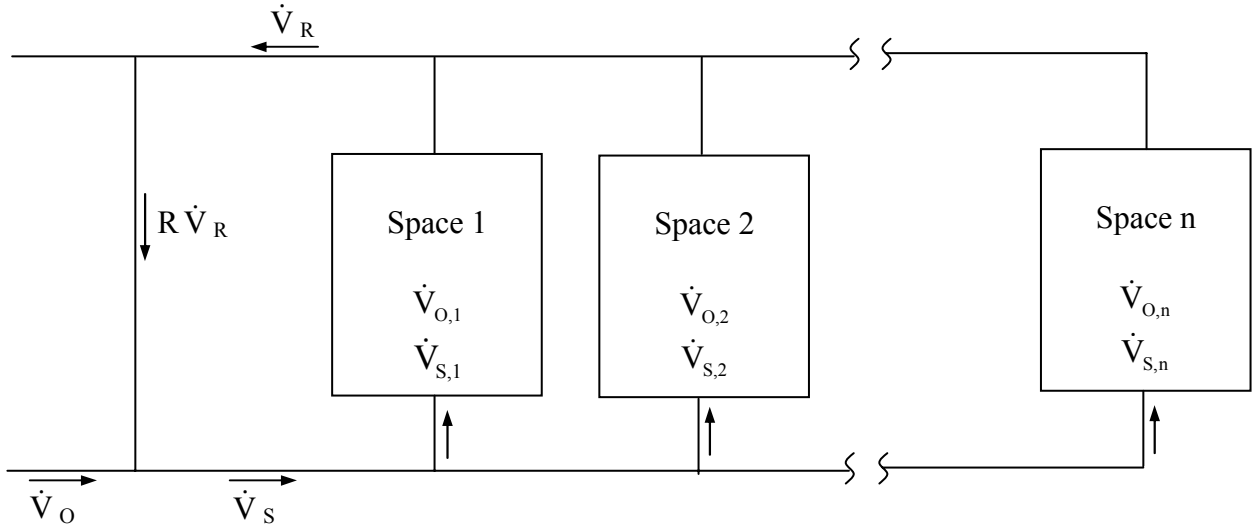
A total supply flow rate for each space $\dot{V}_{S,i}$ is defined from heating/cooling load.

Fraction of outdoor air OA in supply is defined as: $\dfrac{\dot{V}_{O,i}}{\dot{V}_{S,i}}$

A **Critical Space** has outdoor air fraction "**Z**":

$$Z \equiv \left.\frac{\dot{V}_{O,i}}{\dot{V}_{S,i}}\right|_{MAX}$$

Z is the maximum fraction of OA required.



An average fraction of OA for all spaces "**X**" is:

○

Supply OA fraction X => Some spaces under-ventilated

Supply OA fraction Z => { No space under-ventilated
Most spaces over-ventilated
Energy wasted

For over-ventilated space:

- Airflow rate of fresh air is higher than required
- "Unused" fresh air returns to AHU

Define "corrected" OA fraction "**Y**" as:

$$Y \equiv \frac{\dot{V}_{OC}}{\dot{V}_S}$$

Y is OA fraction that accounts for "unused" fresh air.

Fresh air balance:

- Return and supply flows are equal
- Recirculated fraction is R

Critical space is satisfied if:

○

Required OA fraction = Z (critical space)

"Unused" OA in return = (Z-X) $\dot{V}_S$

Recirculated fraction of return:

$$R \equiv \frac{\dot{V}_S - \dot{V}_{OC}}{\dot{V}_S} = 1 - Y$$

Recirculated OA = $R(Z - X)\dot{V}_S = (1 - Y)(Z - X)\dot{V}_S$

Critical space is satisfied if:

$$\dot{V}_{OC} + (1 - Y)(Z - X)\dot{V}_S = Z\dot{V}_S$$

$$Y + (1 - Y)(Z - X) = Z$$

"Multiple Spaces Equation":

○

The equation is used to calculate the corrected fraction of outdoor air that takes into account fraction of the recirculated fresh air. In some cases, the saving are significant if we compare Y and Z.

Example

Four spaces are air-conditioned from a central AHU. The following table gives airflow rates for supply air and fresh air:

Spaces	1	2	3	4	Total
Supply air [cfm]	500	400	600	500	2000
Fresh air [cfm]	200	80	80	75	435
Fresh/Supply	0.40	0.20	0.13	0.15	0.22

Calculate the required OA fraction?

Solution:

Average fraction of OA:
OA fraction for the critical space 1:

Corrected fraction of OA:

The required flow rate of OA is

Notice: X < Y <Z
435 cfm < 540 cfm < 800 cfm

Standard 62-1999 recognizes transients, i.e. high occupancy/low use spaces such as conference rooms. If the peak occupancy is used to determine required flow rate of OA, calculated Y is unnecessary high.

To compensate, may:

- Ventilate based on average occupancy
- Lag ventilation start
- Control OA with CO_2 concentration

Removal of Contaminants

Air Cleaner can take place in ducts or in spaces.

Gas and odor removal

Absorption (solid/liquid absorbers, air washer)
Adsorption (activated charcoal filters)
Chemisorption (by chemical reaction)
e.g.: UV+TiO_2 photocatalytic oxidization of VOCs

Particulate removal by filtering

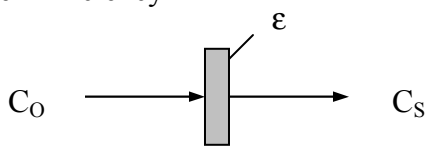
Particles:
Size and shape
Specific gravity
Concentration
Electrical properties

Air cleaners

Types:

- Fibrous media unit filter
- Renewable media filter
- Electronic air cleaners
- Combination air cleaners

Filter Efficiency



Air-flow resistance (loss of total P at a given flow rate) –operating costs

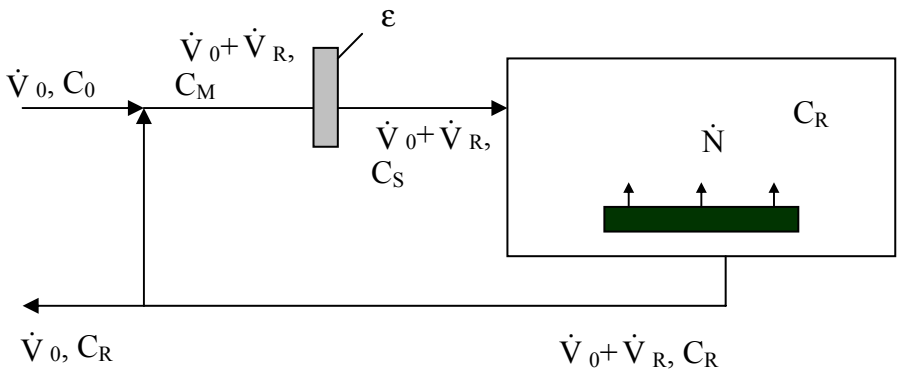
$\Delta P \sim V^2$

Filtration can reduce OA (outdoor air) requirement.
System with recirculation should use filters.

	Media Filtration	HEPA Filters	Activated Carbon Adsorption	UV Photocatalytic Oxidation
Odors			×	×
VOCs			×	×
Bio-Aerosols		×		×
Dust	×	×		

Example

Given $\dot{N}$, C_0, $\dot{V}_0$, $\dot{V}_R$
Find C_R



Solution:

For the room (mass conservation):

○ (1)

For the filter:

○ (2)

For the mixing:

○ (3)

(2)+(3):

$$C_S = \frac{\dot{V}_0 C_0 + \dot{V}_R C_R}{\dot{V}_0 + \dot{V}_R}(1-\varepsilon) \qquad (4)$$

(1)+(4):

$$C_R = \frac{\dot{V}_0 C_0 (1-\varepsilon) + \dot{N}}{\dot{V}_0 + \dot{V}_R \varepsilon} \qquad (5)$$

3.2.2 Thermal comfort

"That condition of mind in which satisfaction is expressed with the thermal environment."

ANSI/ASHRAE Standard 55-1992:
Thermal Environmental Conditions for Human Occupancy

The following gives a few facts about human beings:

- Heat generated from a human body:
  -
- Human body temperature:
  -
- One would wake up from sleep if this person loses 24 W or more. In this case, the skin temperature decreases 2.8 ^{o}C.
- One would feel uncomfortable or sick, if the person's body temperature is 1 K higher.
-
-

Human body energy equation

Human body obeys the first law of thermodynamics. Energy balance for human body:

Heat production – Mechanical work = Heat losses

M - Rate of metabolic heat production (W/m^2 body surface area) (Table 4-1, textbook)
W - Rate of mechanical work
Q - Heat losses
C - Convective heat losses
R - Radiative heat losses
E - Evaporative heat losses
sk - Skin
res - Respiration

Heat production: measured in metabolic or met units.
1 met = 58.2 W/m^2 = 18.4 Btu/(h·ft^2) body surface area

Table 4 Typical Metabolic Heat Generation for Various Activities

	W/m^2	met[a]
Resting		
Sleeping	40	0.7
Reclining	45	0.8
Seated, quiet	60	1.0
Standing, relaxed	70	1.2
Walking (on level surface)		
3.2 km/h (0.9 m/s)	115	2.0
4.3 km/h (1.2 m/s)	150	2.6
6.4 km/h (1.8 m/s)	220	3.8
Office Activities		
Reading, seated	55	1.0
Writing	60	1.0
Typing	65	1.1
Filing, seated	70	1.2
Filing, standing	80	1.4
Walking about	100	1.7
Lifting/packing	120	2.1
Driving/Flying		
Car	60-115	1.0-2.0
Aircraft, routine	70	1.2
Aircraft, instrument landing	105	1.8
Aircraft, combat	140	2.4
Heavy vehicle	185	3.2
Miscellaneous Occupational Activities		
Cooking	95-115	1.6-2.0
Housecleaning	115-200	2.0-3.4
Seated, heavy limb movement	130	2.2
Machine work		
sawing (table saw)	105	1.8
light (electrical industry)	115-140	2.0-2.4
heavy	235	4.0
Handling 50-kg bags	235	4.0
Pick and shovel work	235-280	4.0-4.8
Miscellaneous Leisure Activities		
Dancing, social	140-255	2.4-4.4
Calisthenics/exercise	175-235	3.0-4.0
Tennis, singles	210-270	3.6-4.0
Basketball	290-440	5.0-7.6
Wrestling, competitive	410-505	7.0-8.7

Compiled from various sources. For additional information, see Buskirk (1960), Passmore and Durnin (1967), and Webb (1964).

[a] 1 met = 58.2 W/m^2

Environmental and personal comfort variables

Environmental factors

Air temperature ~
Relative humidity ~
Air velocity near a human body, V ~
Surface temperature of the enclosure and other objects ~

Personal factors

Activity (rate of metabolic)
Clothing (insulation)
1 clo = 0.88 F-ft2-hr/Btu

Determination of individual heat loss/gain

W = Active work and shivering (involuntary work) (W/m^2 body area)

M = Rate of metabolic heat production (W/m^2)
(M is given in Table 4-1 of the textbook. 1 met = 58.2 W/m^2 body area)

$C_{sk} = h_c (T_{cloth} - T_{air}) A_{cloth}/A_{body}$ (W/m^2 body area)

$h_c = 2.38 (T_{cloth} - T_{air})^{0.25}$ when $2.38 (T_{cloth} - T_{air})^{0.25} > 12.1 V^{0.5}$
$h_c = 12.1 V^{0.5}$ when $2.38 (T_{cloth} - T_{air})^{0.25} < 12.1 V^{0.5}$

$$T_{cloth} = 35.7 - 0.0275 (M - W) - R_{cloth} \{(M - W) - 3.05 [5.73 - 0.007 (M - W) - p_v] - 0.42 [(M - W) - 58.15] - 0.0173 M (5.87 - p_v) - 0.0014 M (34 - T_{air})\}$$

$A_{body} = 0.202 m^{0.425} l^{0.725}$ (m^2)
m - Body weight (kg)
l - Height (m)

A_{cloth}/A_{body} = f(Garment insulation value)
$= 1.0 + 0.3 I_{cl}$

R_{cloth} = Cloth thermal resistance (m^2 K/ W)
$R_{cloth} = 0.155 I_{cl}$
1 clo = 0.155 m^2 K/W

I_{cl} can be found in Table 8, Chapter 8 of ASHRAE Fundamentals, 1997.

$R_{sk} = \sigma \varepsilon_{cloth} \varepsilon_{enclosure} [(T_{cloth} + 273)^4 - (T_{enclosure} + 273)^4] A_{cloth}/A_{body}$

σ - Stefan-Boltzmann constant (5.67×10^{-8} W/m^2K^4)
ε - Emittance (a value between 0 and 1)
$R_{sk} = 3.96 \times 10^{-8} [(T_{cloth} + 273)^4 - (T_{enclosure} + 273)^4] A_{cloth}/A_{body}$ (W/m^2)

$E_{sk} = m_{sk} i_{fg}$
$= 3.05 [5.73 - 0.007 (M - W) - p_v]$
$+ 0.42 [(M - W) - 58.15]$ (W/m^2)

p_v - Vapor pressure (kPa)

$C_{res} = m_{res} C_{p,a} (T_{res} - T_{air})$
$= 0.0014 M (34 - T_{air})$ (W/m^2)

$E_{res} = m_{res} i_{fg}$
$= 0.0173 M (5.87 - p_v)$ (W/m^2)

Note: Body energy balance ≠ Always thermally comfort
Thermal comfort = Always body energy balance
Only in a limited range of environmental parameters, thermal comfort can be achieved.

Prediction of thermal comfort

- Predicted Mean Vote (PMV)

		+ 3	hot
		+ 2	warm
		+ 1	slightly warm
PMV	=	0	neutral
		-1	slightly cool
		-2	cool
		-3	cold

○

L - Thermal load on the body
L = Internal heat production - heat loss to the actual environment

○

- Predicted Percentage Dissatisfied (PPD)

$$PPD = 100 - 95 \exp [- (0.03353\ PMV^4 + 0.2179\ PMV^2)]$$

PMV	PPD
0	5%
± 0.5	10%
± 1.0	25%

Example:

The environmental parameters in an office are:
$T_{air} = 25\ ^oC$ $T_{wall} = 25\ ^oC$
$p_v = 2500$ Pa $h_c = 3\ W/m^2K$
$m_{person} = 70$ kg $l_{person} = 1.86$ m
$I_{cl,\ person} = 1.0$ clo (Business suit)

Determine the comfort level in the office.

Solution:

From Table 4-1 of the textbook (or handouts), $M = 60\ W/m^2$
Since T_{air} is 25 ^oC and person does not do physical work, $W = 0$.

$$R_{cloth} = 0.155\ I = 0.155 \times 1.0 = 0.155\ m^2K/W$$

$$A_{body} = 0.202\ m^{0.425}\ l^{0.725}$$
$$= 0.202 \times 70^{0.425} \times 1.86^{0.725} = 1.927\ m^2$$

$$A_{cloth}/A_{body} = 1.0 + 0.3\ I = 1.0 + 0.3 \times 1.0 = 1.3$$

$$T_{cloth} = 35.7 - 0.0275\ (M - W) - R_{cloth}\ \{(M - W)$$
$$- 3.05\ [\ 5.73 - 0.007\ (M - W) - p_v]$$
$$- 0.42\ [\ (M - W) - 58.15] - 0.0173\ M\ (5.87 - p_v)$$
$$- 0.0014\ M\ (34 - T_{air})\}$$
$$= 35.7 - 0.0275 \times (60 - 0) - 0.155\{(60 - 0)$$
$$- 3.05\ [\ 5.73 - 0.007\ (60 - 0) - 2.5]$$
$$- 0.42\ [\ (60 - 0) - 58.15] - 0.0173 \times 60\ (5.87 - 2.5)$$
$$- 0.0014 \times 60\ (34 - 25)\}$$
$$= 26.86\ ^oC$$

$$C_{sk} = h_c\ (T_{cloth} - T_{air})\ A_{cloth}/A_{body}$$
$$= 3 \times (26.85 - 25) \times 1.3$$
$$= 7.254\ W/m^2$$

R_{sk} $= 3.96x10^{-8} [(T_{cloth} + 273)^4 - (T_{enclosure} + 273)^4] A_{cloth}/A_{body}$
$= 3.96x10^{-8} [(26.86 + 273)^4 - (25+273)^4] \times 1.3$
$= 10.23 \text{ W/m}^2$

E_{sk} $= 3.05 [5.73 - 0.007 (M - W) - p_v]$
$+ 0.42 [(M - W) - 58.15]$
$= 3.05 [5.73 - 0.007 (60 - 0) - 2.5]$
$+ 0.42 [(60 - 0) - 58.15]$
$= 9.348 \text{ W/m}^2$

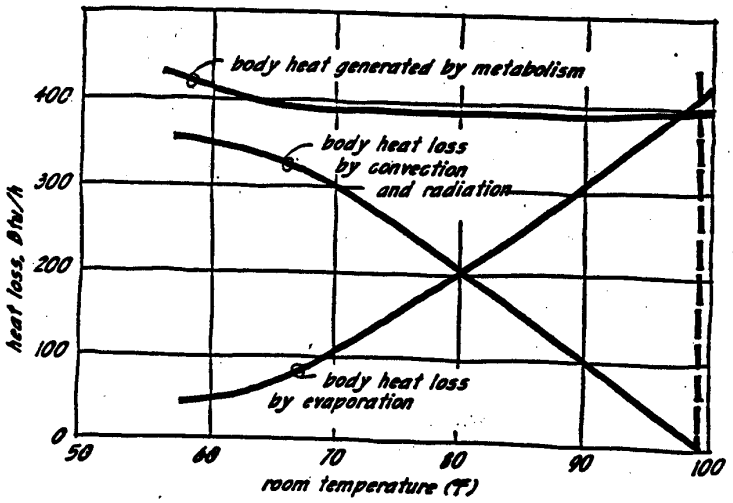
C_{res} $= 0.0014 M (34 - T_{air})$
$= 0.0014 \times 60 (34 - 25)$
$= 0.756 \text{ W/m}^2$

E_{res} $= 0.0173 M (5.87 - p_v)$
$= 0.173 \times 60 (5.87 - 2.5)$
$= 3.50 \text{ W/m}^2$

L $= M - W - [(C_{sk} + R_{sk} + E_{sk}) + (C_{res} + E_{res})]$
$= 60 - 0 - [(7.254 + 10.23 + 9.348) + (0.756 + 3.5)]$
$= 29 \text{ W/m}^2$

PMV $= [0.303 \exp (-0.036 M) + 0.028] L$
$= [0.303 \exp (-0.036 \times 60) + 0.028] \times 29$
$= 1.825$ => Warm

PPD $= 100 - 95 \exp [- (0.03353 PMV^4 + 0.2179 PMV^2)]$
$= 100 - 95 \exp [- (0.03353 \times 1.825^4 + 0.2179 \times 1.825^2)]$
$= 68\%$



ure 2.4: At lower temperatures heat loss by convection and radiation dominate, but ıve 80°F air and surface temperature evaporation begins to dominate. *(After Flynn f Segil, 1970.)*

Table 2.1: Clo Values for Various Men's Clothing Ensembles

Clothing Ensemble	*clo value*
Nude	0
Shorts	0.1
Tropical clothing: shorts, open-neck short-sleeved shirt, light socks and sandals	0.3 - 0.4
Light summer clothing: long light-weight trousers, open-neck short-sleeved shirt	0.5
Light working clothing: athletic shorts, wool socks, open-neck short-sleeved cotton shirt, long cotton trousers	0.6
Light outdoor sportswear: T-shirt, cotton shirt, cotton undershorts, cotton trousers, cotton socks, shoes; and single-ply, light-weight poplin jacket	0.7
Typical business suit	1.0
Typical business suit with cotton topcoat (or heavy wool suit without topcoat)	1.5
U.S. Army standard cold-wet uniform: cotton-wool undershirt and undershorts, wool/nylon flannel shirt, wind-resistant, water-repellent trousers and field coat, cloth mohair and wool coat, and wool socks	1.5 - 2.0
Heavy wool pile ensemble: (Polar weather suit)	3.0 - 4.0

(Reproduced from Fanger, 1982, by permission.)

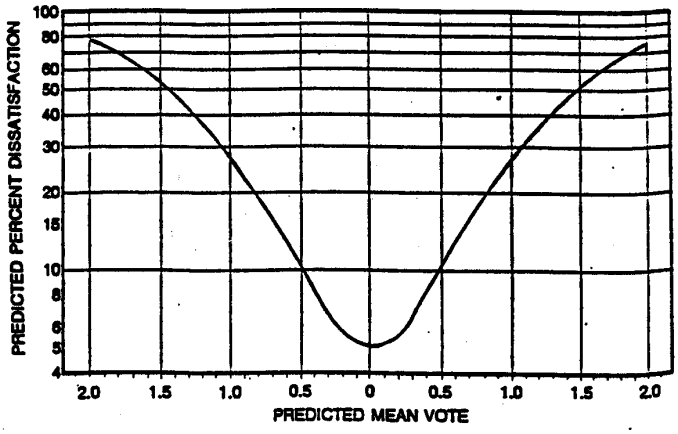


Fig. 11 Predicted Percentage of Dissatisfied (PPD) as Function of Predicted Mean Vote (PMV)

ASHRAE comfort zone

- Operative temperature (T_o)
- Effective temperature (ET*)
- ASHRAE comfort zone

Operative temperature (T_o):

○

T_{mrt} - Mean radiant temperature

$T_{mrt} = \Sigma A_i T_i / \Sigma A_i$

T_i - Surface temperature of enclosure i
A_i - Area of surface i

Effective temperature (ET*):

The temperature of an environment at 50% relative humidity that results in the same total (sensible + latent) heat loss from the skin as in the actual environment. It combines operative temperature and humidity into a single index.

ASHRAE comfort zone (Fig. 4-2):

The "comfort zone" represents combinations of air temperature and relative humidity that most often produce comfort for a seated North American adult in shirtsleeves, in the shade.

Winter: T_o = 20-23.5°C (68-74°F) at 60% relative humidity and T_o = 20.5-24.5°C (69-76°F) at 2°C (36°F) dew point. Slanting side boundaries correspond to ET* of 20°C (68°F) and 23.5°C (74°F).

Summer: To = 22.5-26°C (73-79°F) at 60% relative humidity and T_o = 23.5-27°C (74-80°F) at 2°C (36°F) dew point. Slanting side boundaries correspond to ET* of 23°C (73°F) and 26°C (79°F).

Conditions: Typical summer and winter clothing
Light and primarily sedentary activity (<=1.2 met)
10% dissatisfaction

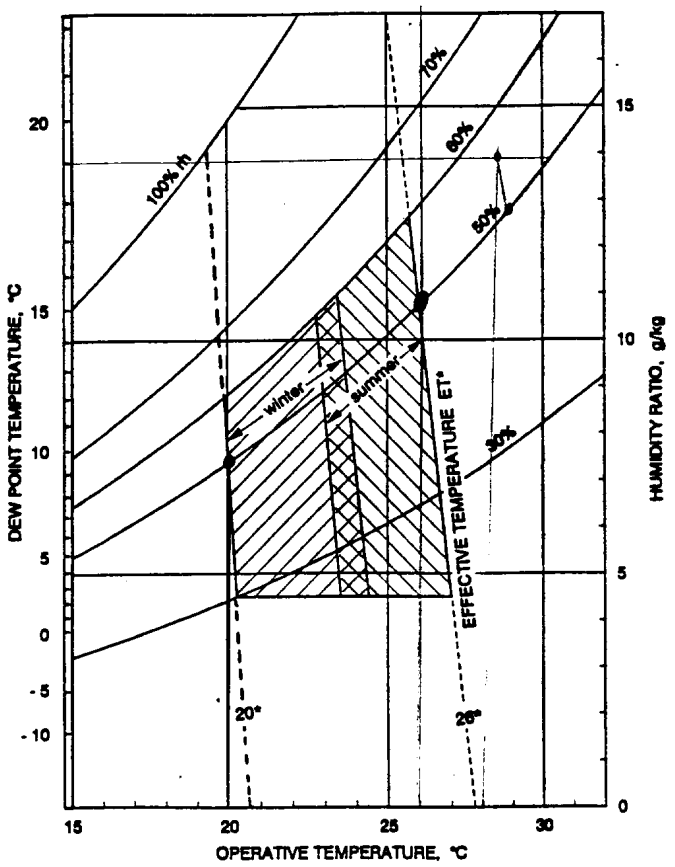


Fig. 5 Standard Effective Temperature and ASHRAE Comfort Zones

Adjustment of the comfort zone

(1). Clothing (e.g., minimum clothing such as briefs T_o = 26-29°C (79-84°F) (Table 4-3)
(2). Air speed V↑ -> To↑ (Fig. 4-3)
(3). Activity (met)↑ -> To↓ (Fig. 4-4)
(4) Adaptation

Draft

Draft is related to air temperature, air velocity, and turbulence intensity. Percentage dissatisfied people due to draft can be expressed as:

$$PD = 3.143\,(34 - T_{air})\,(V - 0.05)^{0.622} + 0.3696\,(34 - T_{air})\,(V - 0.05)^{0.622}\,V\,Tu$$

When V < 0.05 m/s, use V = 0.05.
When PD > 100%, insert PD = 100%.

PD - Percentage dissatisfied people due to draft (%)
V - Air velocity (m/s)
T_{air} - Air temperature (^oC)
Tu - Turbulence intensity (%)

$Tu = 100\ V_{standard\ deviation} / V$ (%)

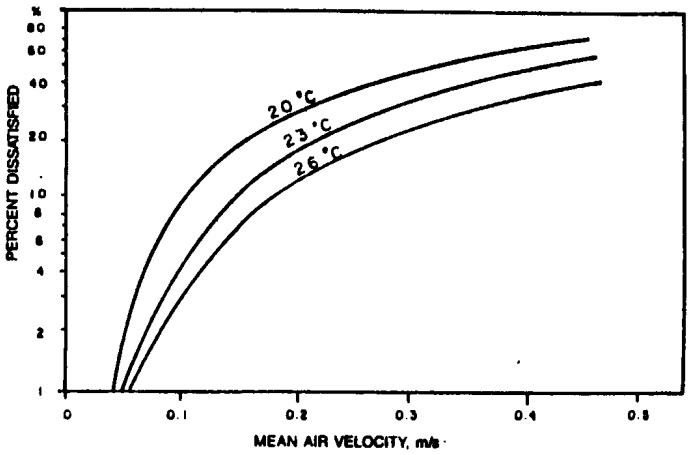


Fig. 13 Percentage of People Dissatisfied as Function of Mean Air Velocity

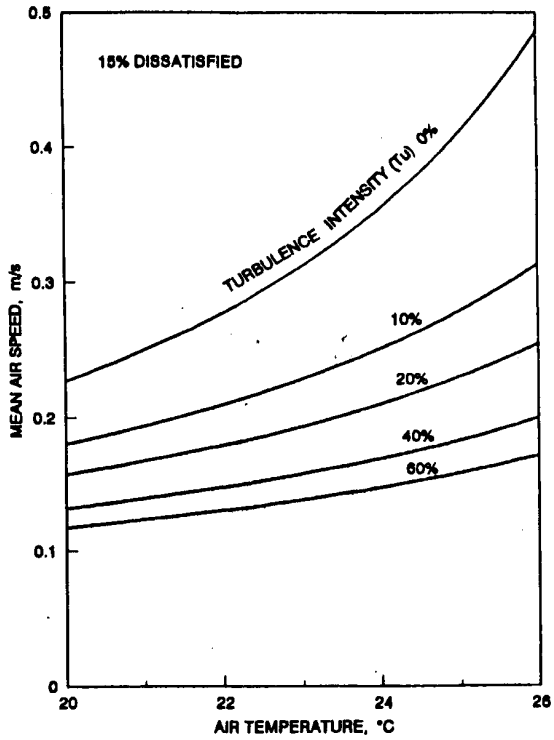


Fig. 14 Draft Conditions Dissatisfying 15% of Population

Asymmetry of thermal radiation

Asymmetry
Warm ceiling (----)
Cool wall (---)
Cool ceiling (--)
Warm wall (-)

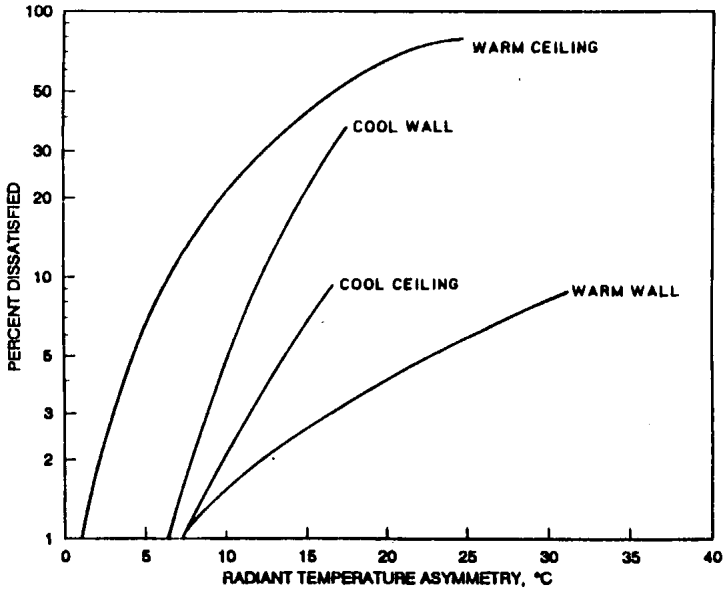


Fig. 12 Percentage of People Expressing Discomfort Due to Asymmetric Radiation

Thermal stratification

Temperature difference < 3-4 K between head and feet.

For large temperature gradients, local warm discomfort can occur at the head, and/or cold discomfort can occur at the feet, although the body as a whole is thermally neutral.

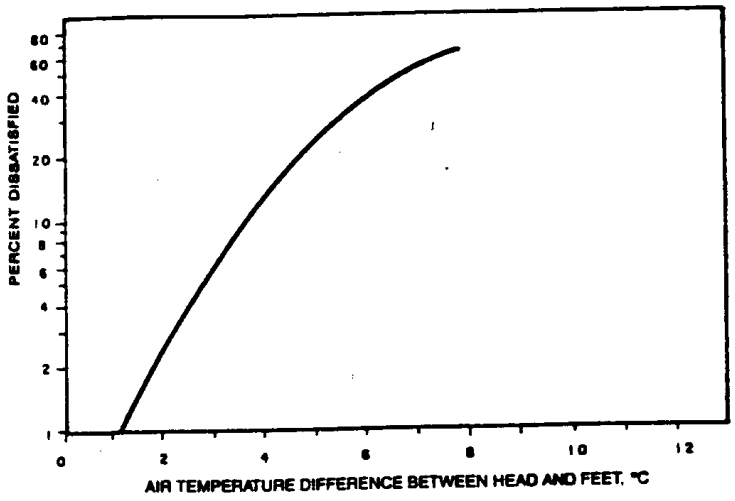


Fig. 15 Percentage of People Dissatisfied as Function of Vertical Air Temperature Difference between Head and Ankles

Warm or cold floors

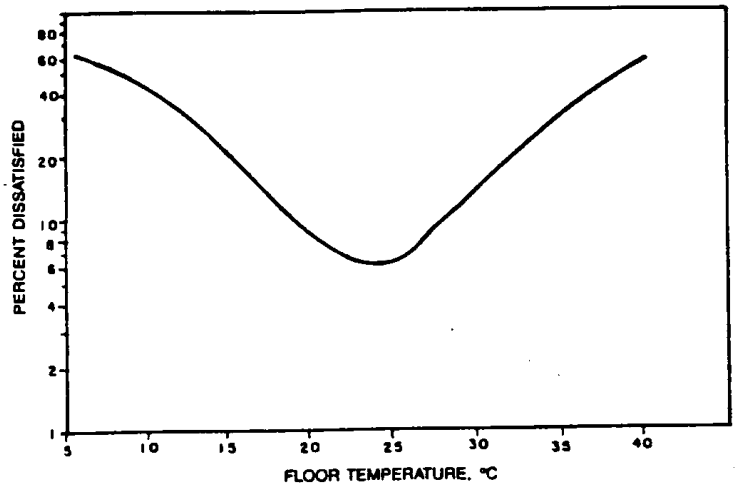


Fig. 16 Percentage of People Dissatisfied as Function of Floor Temperature

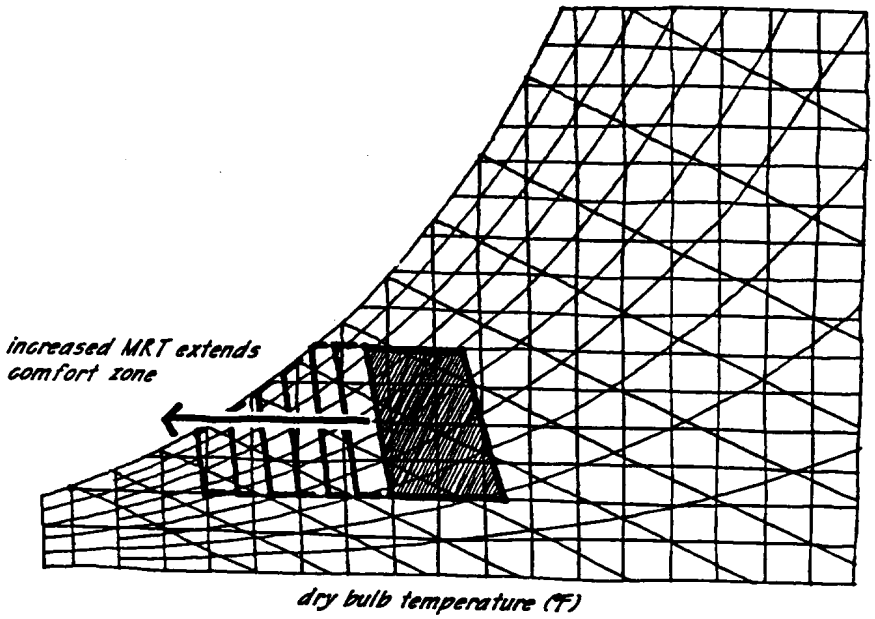


Figure 2.7: Effect of mean radiant temperature on comfort zones. *(After Arends et al., 1980, modified to current ASHRAE comfort zone.)*

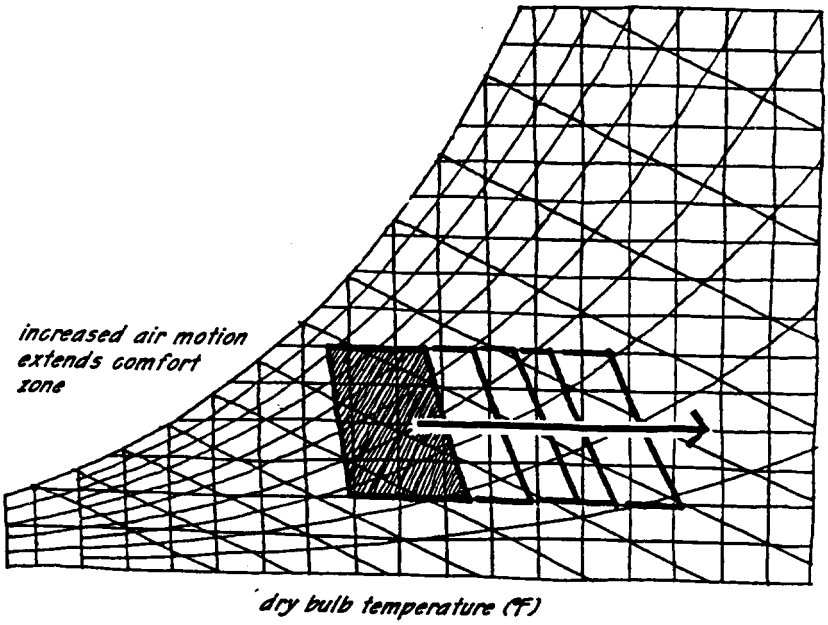


Figure 2.8: Effect of air motion on the summer comfort zone. *(After Arends et al., 1980, modified to current ASHRAE comfort zone.)*

Chapter 4. Air-Conditioning Systems

4.1 Introduction
4.2 Automatic control systems
4.3 All-air systems
4.4 Air-water systems
4.5 All-water systems
4.6 Unitary and hybrid systems
4.7 Summary of different air-conditioning systems

Reading: Chapter 2 of the text book (M&P)
ASHRAE Systems & Equipment Handbook, ASHRAE Applications Handbook
Kreider J.F, and Rabl A. "Heating and Cooling of Buildings – Design for Efficiency"

4.1 Introduction

Purpose of an air-conditioning system is to control indoor air parameters within required thermal comfort and indoor air quality. To achieve required indoor air parameters, the system: heat, cool, humidify, dehumidify and filter outdoor air.

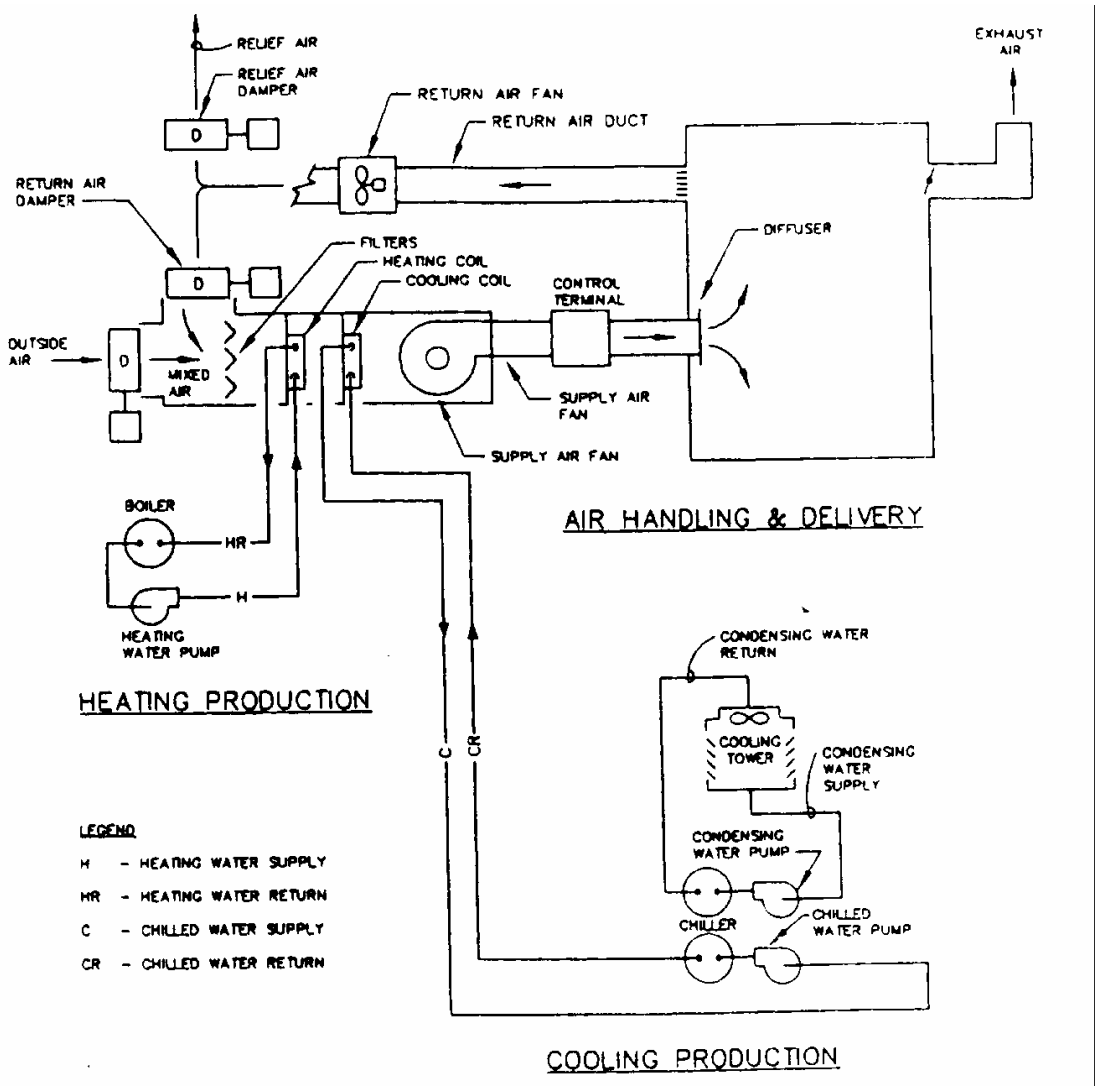
HVAC Subsystems

See Figure on the next page:

End Use

Consumes capacity to condition space air or air stream supplying space.
Air-conditioning systems = air handling systems + ducts + air distribution devices

How to select an air-conditioning system?
- Performance requirements (loads, process)
- Capacity requirements (building types, loads)
- Spatial requirements (building types)
- First costs (location, size of HVAC, investment)
- Operating costs
- Reliability
- Flexibility
- Maintainability



Classification in terms of system:

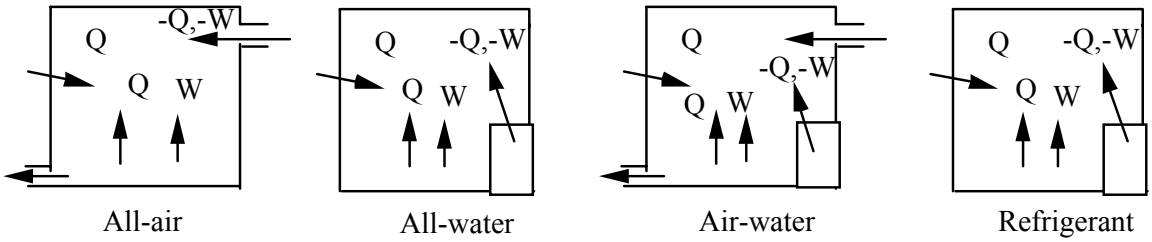
○

Classification in terms of air supplied

○

Classification in terms of load carried:

○



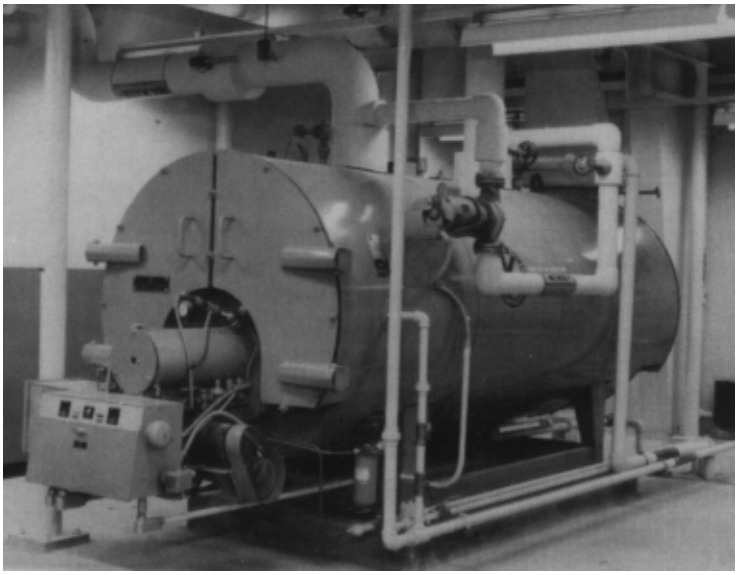
Production

Convert primary energy for heating/cooling. Energy sources:

- Coal
- Natural gas
- Fuel oil
- Biomass

Produce steam and electricity.

Heating production equipment:

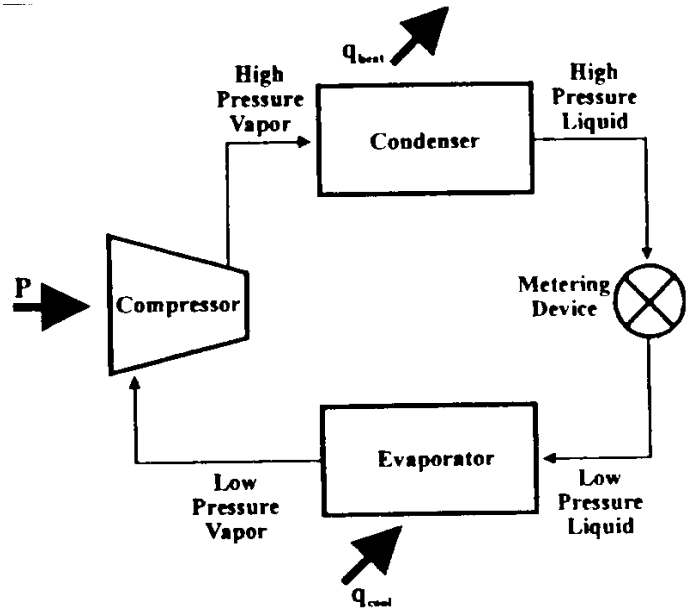


A packed fire-tube boiler. (*Courtesy of Federal Corp., Oklahoma City, OK*)

Cooling production equipment:

Vapor compressor:

- Refrigerant
- Compressor
- Drive (usually electric motor)



Components in a Vapor-Compression Refrigeration Cycle

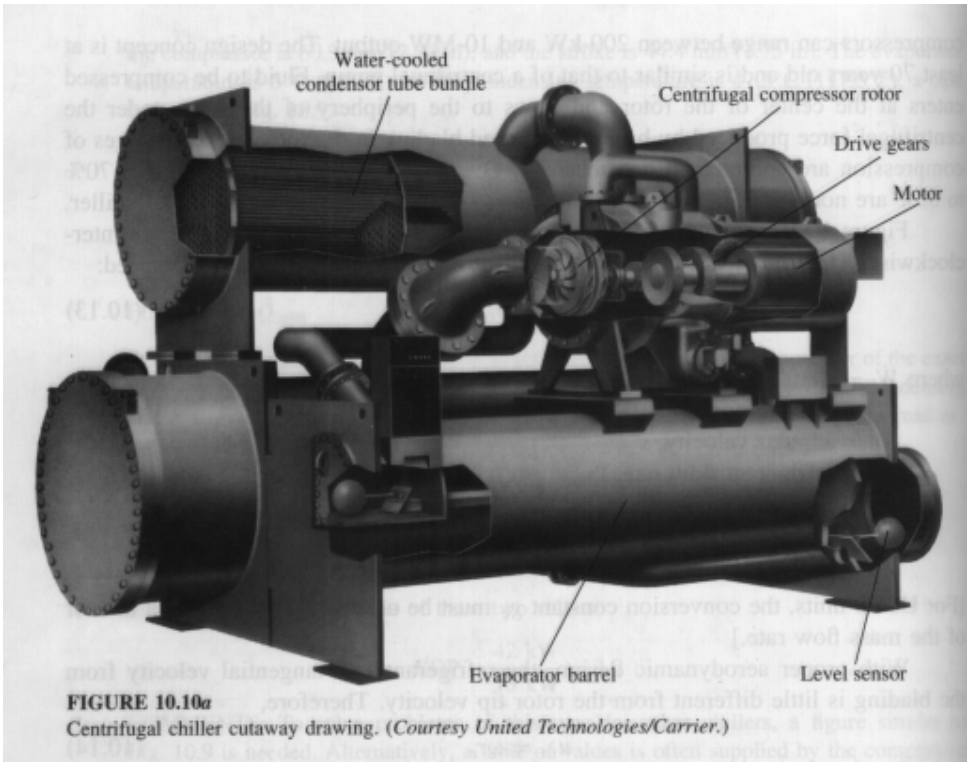


FIGURE 10.10*a*
Centrifugal chiller cutaway drawing. (*Courtesy United Technologies/Carrier.*)

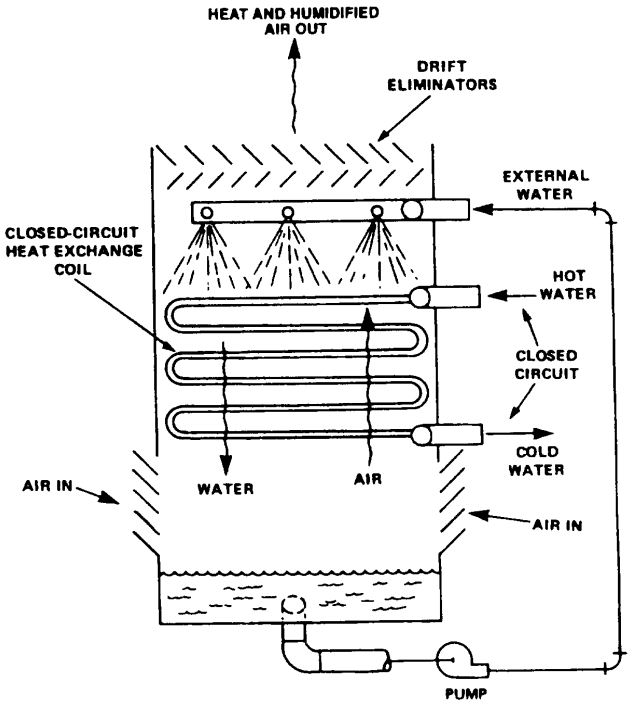
Centrifugal chiller cutaway drawing (*Courtesy United Technology / Carrier*)

Heat rejection:

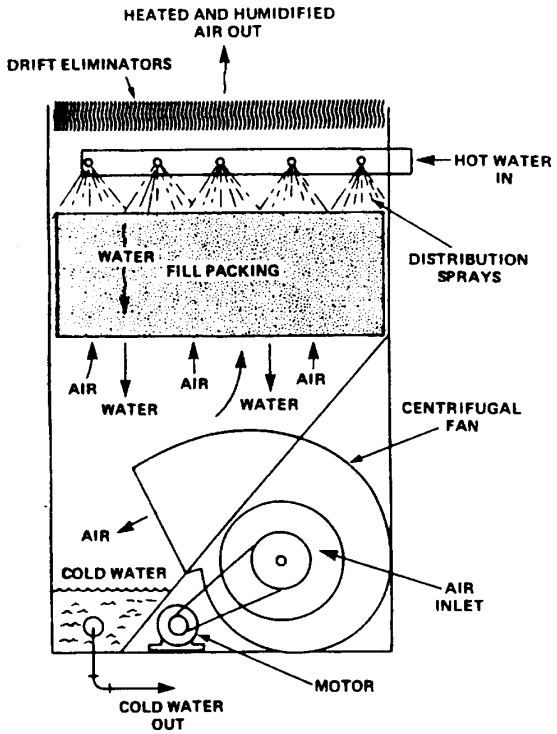
- Disposes of heat from cooling process
- Cooling tower, evaporative condenser, air-cooled condenser
- "Sink" for waste heat: ambient dry bulb, ambient wet bulb, ground, surface water
- Trades offs between cost and COP



Forced-draft cooling tower (*Courtesy Marley Co.*)



Indirect-Contact Evaporative Cooling Tower



Factory-Assembled Counterflow Forced-Draft Tower

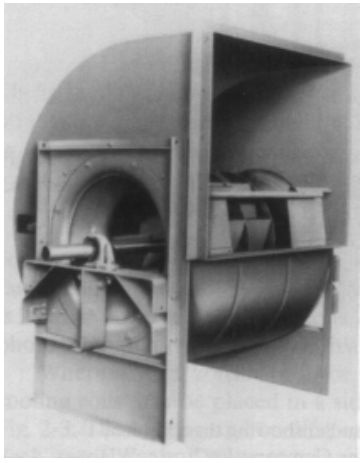
Distribution

Moves capacity from production to use.
Water and steam distribution:

○

Air distribution:

○



A centrifugal fan (*Courtesy of the Train Company, LaCrosse, WI*)

Packed equipment
Air-handling unit: ○

4.2 Automatic Control Systems

HVAC systems are dynamic:

Sized for extreme conditions
Most operation is part load / off-design

Deviation from design => imbalance since Capacity > Load

Without control system, HVAC would overheat or overcool spaces.

Automatic control system
"A system that reacts to a change or imbalance in the variable it controls by adjusting other variables to restore the desired balance."

Modern computer-based systems manage system resources ("supervisory"):

- Reduce energy use
- Identify maintenance problems

Essential components of a control system:

○

Controlled variable is a characteristic of system to be regulated.

"set point" is desired value
"control point" is actual value

"set point"- "control point" ≡ "error" or "offset"

Sensor measures actual value of controlled variable.

Controller modifies action of controlled device in response to error.

Controlled device acts to modify controlled variable as directed by controller.

Example: Water tank level control

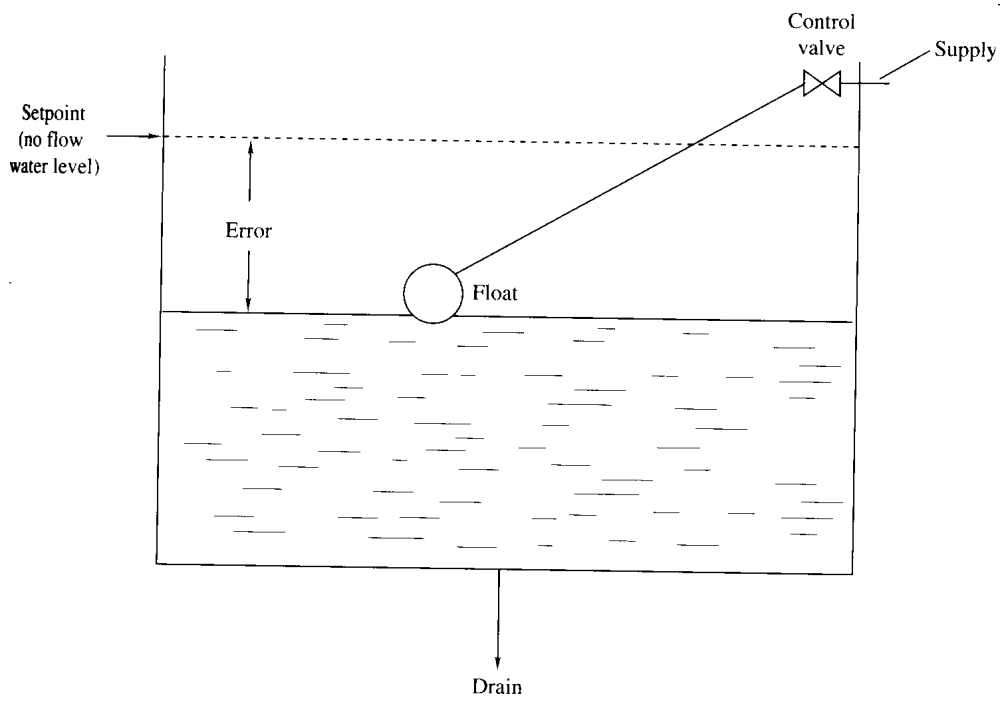


FIGURE 12.1
Simple water-level controller. The set point is the full-water level; the error is the difference between the full level and the actual level.

○

Example: Steam heating coil.

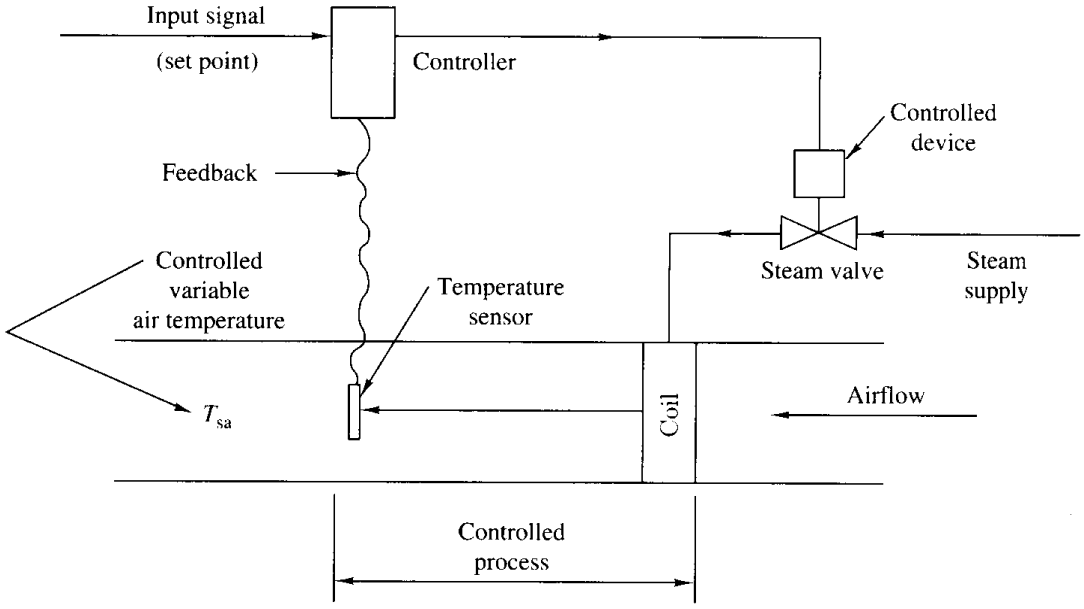


FIGURE 12.2*a*
Simple heating coil control system showing the process (coil and short duct length), controller, controlled device (valve and its actuator), and sensor. The set point entered externally is the desired coil outlet temperature.

○

4.2.1 Control System Types

- Closed loop (feedback) control
- Open loop (feedforward) control

Does sensor measure controlled variable?
If yes the control system is closed loop, if no the system is open loop.

In the closed system, controller responds to error in controlled variable. Previous example of the steam heating coil is a closed loop. In general, HVAC control systems are primarily closed loops.

In the open loop system, there is an indirect link between controller and controlled variable. The system action is based on external variable. The relationship between external variable and controlled variable is assumed. An example of open loop is electric blanket.

4.2.2 Control Action

- Two-position (on-off) control
- Modulating control

Two-position control systems are always at full capacity or off. Best for systems with slow rate of change for controlled variable. This control is common in low cost systems, and it is relatively imprecise.

Example: Two-position control for steam valve in the steam heating coil.

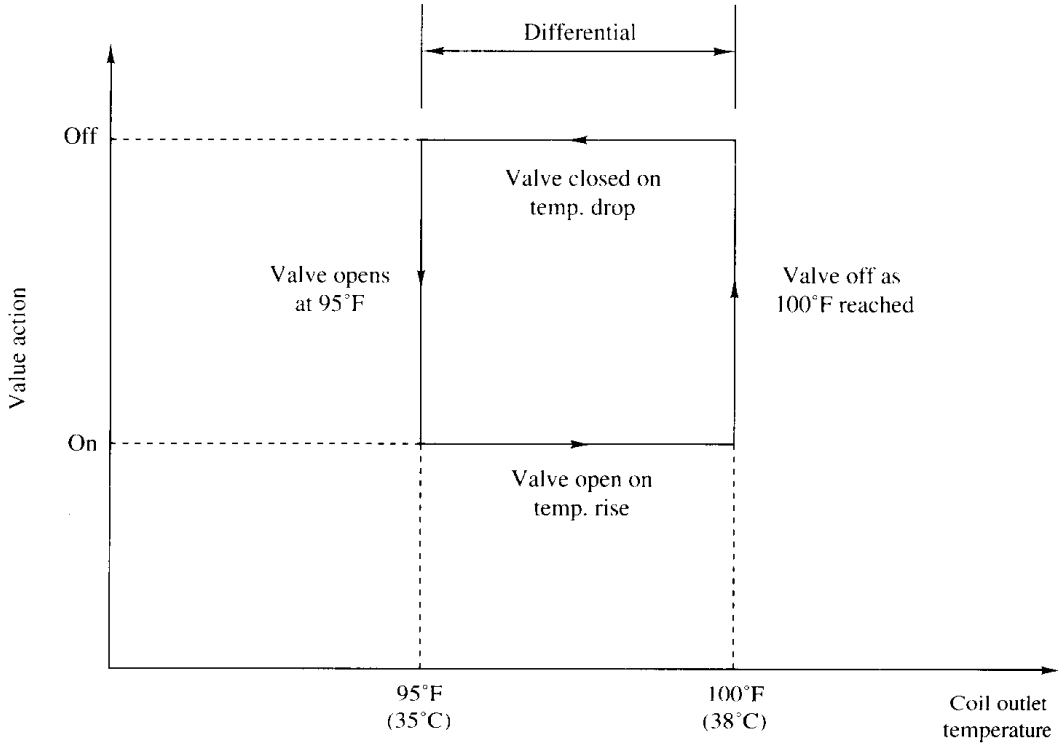


FIGURE 12.3
Two-position (on-off) control characteristic for steam valve in Fig. 12.2*a*.

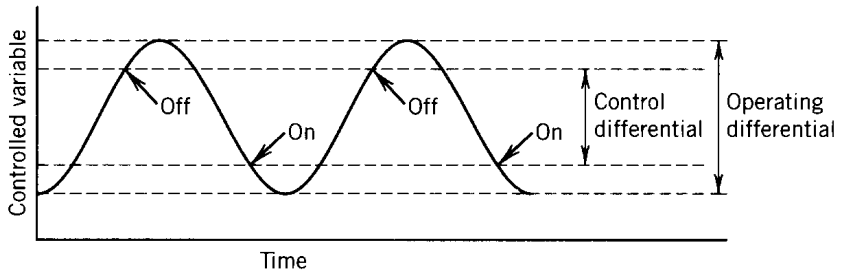


Figure 2-14 Two-position (on–off) control action.

Control differential is difference between "on" and "off" values of controlled variable.
Operating differential is difference between extreme values of controlled variable.

Operating differential > Control differential

Modulating control systems produce continuously variable output over a range. This is finer control system than two-position system, and it is typical in large HVAC systems.

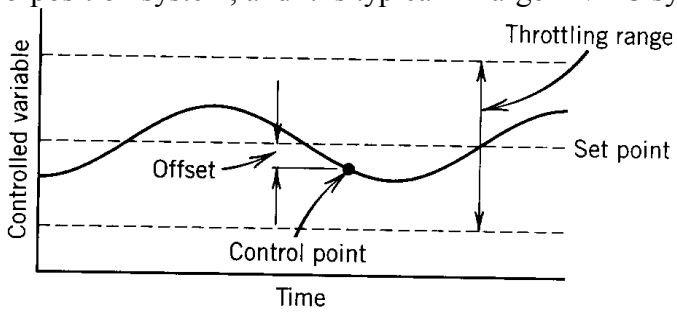


Figure 2-16 Modulating control action.

Throttling range (TR) is a range of input variable over which output varies through its full range. Gain is Δ output per Δ input, and it is usually adjustable.

Proportional control is the simplest modulating action for which the controller output is a linear function of input:

○

where O_P is the proportional controller output, A is the controller output at zero offset, e is the error (offset), and K_P is the proportional gain constant.

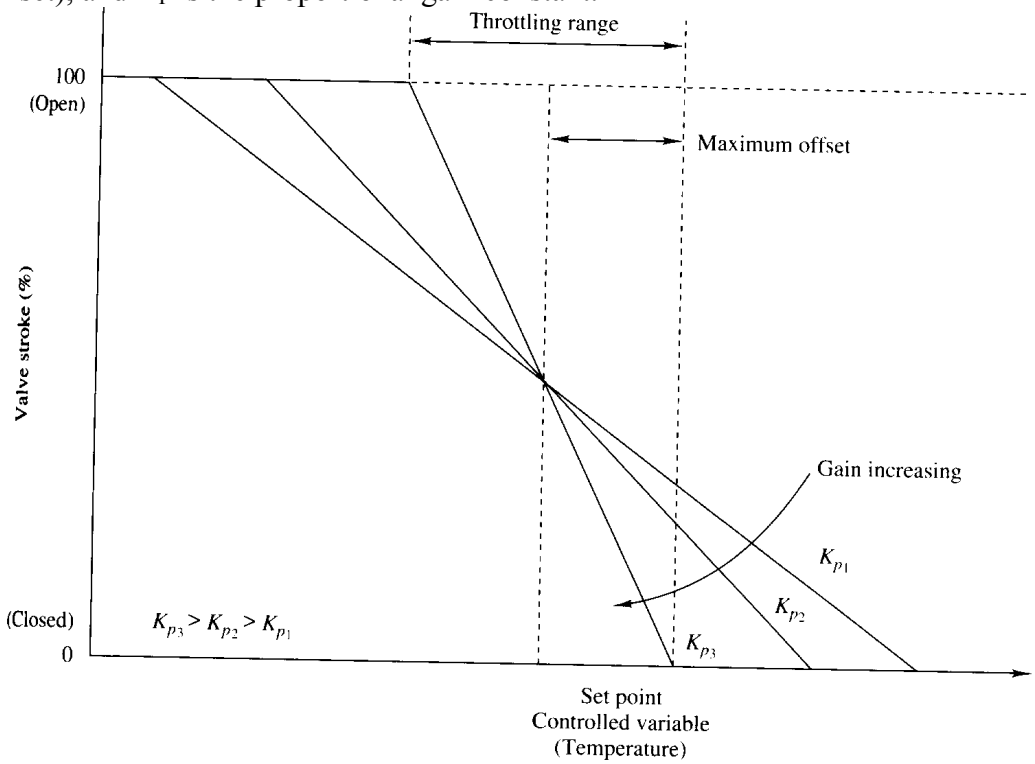


FIGURE 12.4
Proportional control characteristic showing various throttling ranges and the corresponding proportional gains K_p. This characteristic is typical of a heating coil temperature controller.

Smaller TR (larger gain) =>smaller offset. Smaller TR may cause stability problems.

Stability is tendency of a system to find a steady control point after an upset.
Instability is tendency for oscillations to grow.

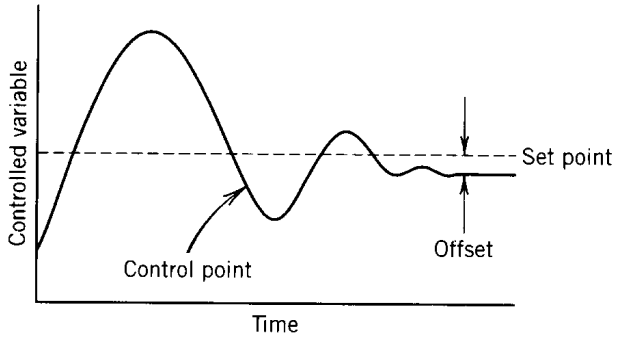


Figure 2-18*a* A stable system under proportional control.

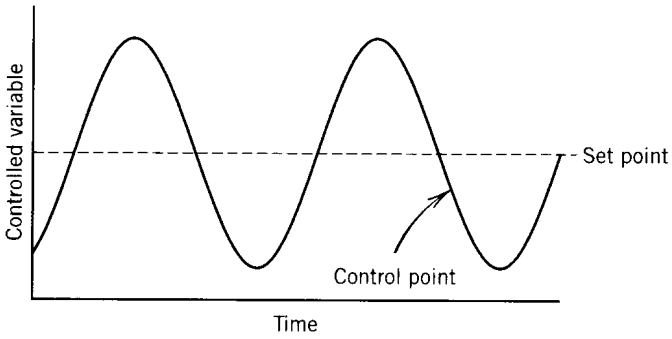


Figure 2-18*b* An unstable system under proportional control.

Proportional plus integral (PI) control is designed to eliminate offset.

$$\text{Proportional} + \text{Factor} \propto \text{integral of offset}$$

○

where O_{PI} is the PI controller output, and K_i is the integral gain constant.
Integral term drives offset to zero.

Examples of PI control in buildings include mixed-air control, duct static pressure control, and coil controls.

Proportional plus integral plus derivative (PID) control further speeds up action of PI control May not be suitable for HVAC that usually do not require rapid control response.

$$\text{Additional control} \propto \text{rate of change of error}$$

○

where O_{PID} is the PID controller output, and K_d is the derivative gain constant.
Example of PID application in buildings is duct static pressure control.

Example: Comparison of P, PI, and PID controller response to input step change

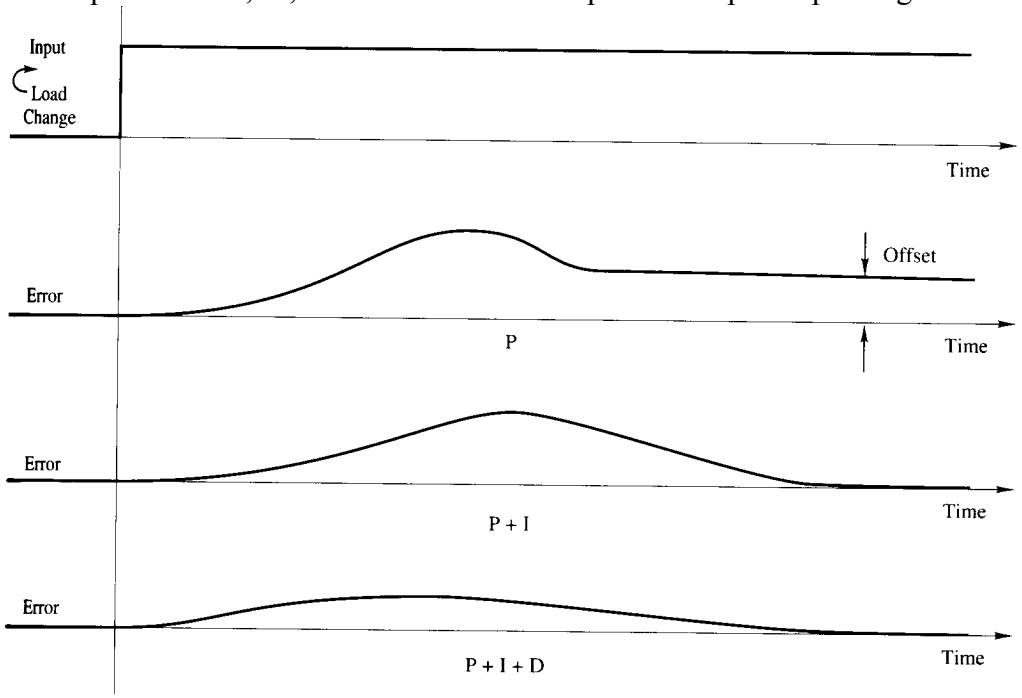
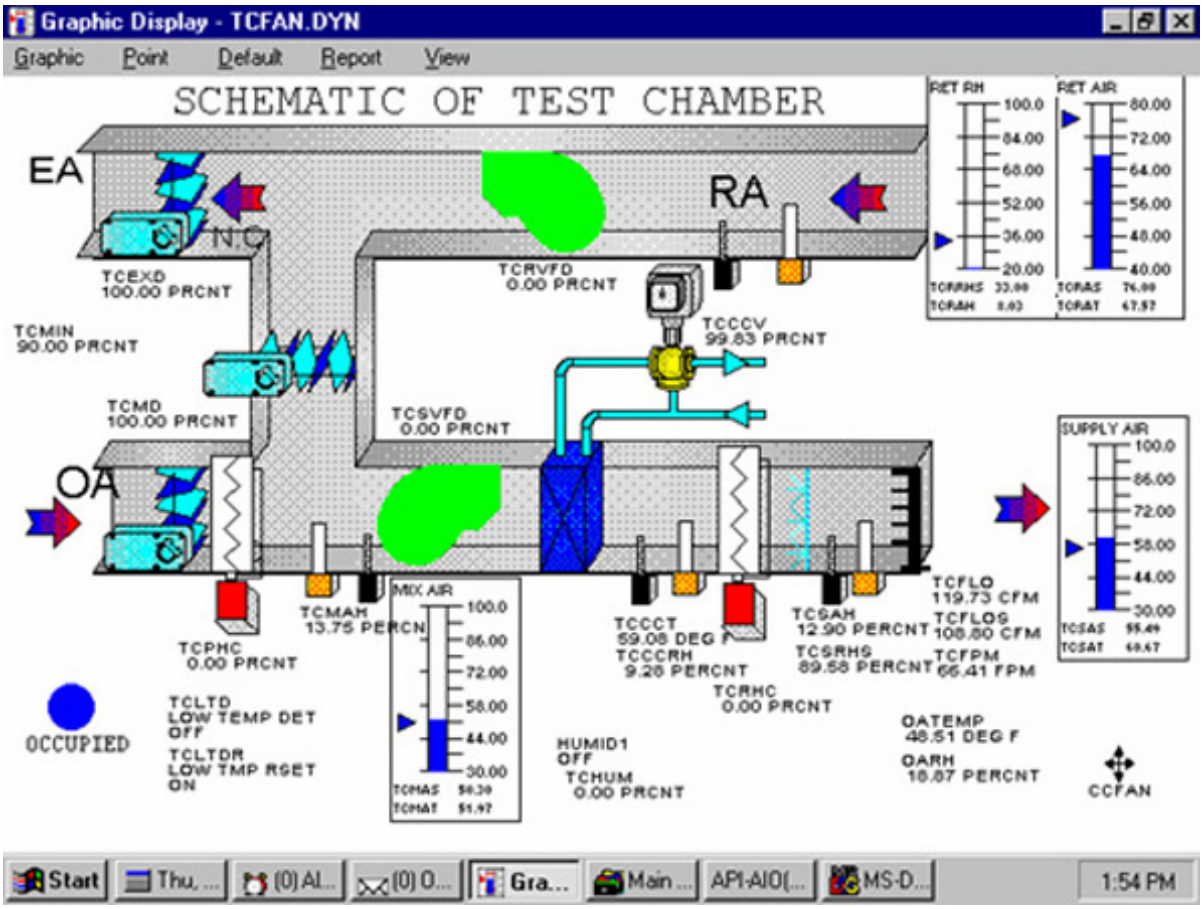


FIGURE 12.5
Performance comparison of P, PI, and PID controllers when subjected to a uniform input step change.

4.2.3 Computers in Control

Software is replacing "mechanical" logic. More sophisticated schemes are possible. Simulation and optimization are possible in real time.

Example: Graphical interface for HVAC system control.



HVAC Systems

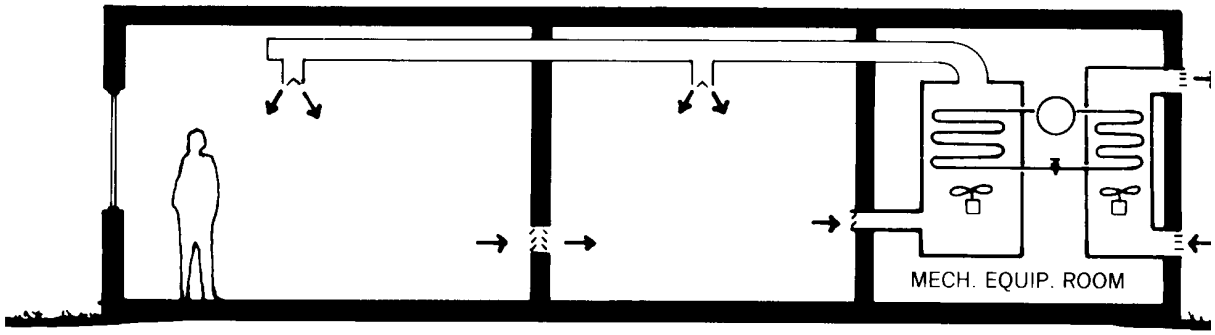


FIGURE 14.9c
Schematic diagram of an all-air system.

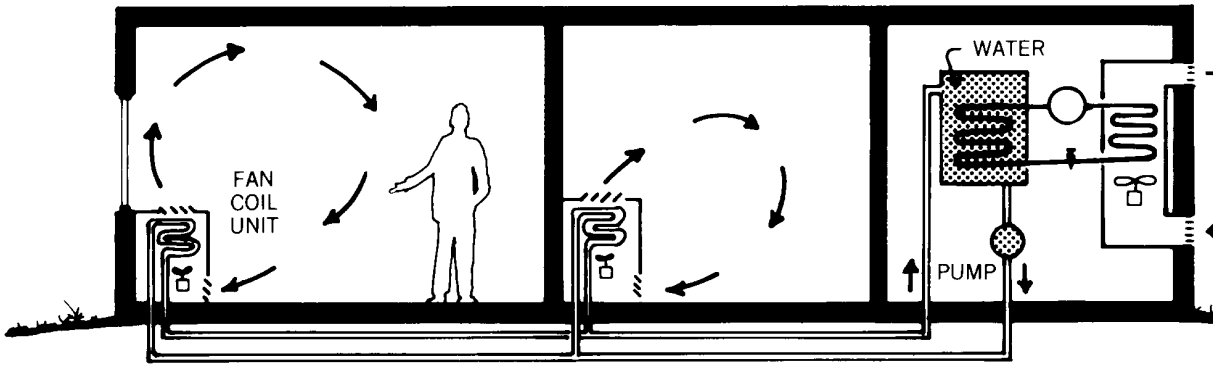


FIGURE 14.9d
Schematic diagram of an all-water system.

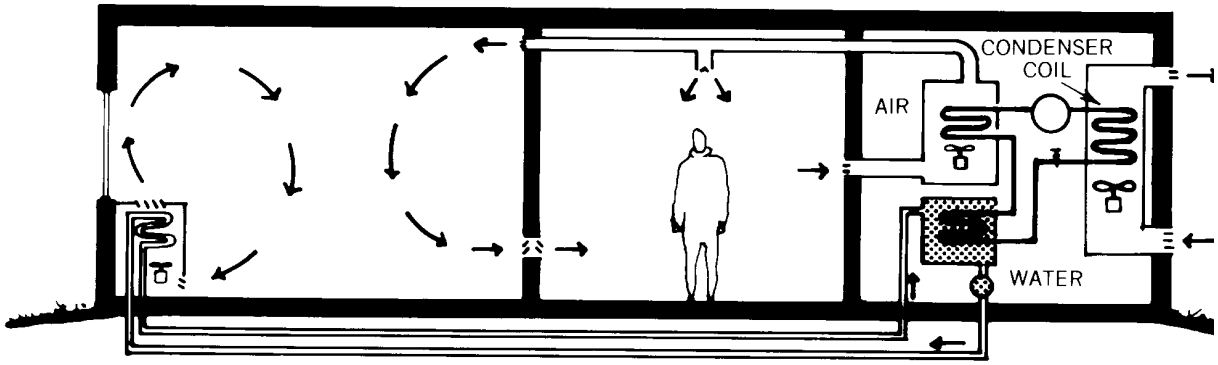


FIGURE 14.9e
Schematic diagram of an air–water system.

Air-handling unit (**AHU**) usually consists of: coil(s), fan(s), filter(s), air-mixing controls, humidifier, and heat recovery. The following figure represents AHU for a single zone all-air system.

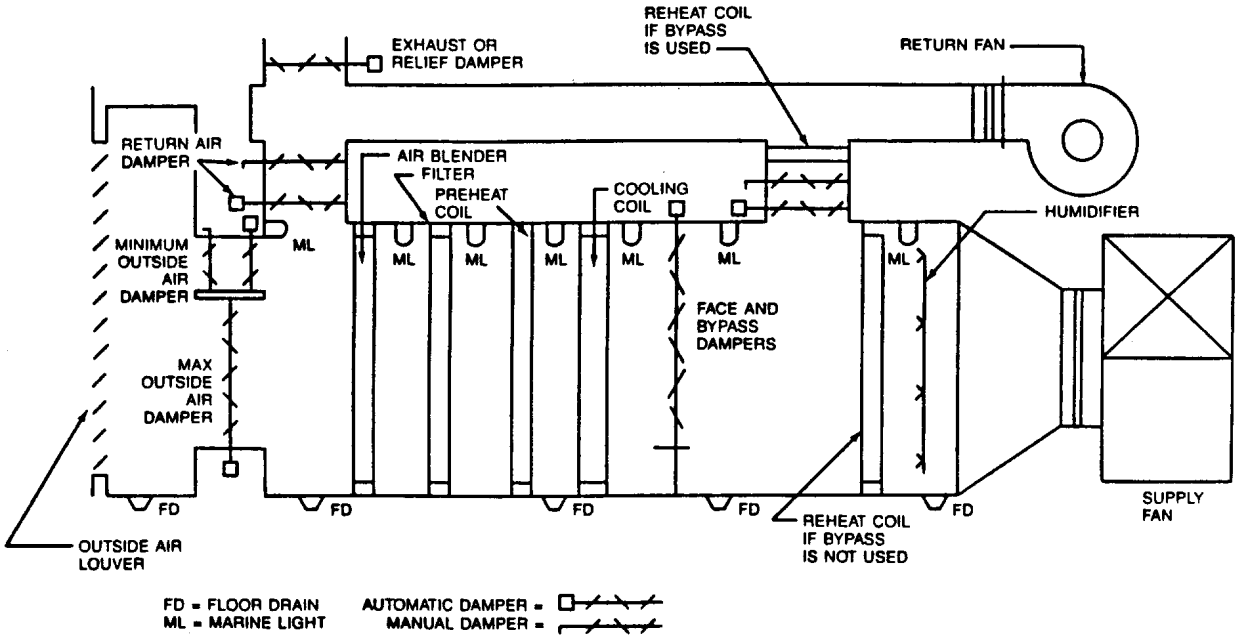


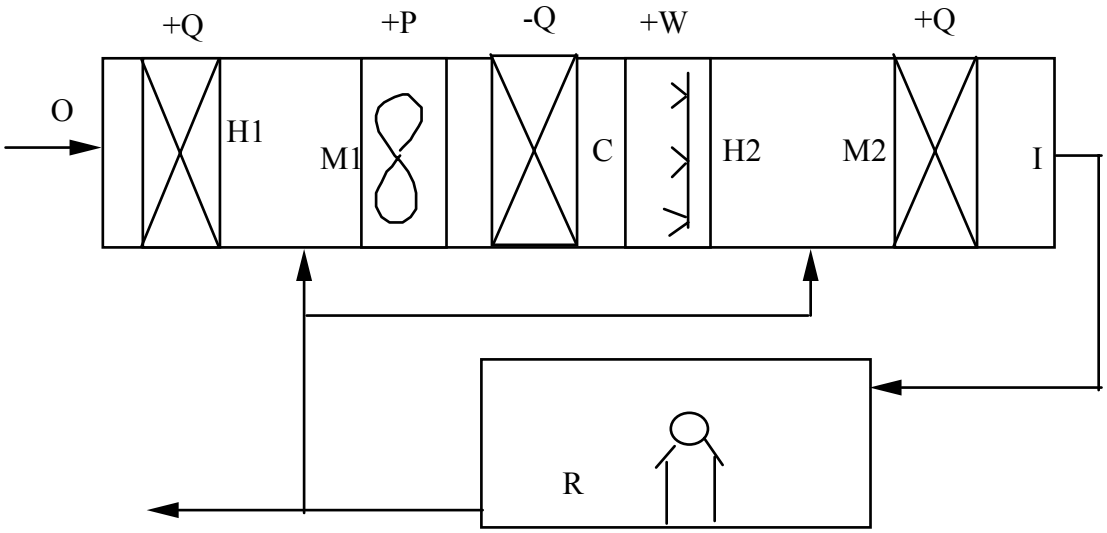
Fig. 1 Equipment Arrangement for Central Systems

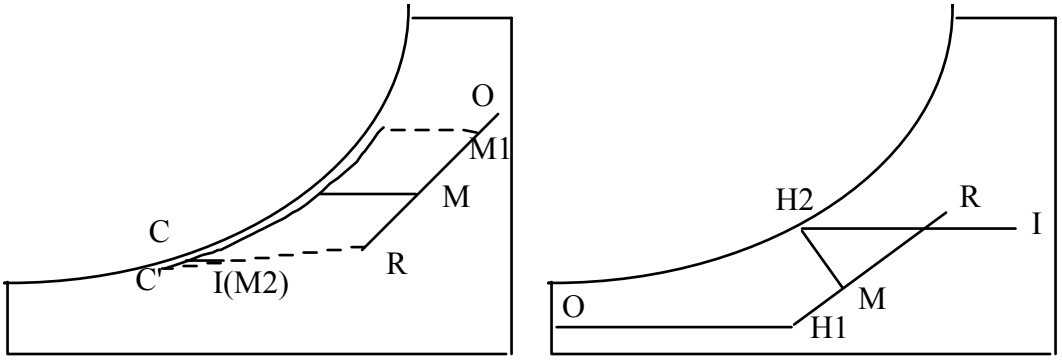
4.3 All-Air Systems

4.3.1 Constant-air-volume systems

○

Use in new systems discouraged by code.





Summer cooling Winter heating

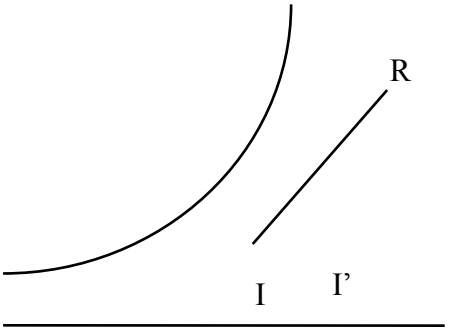
Summer:
Single mixing with room air:
O + R => M (cooling + dehumidification) => C (re-heat) => I (Q + W) => R

Double mixing with room air:
O + R => M1 (cooling + dehumidification) => C' + R => M2 or I (Q + W) => R

Winter:
O (pre-heat) => H1 + R => M (humidification) => H2 (re-heat) => I (Q + W) => R

4.3.2 Variable-air-volume systems

Example:
You turn the fan speed up or down in your car.
$\dot{Q} = \dot{m} (i_R - i_I)$



AHU fan varies power to match loads. Less load => lower fan power.
Pressure in supply ducts is maintained to a fixed value.

Design cooling:
- box is 100% open
- no reheat

Off-design cooling:
- zone temperature drops since cooling load decreases
- box throttles until minimum flow is reached

Dead band:
- no control action
- start reheat at lower limit

Off-design heating:
- minimum primary air
- thermostat increases reheat as space temperature falls

Design heating:
- fully energized

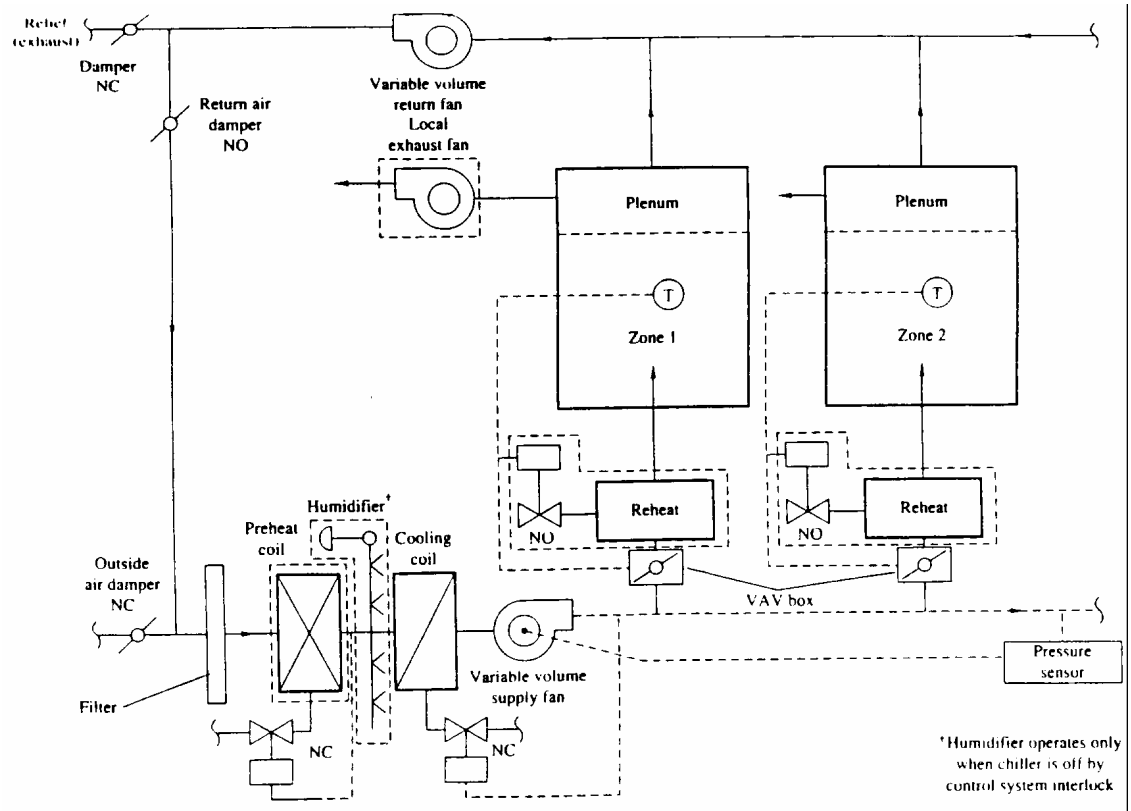


FIGURE 11.23
Variable-air-volume system with optional reheat.

VAV terminals:
- Single-blade dumper (pressure dependent or independent)
- Air valve
- Induction

Primary flow induces secondary flow from plenum.

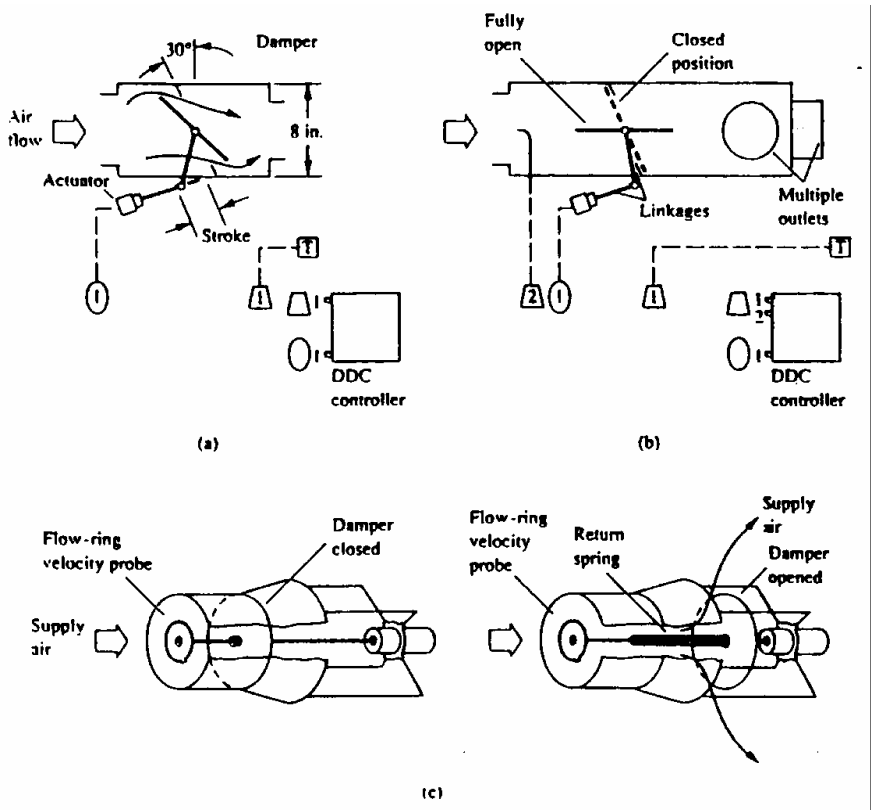


FIGURE 27.3 VAV boxes: (a) single-blade, pressure-dependent; (b) single-blade, pressure-independent; and (c) air valve.

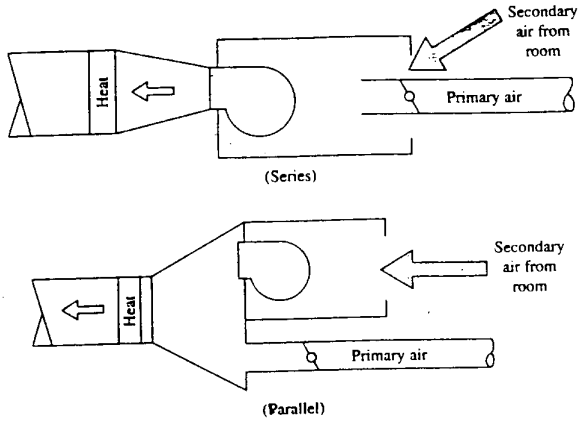


FIGURE 11.24b
Schematic diagrams of series and parallel fan-powered VAV box.

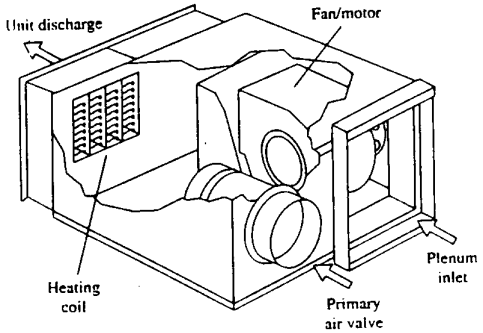


FIGURE 11.24c
Cutaway drawing of parallel VAV box. (*Courtesy of Titus, Inc.*)

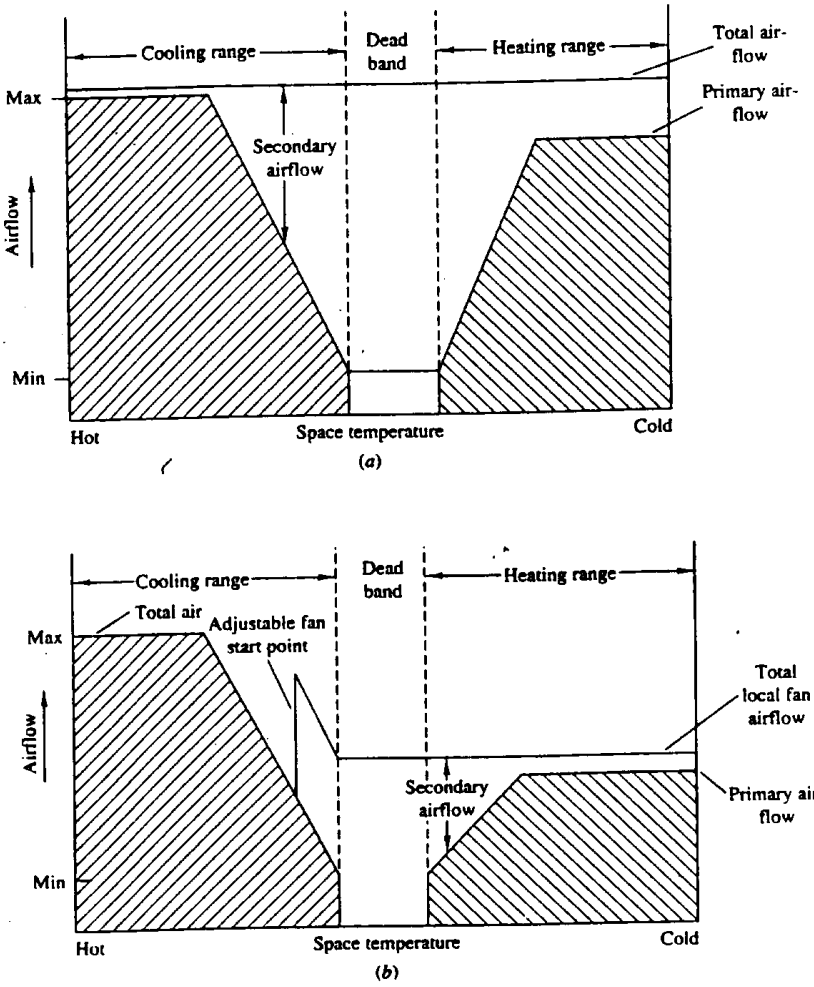


FIGURE 11.25
(*a*) Flow characteristics for series fan-powered VAV boxes. Cross-sectioned areas represent primary airflow. Total box airflow is constant. (*b*) Flow characteristics for parallel fan-powered VAV boxes. Cross-hatched areas represent primary airflow. Total box airflow is higher during cooling than heating. Fan operates only at zone temperatures below the adjustable set point.

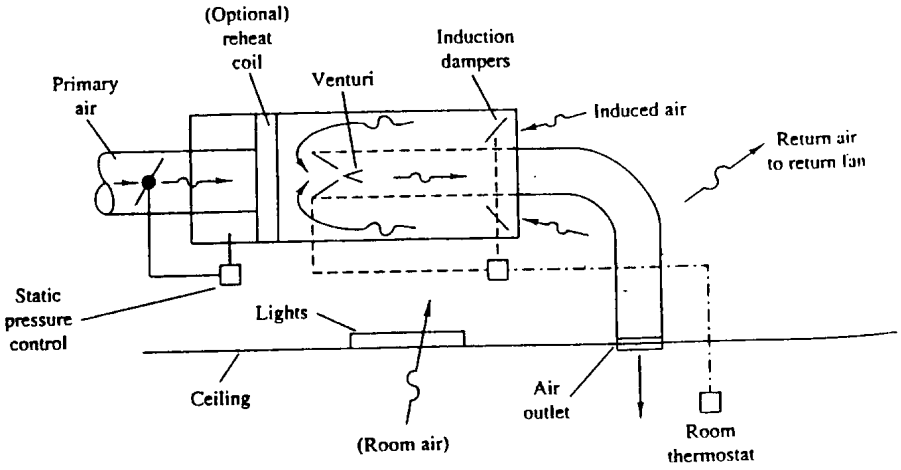


FIGURE 11.24*a*
Schematic diagram of induction VAV box.

Fan –powered series: Fan is always on and space flow is constant. Damper controls supply of primary air. Perimeter zones may need baseboard or fan-coil units.
Fan-powered parallel: Fan injects plenum air to reheat. Supply pressure drives primary flow that is controlled by dumpers. Variable space flow => less fan energy.

Advantages of VAV:

○

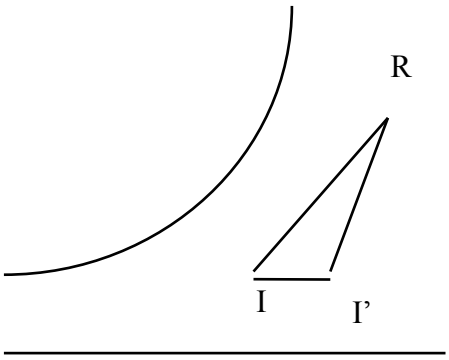
Disadvantages and problems of VAV for off design (low flow rate):

○

4.3.3 Re-heat systems (CAV/RH)

Heating coil (reheat) inserted in the zone supply.

○



Fixed supply airflow rate as well as heating and cooling coil temperatures. Capacity controlled by terminal reheat coil.

Summer:

Cooling coil lowers T_{SA} to set point. Reheat coil adds heat to satisfy thermostat. Typical temperatures for cooling coil are 13°C (55°F). Reheat temperature for full load is 13°C (55°F), when reheat turns off. For this process energy is wasted by overcool & reheat.

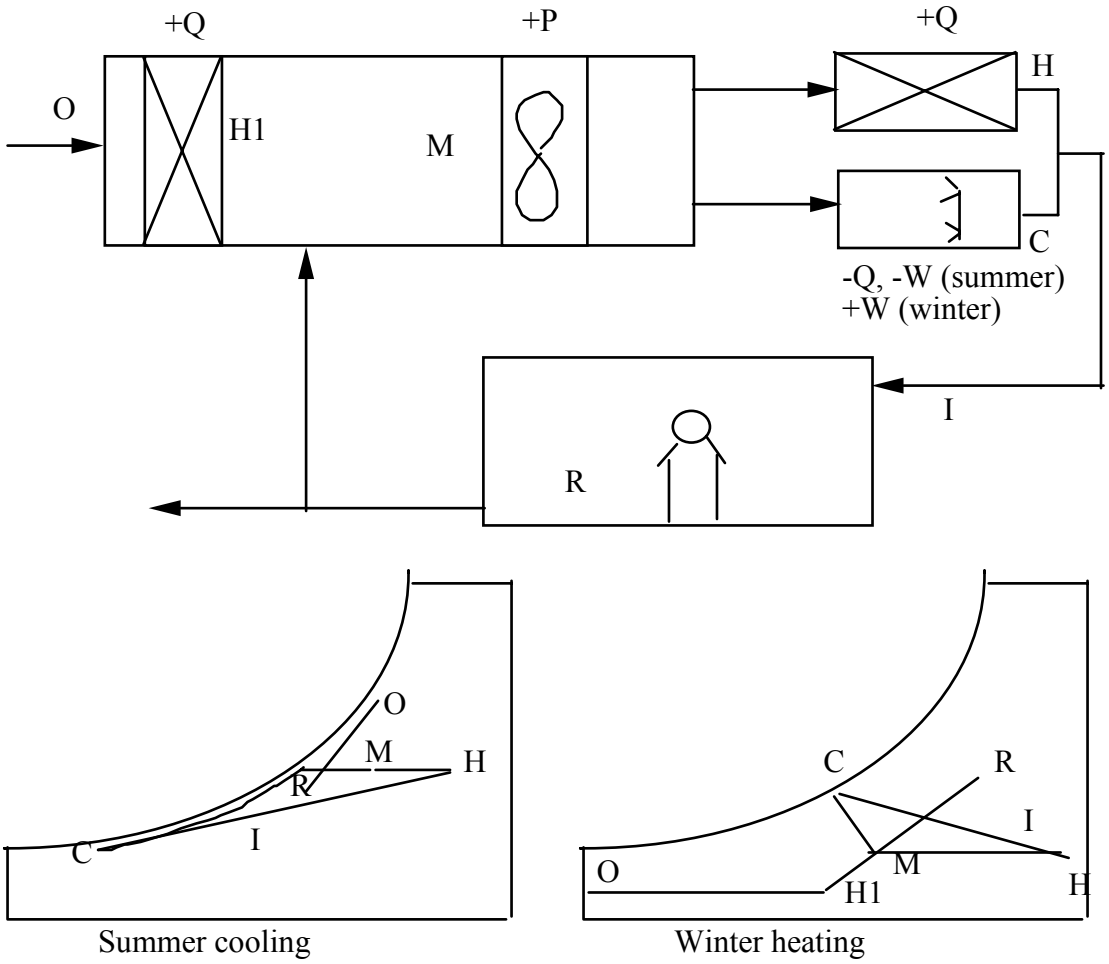
Winter:

Preheat coil raises T_{SA} to set point. Reheat coil adds heat to satisfy thermostat. Typical temperatures for preheat coil are 13°C (55°F), and reheat under full load are 38°C (100°F). No wasted energy.

4.3.4 Dual-duct systems

The system mixes hot and cold air to satisfy zone thermostat. Cold and hot air streams distributed in separate ducts. This is variation of CAV/RH system.

- High duct cost
- Large plenum space required
- Unlimited number of zones



Example:

Design a dual-duct system for the classroom at PSU (Use the data from previous example). Assume cooling coil could reach a relative humidity of 90%.

Summer cooling processes:

O + R => M Hot duct: M => H
Cold duct: M=> C C + H => I

Given:

O: i_O =68.15 kJ/kg, W_o =14.4 g/kg
R: i_R =50.72 kJ/kg, W_R =10 g/kg
I: i_I = 37.91 kJ/kg, W_I = 9 g/kg
M: i_M = 54.95 kJ/kg
Total cooling load = 5000 W
m_a =0.396 kg/s
Fresh air: 80 L/s

Find: Fan, cooling and heating coil capacities.

Total air supplied:
$\dot{m}_a$ = 0.396 kg/s

Fan capacity:

Fresh outdoor air:
$\dot{m}_o$ = 80 L/s = 0.080 m^3/s x 1.2 kg/s = 0.096 kg/s

W_I = 9 g/kg_a, W_R = 10g/kg_a, W_o = 14.4 g/kg_a

$W_H = W_M = (\dot{m}_R W_R + \dot{m}_o W_o)/\dot{m}_a = [(\dot{m}_a - \dot{m}_o) W_R + m_o W_o]/\dot{m}_a$
= [(0.396 - 0.096) x 10 + 0.096 x 14.4)]/0.396 = 11 g/kg_a

From the analysis in the psychrometric chart, no heating in the hot duct is needed. Then,
$i_H = i_M$ = 54.95 kJ/kg

Heating coil capacity:

From psychrometric chart:
i_C = 36.5 kJ/kg
$m_H + m_C = m_a$ $m_H + m_C = 0.396$
$m_H i_H + m_C i_C = m_a i_I$ m_H 54.95 + m_C 36.5 =0.396 x 37.91
m_H = 0.03 kg/s
m_C = 0.366 kg/s

Cooling coil capacity:

T_C = 13.5 ^oC is lower than that in the previous example.

The design should be continued for winter condition as well. Then the equipment capacities can be determined.

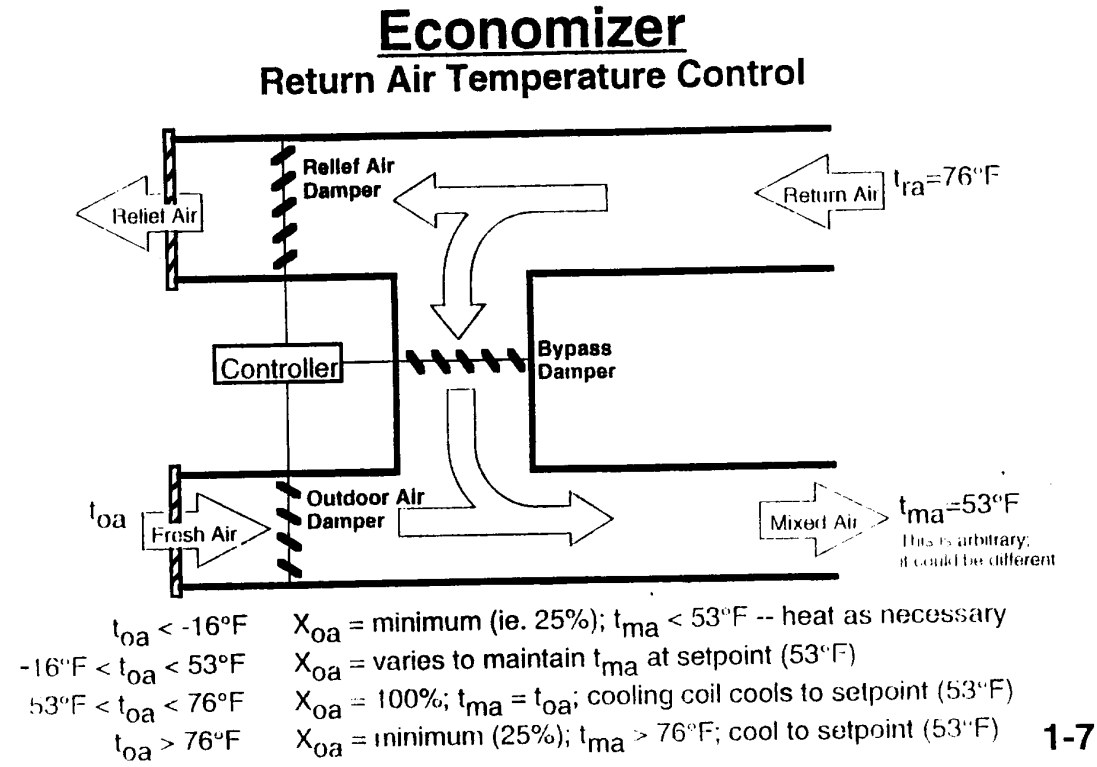
4.3.5 Air-Side Economizer

Air-side economizer uses outside air (OA) to offset mechanical cooling.
AHU mixing dampers vary OA flow from minimal required flow rate (for people in a space) up to 100% OA.

Different control systems for the economizer:

-

Return-Air Temperature Economizer

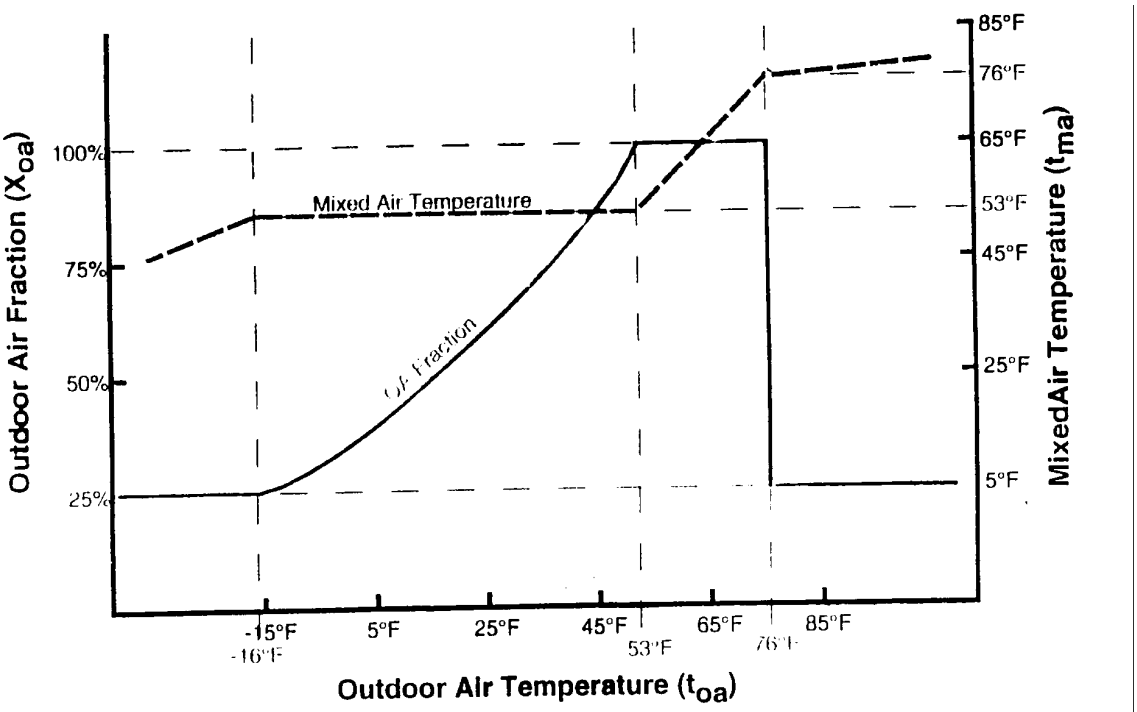


Control action:

$T_{OA}<T_{RA}$ => at min OA preheat coil keeps $T_{MA}=T_{SA}$; increase OA to maintain $T_{MA}=T_{SA}$

$T_{OA}=T_{RA}$ => 100% OA

$T_{OA}>T_{RA}$ => minimum ventilation OA and cooling coil keeps $T_{MA}=T_{SA}$



Variation of OA Fraction and MA Temperature with OA Temperature
(Return Air Temperature Economizer Cycle) 1-8

Advantage: energy savings.

Disadvantages: high VAV energy fan, large OA ducts, humidity control, expensive control system, complexity, and maintenance.

4.4 Air-Water Systems

Air and water are distributed to spaces. Since $(C_p \rho)_{water} > (C_p \rho)_{air}$, air is supplied for better air quality while water is used to remove heating/cooling load.

$$Q = C_p \rho(T_{return} - T_{supply})$$

Primary air has constant volume ≥ minimum OA required for ventilation. In winter, primary air is heated space temperature and humidifies. In summer, primary air is cooled to dehumidify.

Secondary air is passing through water coil (heat exchanger) before mixing with primary air. Central plant makes hot or cold water that is distributed via piping system to the water coil. The water coil heats/ cools to control space temperature, and does not control humidity.

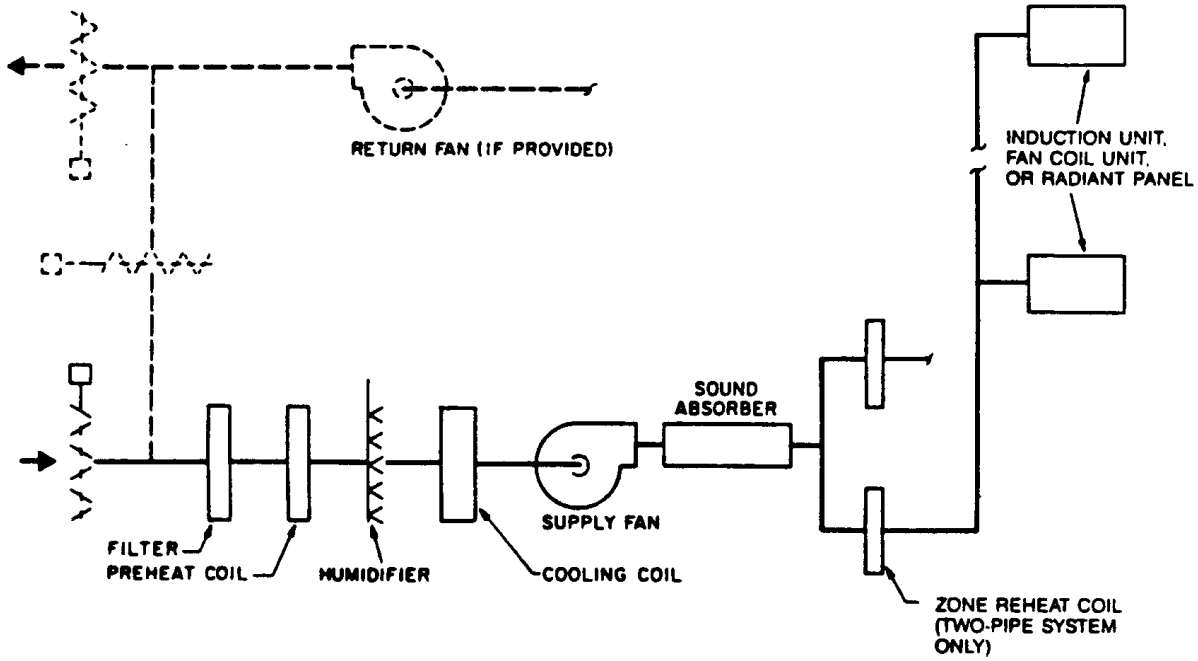
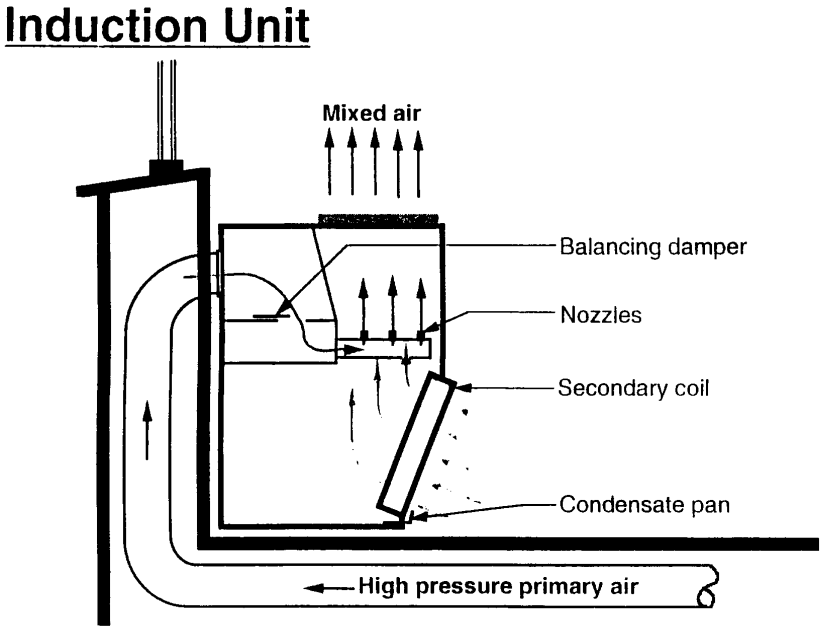


Fig. 2 Primary Air System

4.4.1 Air-water induction systems

Air is supplied with high pressure for induction. High pressure produces high velocities of primary air, and therefore secondary air is induced over water (secondary) coil. No fan needed.



2-4

4.4.2 Fan-coil systems

The systems can be with air-water or all water. They can be further divided into

- Two-pipe: Either hot or cold water
- Three-pipe: Two supplies and one for common return
- Four-pipe: Two for supply and the other two for return.

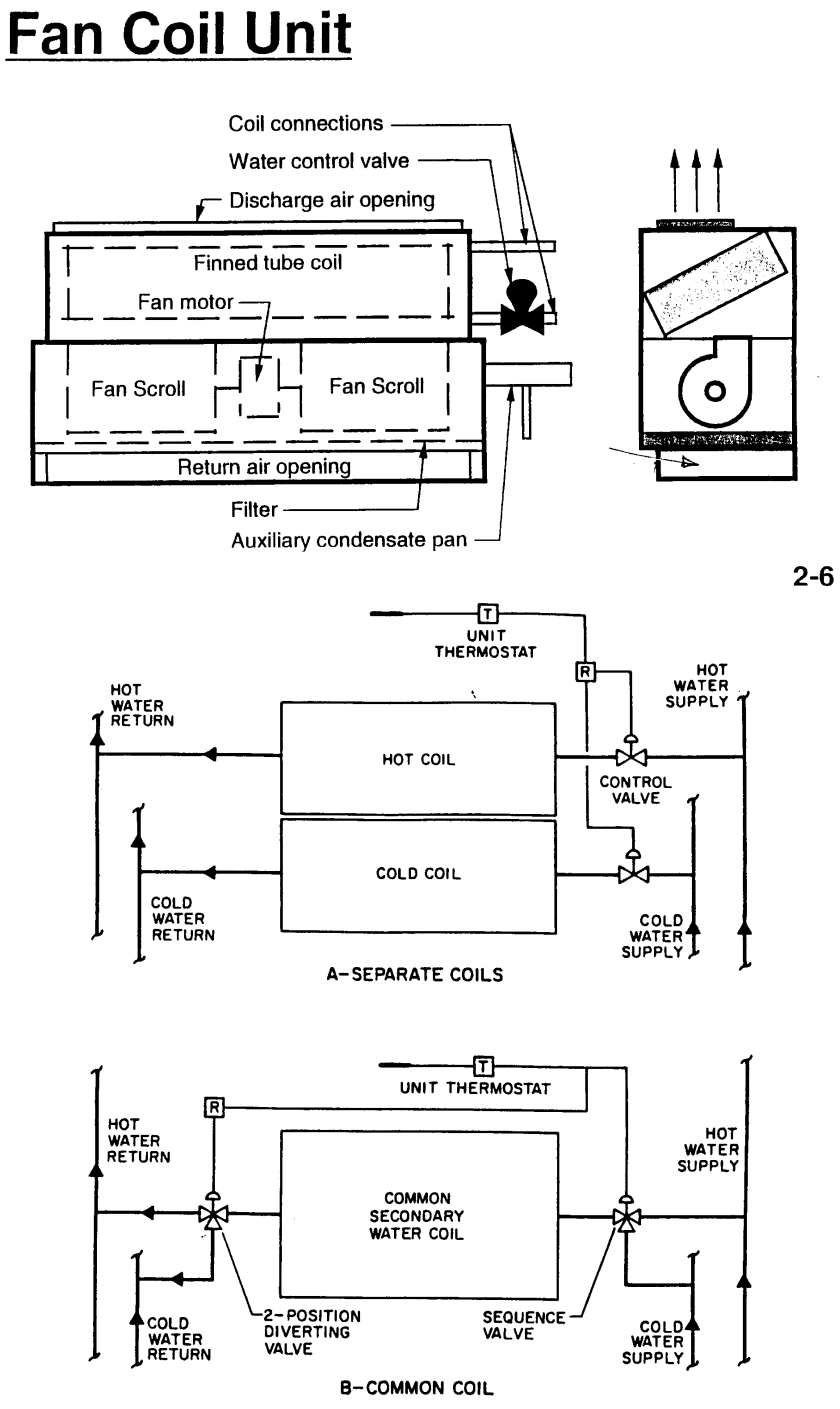


Fig. 9 Four-Pipe System Room Unit Control

Fan coil unit consists of:

New fan coils usually have separate coils for heating and cooling that increases first cost compared to the units with a single coil. However, four-pipe system is more flexible than two-pipe system, and does not require "changeover" or zone reheat.

The fan coil unit is flexible, can condition w/o primary air and has better filtration than the induction unit. Primary air is directly supplied to space if the system is air-water.

4.5 All-Water Systems

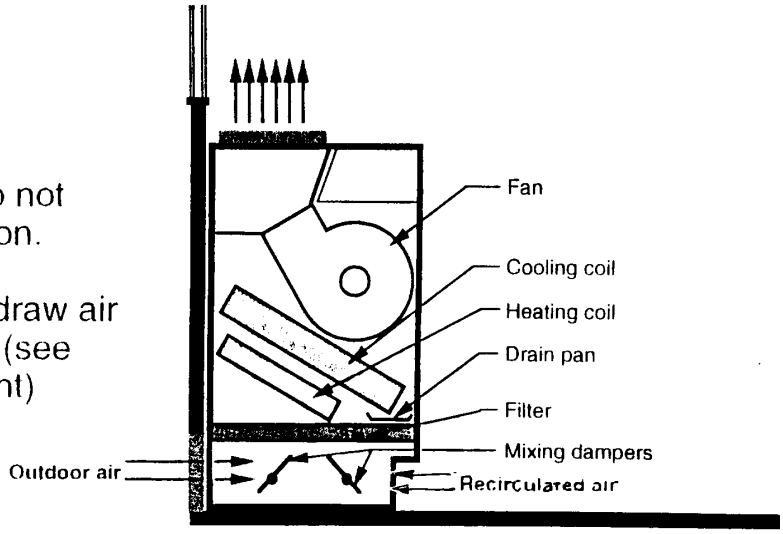
All air-conditioning is achieved by water-air heat exchanger at terminal. Examples:

- Fan-coil
- Unit ventilator
- Radiant panels

All-Water Systems

Fan coil units do not provide ventilation.

Unit ventilators draw air in through walls (see schematic at right)



This system may be only for heating:

Fan coils have no OA, while unit ventilator has OA intake. Infiltration is a mechanism that provides fresh air in spaces with no OA.

Advantages: small/no plenum, individual control, and simple retrofit.

Disadvantages: high maintenance, condensate in occupied space, poor humidity control, and mediocre ventilation control.

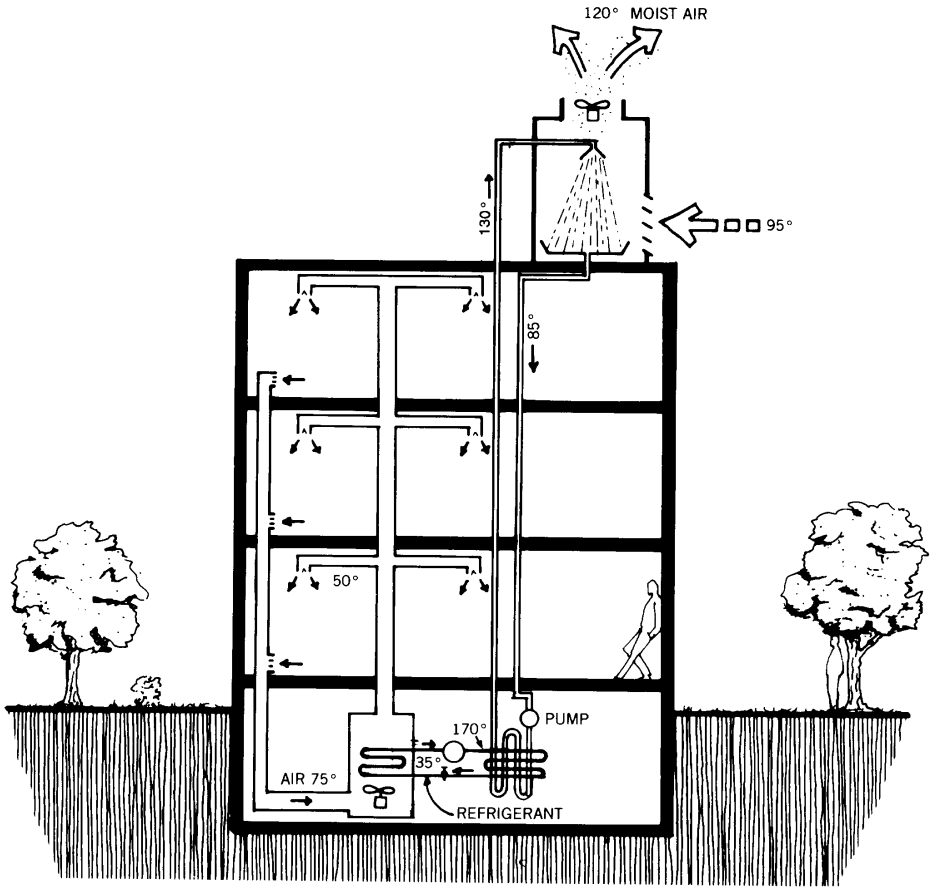
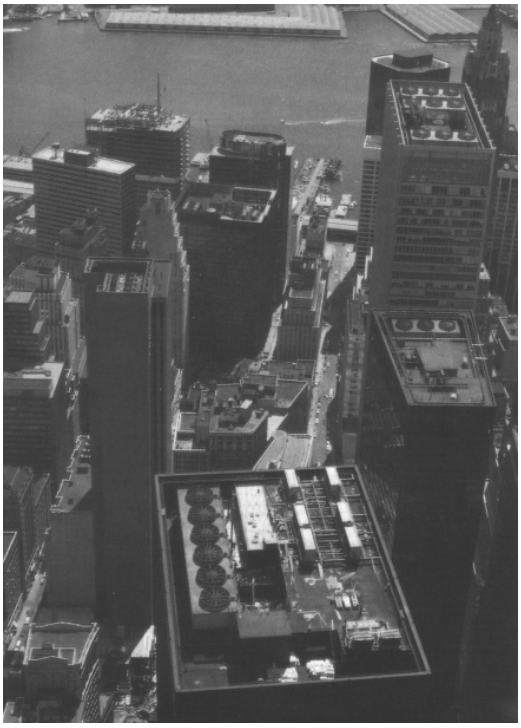


FIGURE 14.9g
A cooling tower cools water by evaporation. This "cooling" water is then used to cool the condenser coil located elsewhere.



4.6 Unitary and Hybrid Systems

Unitary systems are complete “packed” A/C units. Examples:

○

4.6.1 Incremental units

Examples are motel units and larger single zone units. They are full heating, cooling and air handling systems with heating coils, cooling coils, refrigerator, and fans, etc.

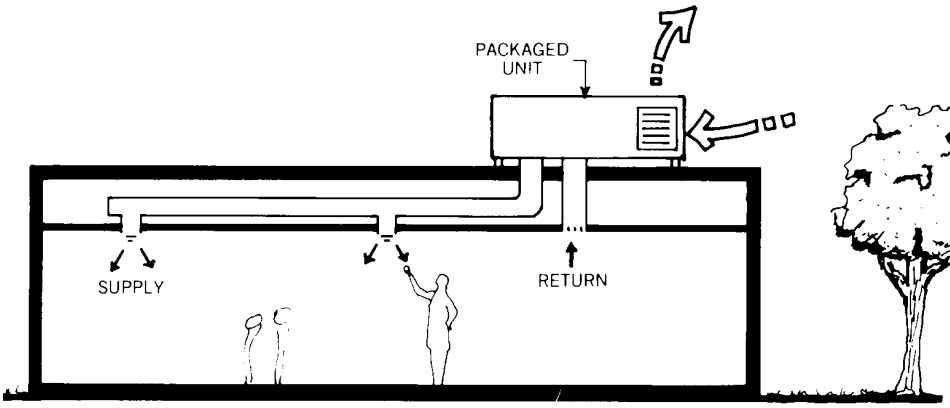


FIGURE 14.10b
Packaged units can contain both heating and cooling equipment.

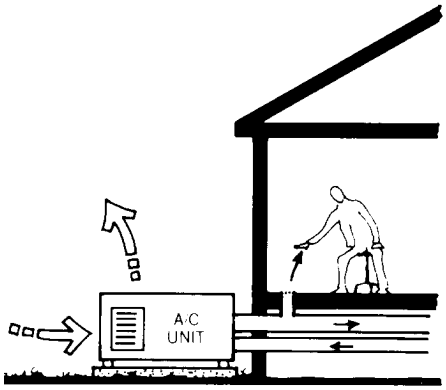


FIGURE 14.10d
Packaged unit designed for crawl space construction.

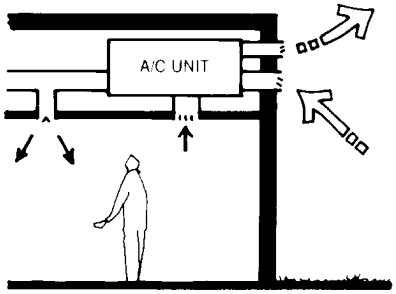


FIGURE 14.10e
Packaged unit designed for placement above suspended ceiling.

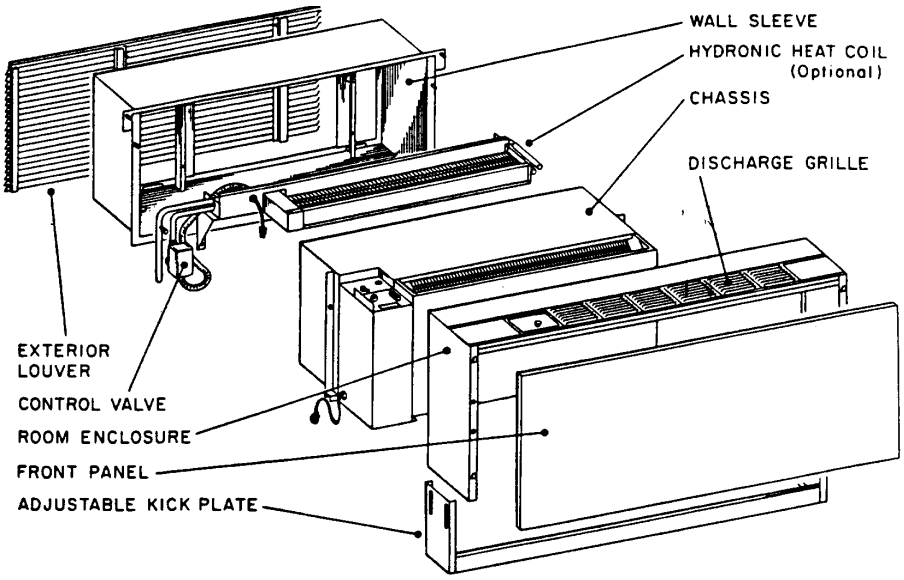
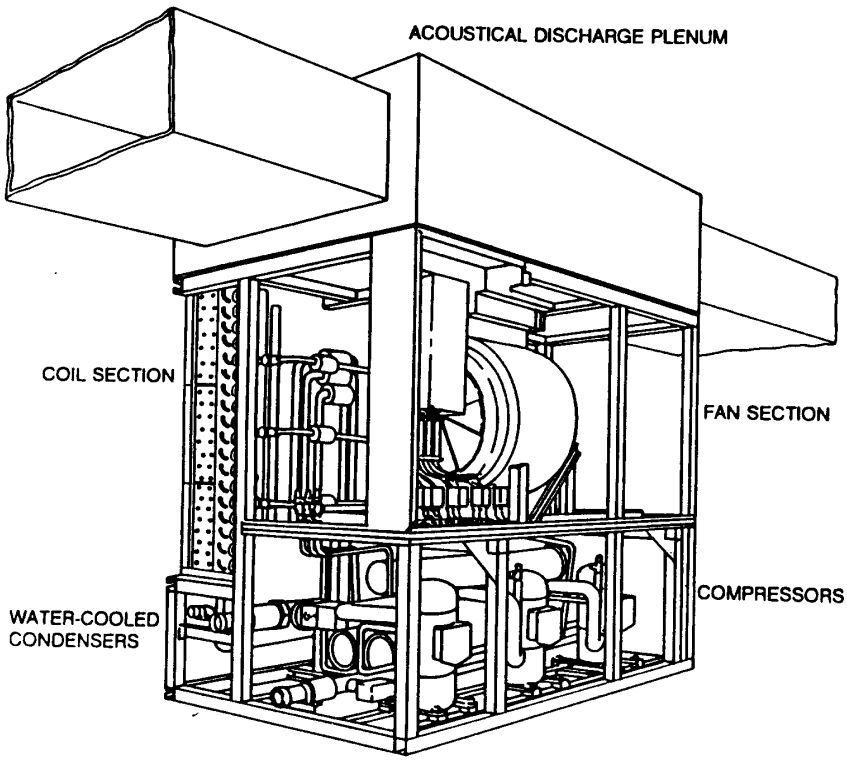


Fig. 1 Packaged Terminal Air Conditioner with Heat Section Separate from Cooling Chassis



ʒ. 5 Commercial Self-Contained Unit with Discharge Plenum

4.6.2 Heat pumps

- Air-to-air heat pumps
- Water-to-air heat pumps (water serves as heat source)

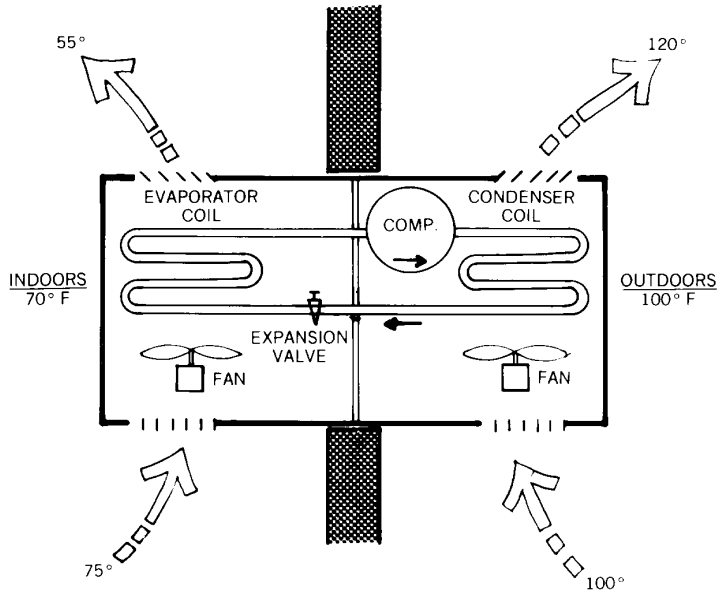


FIGURE 14.8a
A simple through the wall air conditioner unit essentially consists of a compressive refrigeration machine.

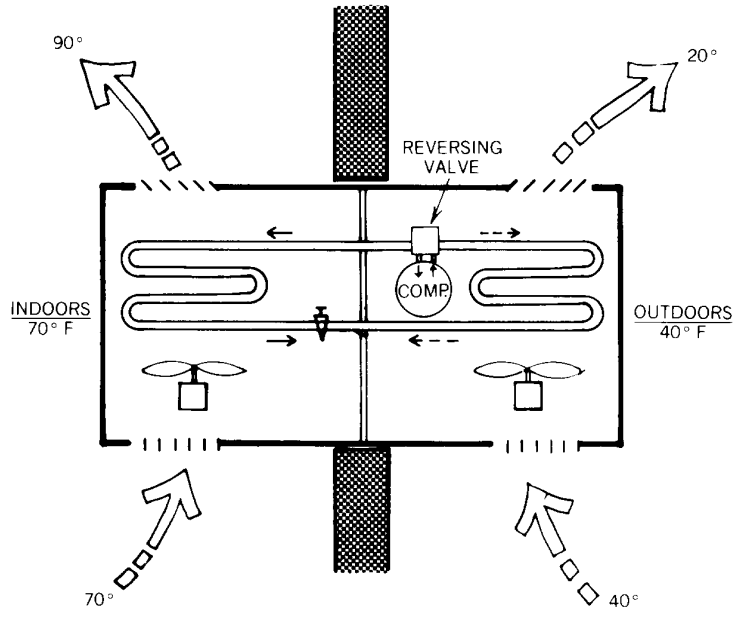


FIGURE 14.8b
In a "heat pump" the reversing valve allows the refrigerant to flow in either direction. In the winter condition shown, the outdoor coil becomes the evaporator and the indoor coil the condenser.

Heat source/sink:

○

Air source - low cost, and it is least efficient. Water (ground) source – high cost, and it is more efficient than air.

4.7 Heat Recovery

Heat recovery is utilization of "waste" energy streams. Sources for heat recovery are:

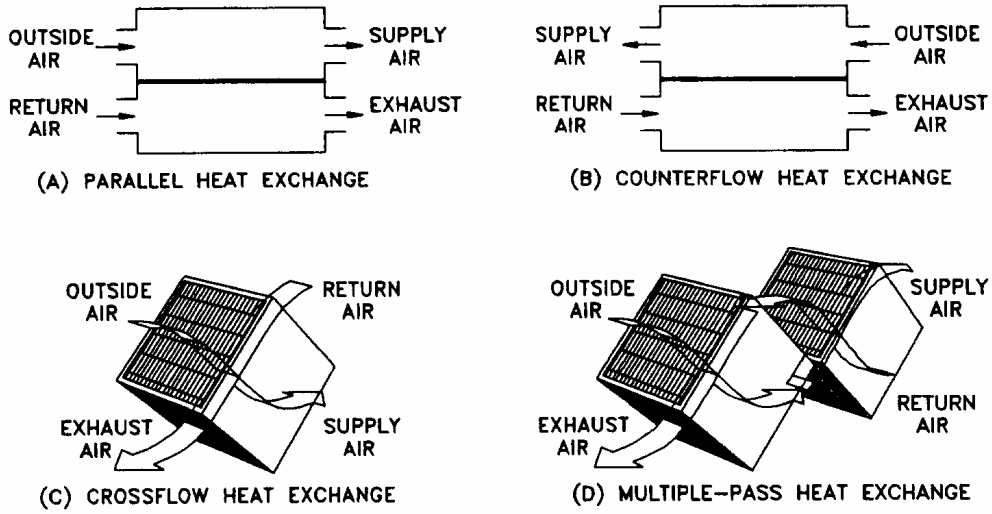
- Relief / exhaust air
- Combustion gases
- Coolant stream

The recovered heat is used to:

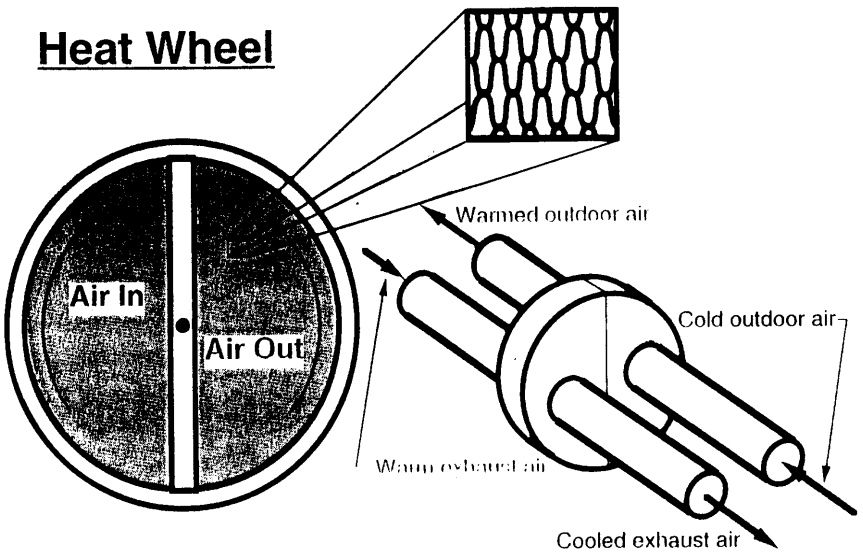
○

Air-to-Air Heat Recovery System

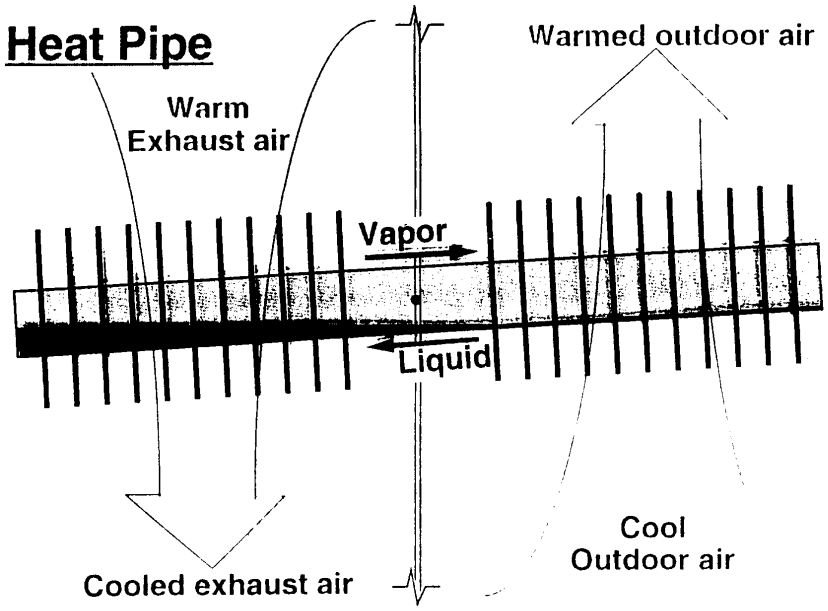
- Air-to-air HX (heat exchanger)
- Heat wheel
- Heat pipe
- Heat pump



Air-to-air type of heat recovery system



2-13



2-14

Potential obstacles for heat recovery:

- Small ΔT => large heat exchanger (very expensive)
- Separation of source and end use
- Non-coincident loads
- "Parasitic" energy

4.7 Summary of Different Air-Conditioning Systems

System	Advantages	Disadvantages
All Air	• Central equipment location • No piping in occupied area • Use of outside air (free cooling) • Easy seasonal change • Heat recovery possible • Closest operating conditions	• Duct clearance • Large ducts - space • Air balancing difficulties
Air Water	• Individual room control • Separate secondary heating/cooling • Less space for ducts • Smaller HVAC central equipment • Central filter, humidification	• Changeover if only two pipes • Operating complex if two pipes • Control is numerous • Fan coil clearance problem • No-shut off for primary air • High pressure for induction • Four pipe system is too expensive
All Water	• Less space • Locally shutoff (individual control) • Quick pull down • Good for existing buildings	• More maintenance in occupied area • Coil cleaning difficulties • Filter • Open window for IAQ
Uni-tary	• Individual room control • Simple and inexpensive • Independent of other buildings • Manufacturer made it ready	• Limited performance • No humidity control (general) • More energy (low efficiency) • Control of air distribution • Filter • Overall appearance

Chapter 5 Heat Transmission in Building Structures

5.1 Basic heat transfer modes
5.2 Construction assemblies

Readings: Chapter 5 of the text
1997 ASHRAE Handbook - Fundamentals, 3.1-3.16, 29.1-29.12
F.P. Incropera and D.P. DeWitt. "Fundamentals of Heat and Mass Transfer"

5.1 Basic Heat Transfer Modes

- Conduction
- Convection
- Thermal radiation

5.1.1 Conduction

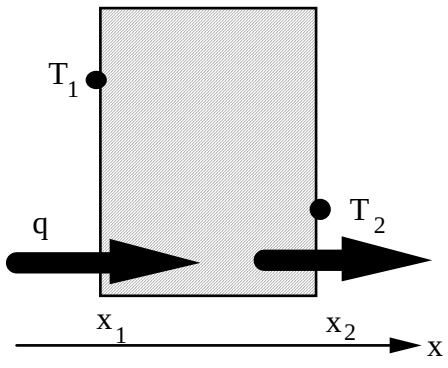
Fourier's Law:

$$q = -k\frac{dT}{dx}$$

where q = heat flux (W/m^2)
k = thermal conductivity (W/m K) (material property, **Table 5-1** of the text)
dT/dx = temperature gradient (K/m)

- Heat transfer in a single-layer wall

In most practical uses, k is approximated as constant. For steady-state heat transfer, q is constant. Then the equation can be integrated



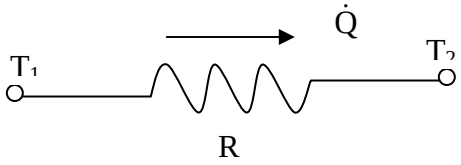
$$\int_{x1}^{x2} q\,dx = \int_{T1}^{T2} -k\,dT$$

to yield

$$\dot{q}\ (x_2 - x_1) = - k\ (T_2 - T_1)$$

or

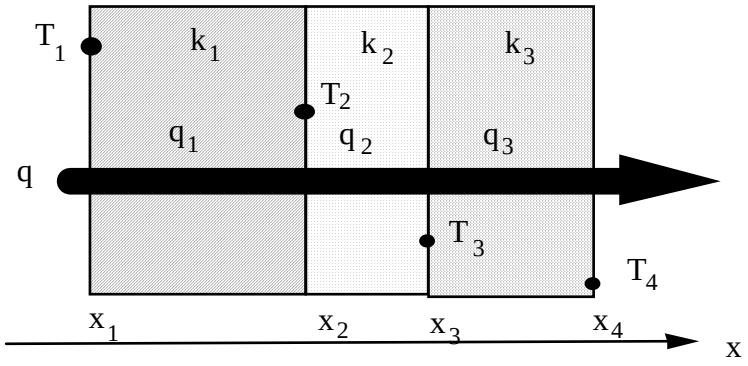
$$\dot{q} = -\frac{k(T_2 - T_1)}{(x_2 - x_1)} = -\frac{(T_2 - T_1)}{R}$$

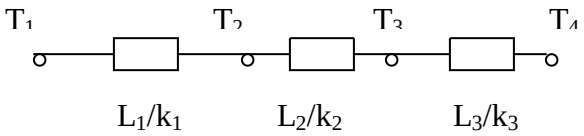


where

$$R = \frac{x_2 - x_1}{k} = \text{thermal resistance (K m}^2\text{/W) or (hr ft °F/Btu).}$$

- Heat transfer in a multi-layer wall





$$q_1 = -\frac{(T_2 - T_1)}{R_1} \qquad R_1 = \frac{x_2 - x_1}{k_1}$$

$$q_2 = -\frac{(T_3 - T_2)}{R_2} \qquad R_2 = \frac{x_3 - x_2}{k_2}$$

$$q_3 = -\frac{(T_4 - T_3)}{R_3} \qquad R_3 = \frac{x_4 - x_3}{k_3}$$

Since

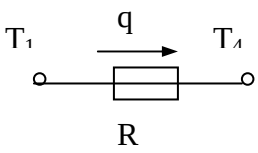
$$q_1 = q_2 = q_3 = q$$

Re-arrange the equations

$q_1R_1 = -(T_2 - T_1)$
$q_2R_2 = -(T_3 - T_2)$ so that $q(R_1 + R_2 + R_3) = -(T_4 - T_1)$
$q_3R_3 = -(T_4 - T_3)$

Or

$$q = -\frac{(T_4 - T_1)}{R_1 + R_2 + R_3} = -\frac{T_4 - T_1}{R}$$



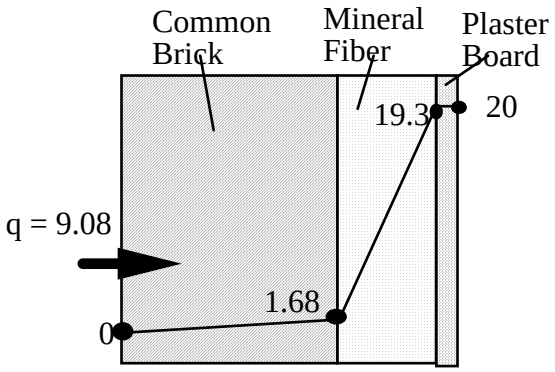
Here

$R = R_1 + R_2 + R_3$ is the total thermal resistance due to conduction.
$C = 1/R$ thermal conductance (W/m^2 K).

Example 5.1

An exterior wall of an PSU classroom consists of 0.24 m thick face brick, 0.09 m thick mineral fiber, and 0.013 m thick plasterboard. If the exterior surface temperature of the wall is 0 ^oC and the interior surface 20 ^oC. Determine the heat flux and temperature distribution in the wall.

Solution:

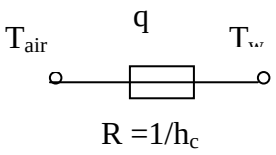


From Table 5-1, k_{brick} = 1.30 W/m K, R_{fiber} = 1.94 m^2 K/W, R_{board} =0.078 m^2 K/W.

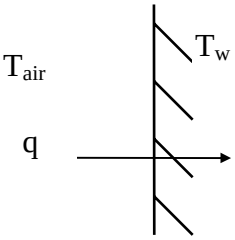
5.1.2 Convection

Newton's Law:

$$q = h_c (T_{air} - T_w)$$



where q = heat flux due to convection (W/m^2)
h_c = convective heat transfer coefficient (W/m^2K)
T_{air} = bulk fluid (air) temperature (^oC)
T_w = surface temperature (^oC)



Newton's Law can be written as

$$q = \frac{T_{air} - T_w}{R} \qquad \text{with} \qquad R = \frac{1}{h_c}$$

The h_c is related to flow state. h_c = 3 ~ 6 W/m^2K for natural convection and 6 ~ 35 W/m^2K for air flow in and around buildings.

5.1.3 Radiation

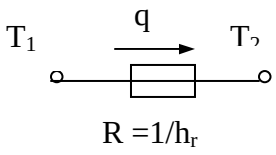
Stefan-Boltzmann's Law:

$$q = \sigma \varepsilon_1 \varepsilon_2 (T_1^4 - T_2^4)$$

where q = heat flux due to radiation (W/m^2)
σ = Stefan-Boltzmann constant (= 5.673×10^{-8} $W/m^2 K^4$)or (= 0.1713×10^{-8} $Btu/h.ft^2 R$)
ε = surface emittance (-)
T = surface temperature (K)

The equation can be further written as:

$$q = \sigma \varepsilon_1 \varepsilon_2 (T_1^3 + T_1^2 T_2 + T_1 T_2^2 + T_2^3)(T_1 - T_2)$$

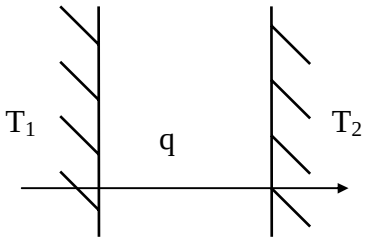


When $T_1 \sim T_2$, the equation can be approximated as:

$$q = 4\sigma \varepsilon_1 \varepsilon_2 \overline{T}^3 (T_1 - T_2)$$
$$= h_r (T_1 - T_2)$$

where $\overline{T} = \frac{T_1 + T_2}{2}$

and the radiative heat transfer coefficient $h_r = 4\sigma \varepsilon_1 \varepsilon_2 \overline{T}^3$.



Example 5.2

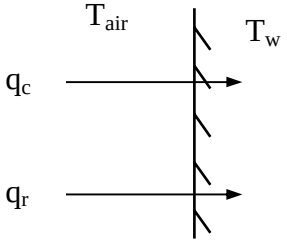
Determine h_r in a typical classroom at PSU.

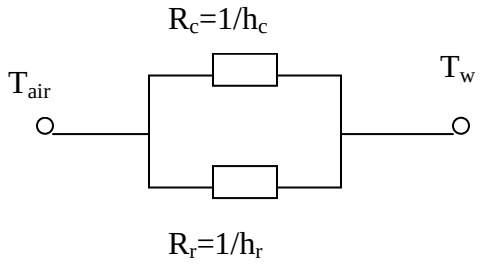
Overall heat transfer coefficient $h = h_c + h_r$
Overall film resistance $R = 1/h = 1/(h_c + h_r)$
(See Table 5-2 for values)

Radiative heat transfer between two arbitrary surfaces is

$$q = \frac{\sigma(T_1^4 - T_2^4)}{\frac{1-\varepsilon_1}{A_1\varepsilon_1} + \frac{1}{A_1F_{12}} + \frac{1-\varepsilon_2}{A_2\varepsilon_2}}$$

6.1.4 Combined convection and radiation



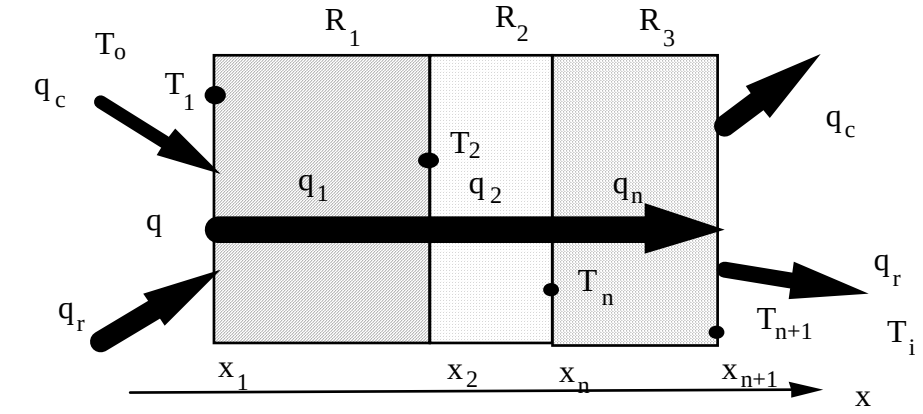


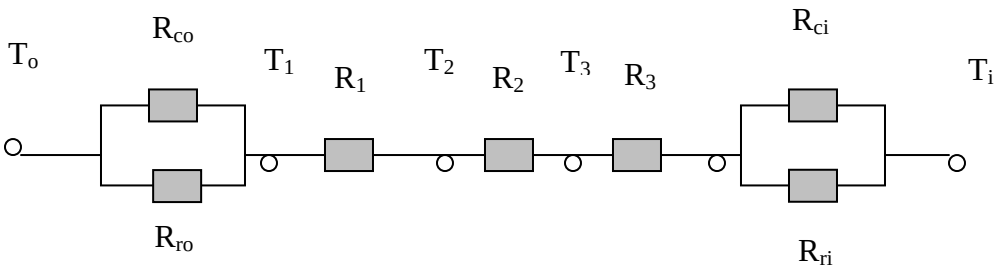
Overall film resistance $R = \frac{R_c R_r}{R_c + R_r}$

$$q = \frac{T_{air} - T_w}{R}$$

5.1.5 Overall heat transmission

It is necessary to consider the heat transfer from inside air to outside air. The heat transfer processes include conduction, convection, and radiation simultaneously.





$q R_o = T_o - T_1$
$q R_1 = T_1 - T_2$
... ...
$q R_n = T_{n+1} - T_n$
$q R_i = T_{n+1} - T_i$

+) $q[R_o + (\sum_{j=1}^{n} R_j) + R_i] = T_o - T_i$

$$q = \frac{T_o - T_i}{R_o + (\sum_{j=1}^{n} R_j) + R_i}$$

$R = R_o + (\sum_{j=1}^{n} R_j) + R_i$ (Overall thermal resistance)

$U = 1/R$ (Overall heat transfer coefficient)

Example 5.3

Continue to work on example 6.1 to determine the outdoor and indoor air temperatures.

Solution:

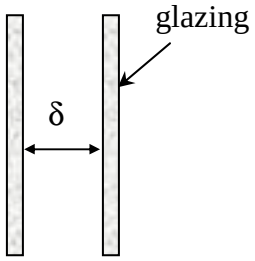
○

5.2 Construction Assemblies

5.2.1 Air spaces and windows

- Air spaces

Air conduction resistance



$$R = \delta/\lambda$$

where R = air conduction resistance
δ = air layer thickness
λ = thermal conductivity (0.023 W/m K for air)

"Ideal case" for 0.02 m thick of air layer

$$R = \delta/\lambda = 0.02/0.023 = 0.87 \text{ m}^2 \text{ K/W}$$

Because of convection and radiation effect, the actual R is between 0.1 to 0.7 m^2K/W.

- Windows

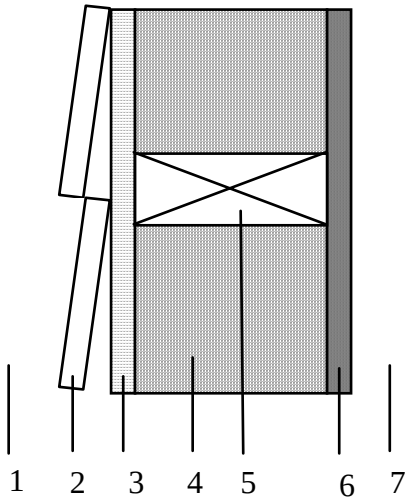
Single glazing	U = 5.5 W/m^2 K	R-1 (ft^2 ^oF h/Btu)
Double glazing	U = 2.7 W/m^2 K	R-2
Triple glazing	U = 1.8 W/m^2 K	R-3

5.2.2 Walls

The following conditions are assumed in calculating the design R-values:
1. Equilibrium or steady-state heat transfer, disregarding effects of heat storage
2. Surrounding surfaces at ambient air temperature

Example 5.4

Calculate the U-factor of the 38 mm by 90 mm stud wall. The studs are at 400 mm on center. There is 90-mm mineral fiber batt insulation (R = 2.30 K m^2/W) in the stud space. The inside finish is 13-mm gypsum wallboard; the outside is finished with rigid foam insulating sheathing (R = 0.70 K m^2/W) and 13-mm by 200-mm wood bevel lapped siding. The insulated cavity occupies approximately 75% of the transmission area and the stud 25%.



1. Outside surface
2. Wood bevel lapped siding
3. Sheathing
4. Mineral fiber batt insulation
5. Wood stud
6. Gypsum wallboard
7. Inside surface

Solution:

Element	R (Insulated cavity)	R (Studs)
1. Outside surface, 3.4 m/s wind	0.03	0.03
2. Wood bevel lapped siding	0.14	0.14
3. Rigid foam insulating sheathing	0.70	0.70
4. Mineral fiber batt insulation	2.30	-
5. Wood stud, 38 mm by 90 mm	-	0.63
6. Gypsum wallboard, 13 mm	0.10	0.10
7. Inside surface, still air	0.12	0.12
	R_1 = 3.39	R_2 = 1.72

- Thermal bridges

Thermal conductivity of metals is a thousand times higher than that of insulation material. The less loss through the metal conduction is considerable.

Example 5.5

The thermal conductance of a 200 mm thick masonry wall is 0.4 W/m^2 K. If there is a ϕ =20 mm aluminum bar through the wall and thermal conductivity is 220 W/m K, compare the heat loss through 1 m^2 of the masonry wall and the aluminum bar.

Solution

○

5.2.3 Other assemblies

- Below-grade walls

$$Q = U\ A\ (T_{ground} - T_i)$$

where Q = heat flow rate
U = U-factor of basement wall (Table 5-8 in the textbook)
A = area of basement wall below grade
T_{ground} = ground surface temperature
T_i = inside air temperature

$$T_{ground} = T_{avg} - Amp$$

where T_{avg} = average winter temperature (Table 5-10 in the textbook)
Amp = amplitude of ground temperature variation (Figure 5-7 in the textbook)

- Below-grade floors

$$Q = U\ A\ (T_{ground} - T_i)$$

where U = U-factor (Table 5-9 in the textbook)
A= basement floor area

- Slab-on-ground construction

$$Q = U'\ P\ (T_o - T_i)$$

where U' = heat transmission loss per linear meter of slab edge (Table 5-11 in the textbook)
P = perimeter of slab
T_o = outdoor design temperature

Chapter 6. Ventilation and Infiltration

6.1 Ventilation
6.2 Infiltration

Readings: Chapter 7 of the textbook, pp. 194-204.
ASHRAE 1997 Fundamentals Handbook, Chapters 15, 25

6.1 Ventilation

6.1.1 Definition

- Ventilation: intentional and controllable air exchange between indoor and outdoor air
- Infiltration: unintentional and uncontrolled air exchange between indoor and outdoor air

- Forced ventilation: powered ventilation
- Natural ventilation: unpowered ventilation

6.1.2 Forced convection

Forced ventilation is used to remove/add heat in a space to maintain appropriate thermal conditions.

Sensible heat exchange:

○

where $\dot{Q}_s$ = sensible heating/cooling load(W)
$\dot{V}$ = airflow rate (m_3/s)
ρ = air density (kg/m^3)
C_p = specific heat of air (J/kg K)
T_i = indoor air temperature (^oC)
T_s = supply air temperature (^oC)

Latent heat exchange:

○

where $\dot{Q}_l$ = latent heating/cooling load (W)
$\dot{V}$ = airflow rate (m^3/s)
ρ = air density (kg/m^3)
i_{fg} = latent heat of water vapor at room air temperature (J/kg K)
W_i = indoor air humidity ratio (kg_v/kg_a)
W_s = supply air humidity ratio (kg_v/kg_a)

V must be the same for the sensible and latent heat exchange.

Forced ventilation is also used to maintain an acceptable indoor air quality.

○

where C_i = contaminant concentration of indoor air
C_s = contaminant concentration of supply air
S = contaminant source
$\dot{V}$ = volume airflow rate

Example 6.1

An PSU classroom with 10 occupants has 4000 W sensible cooling load under summer design conditions. If T_i = 25 ^oC, T_s = 15 ^oC, C_s = 300 ppm, and S_{CO2} = 0.30 L/(min/person), determine the ventilation rate.

Solution:
For comfort:

○

For air quality:

○

Select greater one to meet the demand for both comfort and air quality:

$\dot{V}$ = 1200 m^3/h

6.1.3 Natural convection

Driving force of natural convection:
- pressure difference caused by wind
- air density difference due to buoyancy (stack effect)
- pressure difference caused by appliance operations (combustion devices, hood, etc.)

$$\Delta p = p_o + p_w - p_{ir} + \Delta p_s$$

where Δp = pressure difference between outdoors and indoors at location (Pa)
p_o = static pressure at reference height of undisturbed flow (Pa)
p_w = wind pressure at location (Pa)
p_{ir} = interior static pressure at reference height (Pa)
Δp_s = pressure difference due to buoyancy (Pa)

○

where C_p = surface pressure coefficient
ρ = air density (kg/m^3)
V = wind speed (m/s)

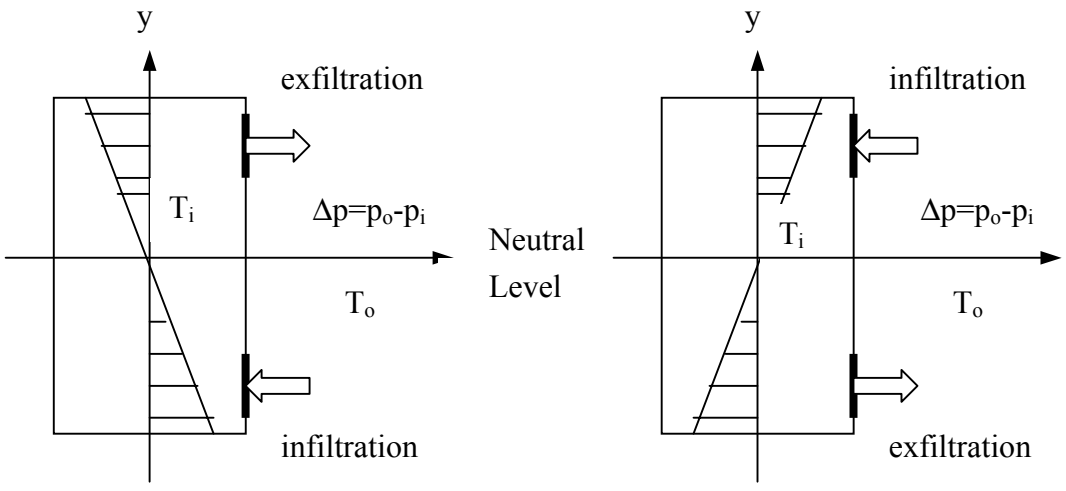
$$C_p = \ln [1.248 - 0.703 \sin (a/2) - 1.175 \sin^2 (a) + 0.131 \sin^3 (2aG) + 0.769 \cos (a/2) + 0.07 G^2 \sin^2 (a/2) + 0.717 \cos^2 (a/2)]$$

where a = angle between wind direction and outward normal of wall under consideration
G = natural log of ratio of wall width under consideration to adjacent wall

○

where g = gravity (9.81 m/s^2)
h_{NPL} = height of natural pressure level

Buoyancy pressure (Δp_s)



Winter ($T_o<T_i$) Summer ($T_o>T_i$)

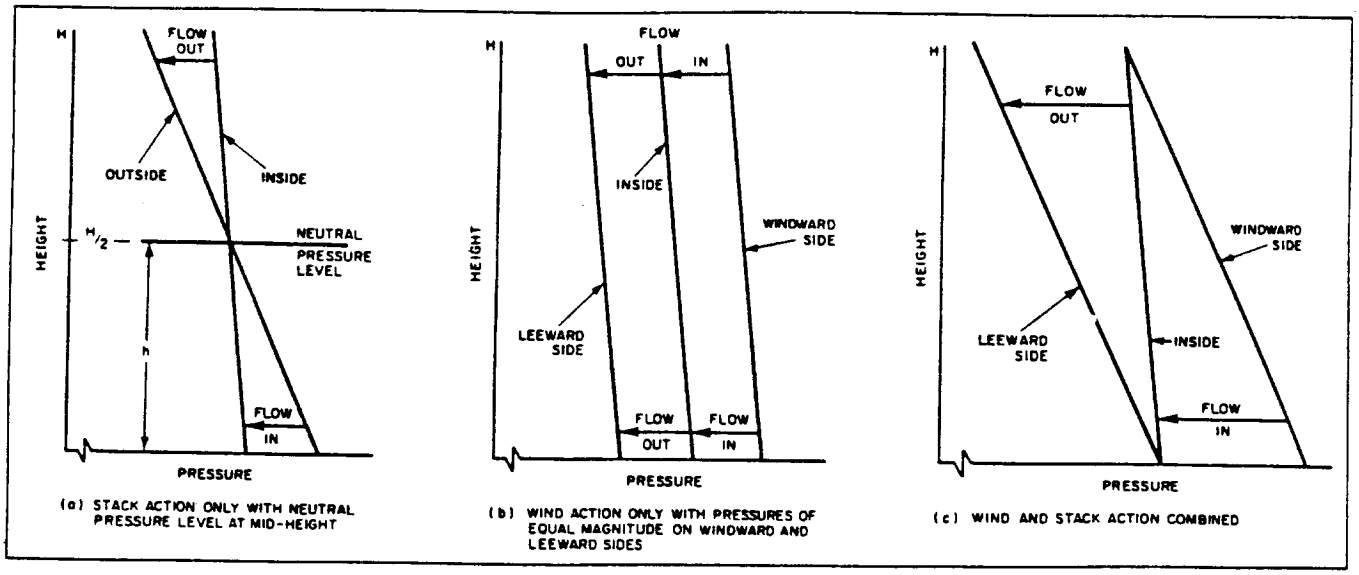


Fig. 3 Distribution of Inside and Outside Pressures over Height of Building

- Airflow through large openings:

$$Q = C_D A \sqrt{\frac{2\Delta p}{\rho}}$$

where Q = airflow rate through the opening (m^3/s)
C_D = discharge coefficient (-)
A = cross sectional area of the opening (m^2)
ρ = air density (kg/m^3)
Δp = pressure difference across opening (Pa)

- Flow caused by wind:

$$Q = C_v A V$$

where C_v = effectiveness of openings (-)
(0.5 to 0.6 for perpendicular winds and 0.25 - 0.35 for diagonal winds)
A = opening area (m^2)

- Flow caused by thermal forces:

$$Q = k A [2g \Delta h_{NPL} (T_i - T_o)/T_i]^{0.5}$$

where k = discharge coefficient ($k = 0.40 + 0.0045 |T_i - T_o|$)
Δh_{NPL} = height from midpoint of lower opening to NPL (m)

6.2 Infiltration

Infiltration is from combined effect of buoyancy and wind. Infiltration rate:

○

where Q = infiltration rate (L/s)
L = effective leakage area (cm^2)
C_s = stack coefficient ($L^2\ s^{-2}\ cm^{-4}\ K^{-1}$)
ΔT = average indoor-outdoor temperature difference (K)
C_w = wind coefficient ($L^2\ s^{-4}\ cm^{-4}\ K^{-1}\ m^{-2}$)
V = average wind speed measured at local weather station (m/s)

Table 5 Example of Calculation of Building Effective Air Leakage Area Based on Component Leakage Areas

Component	Description	Size or Number	× A_L per unit	= A_L, cm^2
Sills	Uncaulked	43.2 m	4.0 cm^2/m	173
Electrical outlets		20	0.5 cm^2 ea	10
Windows	Sliding	13.1 m^2	4.0 cm^2/m^2	75
Framing		13.1 m^2	1.7 cm^2/m^2	
Exterior doors	Single	5.7 m^2	7.7 cm^2/m^2	54
Framing		5.7 m^2	1.7 cm^2/m^2	
Fireplace	Without damper	1	350 cm^2 ea	350
Penetrations	Pipes	7	6.0 cm^2 ea	42
Heating ducts	Ducts untaped, in basement	1	144 cm^2 ea	144
	Calculated total building air leakage area A_c =			848 cm^2

Table 6 Stack Coefficient C_s

	House Height (Stories)		
	One	Two	Three
Stack coefficient	0.000 145	0.000 290	0.000 435

Table 7 Local Shielding Classes

Class	Description
1	No obstructions or local shielding
2	Light local shielding; few obstructions, few trees, or small shed
3	Moderate local shielding; some obstructions within two house heights, thick hedge, solid fence, or one neighboring house
4	Heavy shielding; obstructions around most of perimeter, buildings or trees within 10 m in most directions; typical suburban shielding
5	Very heavy shielding; large obstructions surrounding perimeter within two house heights; typical downtown shielding

Table 8 Wind Coefficient C_w

Shielding Class	House Height (Stories)		
	One	Two	Three
1	0.000 319	0.000 420	0.000 494
2	0.000 246	0.000 325	0.000 382
3	0.000 174	0.000 231	0.000 271
4	0.000 104	0.000 137	0.000 161
5	0.000 032	0.000 042	0.000 049

Example 6.2

Estimate the infiltration at design conditions for a two-story house in State College. The house has an effective leakage area of 500 cm^2, a volume of 340 m^3, and is surrounded by a thick hedge (shielding class 3).

Solution:

Under winter design conditions: T_o = -13 °C, V = 8 m/s, T_i = 22 °C.

From A_Tables 25.6 and 25.8 of the handouts, C_s = 0.000290, C_w = 0.000231

○

Air change rate = 284 / 340 = 0.84 ACH.

Air exchange method:

extremely low	-	0.1 ACH (air change rate per hour)
low	-	0.5 ACH
normal	-	1 ACH
high	-	2 ACH
extremely high	-	3 ACH

Crack method:

$$Q = A\ C\ \Delta p^n$$

where A = effective leakage area of cracks (m^2)
C = flow coefficient
Δp = pressure difference between outdoor and indoor (Pa)

Chapter 7 Solar Radiation

7.1 Solar radiation
7.2 Heat gain through fenestrations

Readings: Chapter 6 of the textbook
ASHRAE 1997 Fundamentals Handbook, Chapter 29.14-49

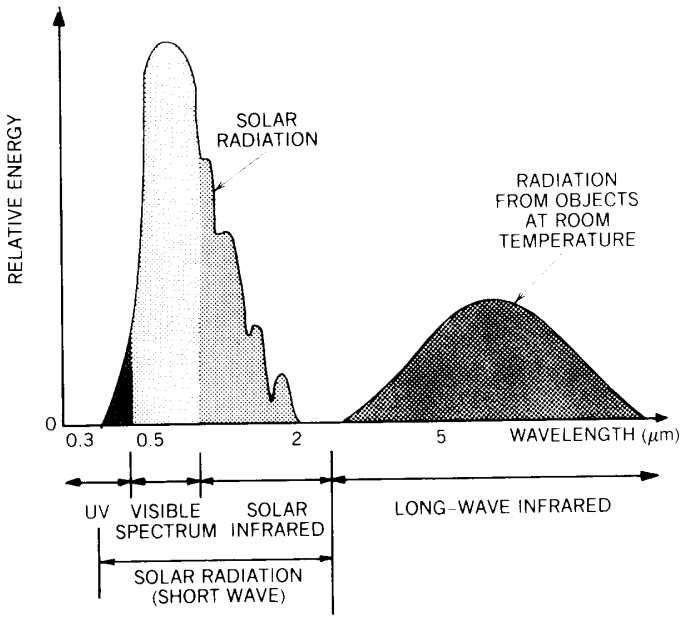
7.1 Solar radiation

7.1.1 Solar radiation

The earth's orbit is elliptical. The sun's surface temperature is approximately 6,000°C (10,000°F). The radiant heat flux outside the atmosphere of the earth is 2200 W/m^2.

The extraterrestrial radiant flux, E_o, is
1418 W/m^2 on January 4
1325 W/m^2 on July 5
1360 W/m^2 as mean value
The solar radiation at sea level is 1370 W/m^2.

The energy spectrum:
0.29 - 0.40 μm (ultraviolet) 9%
0.40 - 0.70 μm (visible) 39%
0.70 - 3.50 μm (infrared) 52%
Peak value: 0.50 μm



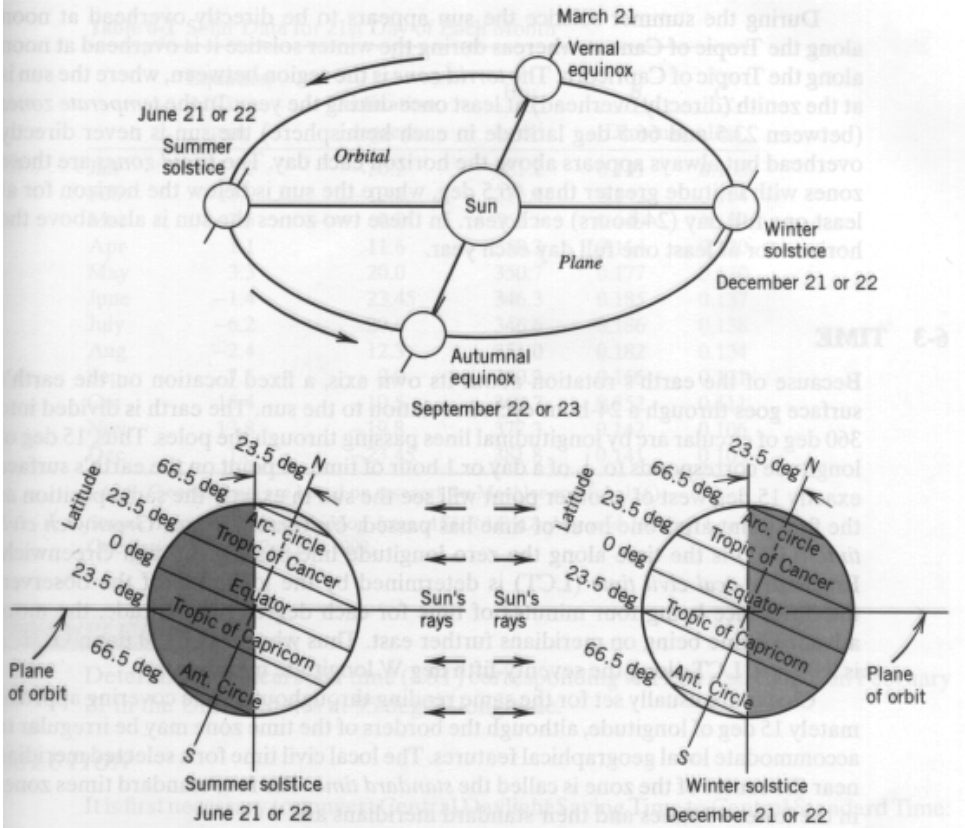


Figure 6-1 The effect of the earth's tilt and rotation about the sun.

7.1.2 Solar time

Apparent solar time (Local solar time) is determined by a sundial (The sun's position in the sky).

15° longitude → 1 hour
1° longitude → 4 min

Mean time (Standard time): GCT = Greenwich Civil Time (0° longitude)

Local Civil Time (LCT): Determined by longitude.
Local Stand Time: Through the center of a time zone

EST = Eastern Standard Time (75° W longitude)
CST = Central Standard Time (90° W longitude)
MST = Mountain Standard Time (105° W longitude)
PST = Pacific Standard Time (120° W longitude)

Equation of time is a factor of earth orbital velocity. Due to non-symmetry of earth's orbit, earth's rotational speed is irregular (T_Table 6-1 or A_Table 8).

Local solar time = Local standard time
+ Equation of time
+ 4 (Local standard time meridian - local longitude) [Minutes]

Local Solar Time (LST): Time measured by position of sun

Daylight Saving Time (1 hr advance in summer)

Example 7-1

Longitude of a PSU classroom is 78° W. Determine the local solar time at EST 12:00 noon on February 21.

Solution:

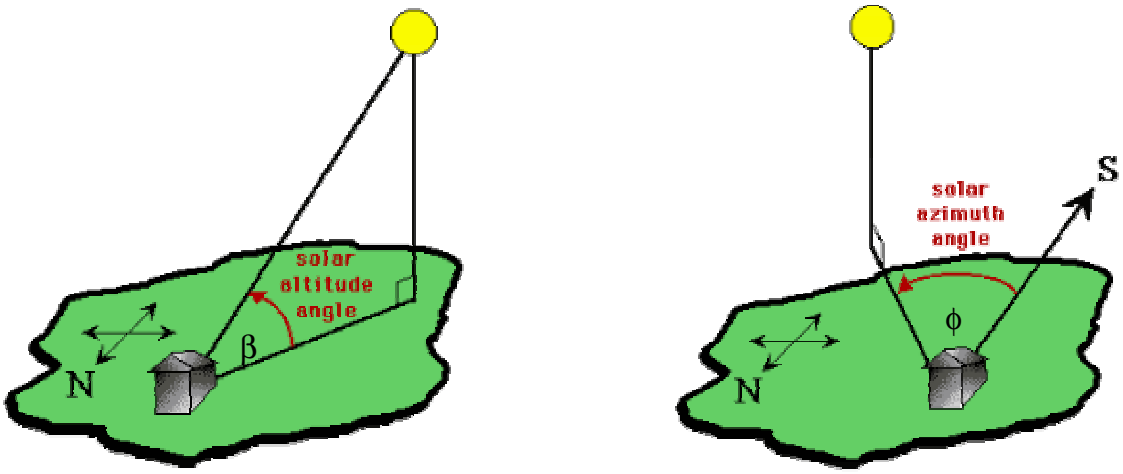
EST is for longitude of 75° W. From T_Table 6-1 or A_Table 8, equation of time is about -14 minutes. Then

Local solar time = Local standard time + Equation of time
+ 4 (Local standard time meridian - local longitude)
= 12:00 - 0:14 + 0:4 (75 - 78)
= 11:34 am.

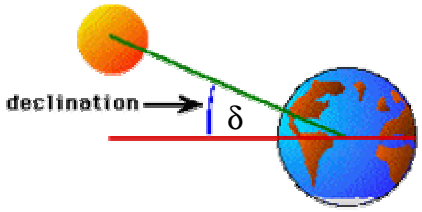
7.1.3 Solar angles

- Sun's position

The sun's position can be determined by two angles:
Solar altitude, β
Solar azimuth from the south, ϕ



$$\sin\beta = \cos L \cos\delta \cos H + \sin L \sin\delta$$
$$\cos\phi = (\sin\beta \sin L - \sin\delta)/(\cos\beta \cos L)$$



where L = local latitude
δ = solar declination (T_Table 6-1, A_Table 8)
H = local solar time expressed as the hour angle

Example 7-2

Continue from example 7-1. Latitude of the PSU classroom is 41° N. Find the sun's position.

Solution

$L = 41^o$
$H = 11:34 \text{ am} - 12:00 \text{ (true south)} = -26 \text{ minutes}$
$= -26/(60 \times 24) \times 360^o = -6.5^o$

$\delta = -10.8^o$ (A_Table 6-1, T_Table 8)

$\sin\beta = \cos L \cos\delta \cos H + \sin L \sin\delta$
$= \cos 41^o \cos(-10.8^o) \cos(-6.5^o) + \sin 41^o \sin(-10.8^o)$
$= 0.6136$
$\beta = 37.85^o$

$\cos\phi = (\sin\beta \sin L - \sin\delta)/(\cos\beta \cos L)$
$= (\sin 37.85^o \sin 41^o - \sin(-10.8^o))/(\cos 37.85^o \cos 41^o)$
$= 0.9899$
$\phi = 8.1^o$

- Incident angle

The incident angle θ of a surface:

$$\cos\theta = \cos\beta \cos\gamma \sin\Sigma + \sin\beta \cos\Sigma$$

where θ = angle between the sun's rays and the normal to the surface
$\gamma = \phi - \psi$
ψ = surface azimuth defined as:

Orientation	N	NE	E	SE	S	SW	W	NW
ψ	180°	-135°	-90°	-45°	0°	45°	90°	135°

Σ = tilt angle of surface (angle between the normal to the surface and the normal to the horizontal surface)

$\Sigma = 0^o$ for a horizontal surface and 90° for a vertical surface

Example 7-3

Continue from Example 7-2. Find the incident angle of a vertical window facing east.

Solution

$\psi = -90^o$
$\Sigma = 90^o$
$\gamma = \phi - \psi = 8.1^o - (-90^o) = 98.1^o$

$\cos\theta = \cos\beta\cos\gamma\sin\Sigma + \sin\beta\cos\Sigma$
$= \cos 37.85^o \cos 98.1^o \sin 90^o + \sin 37.85^o \cos 90^o$
$= -0.1113$
$\theta = 96.39^o$ - No solar on the surface

7.1.4 Solar radiation with a clear sky

Solar radiation with a clear sky consists of three parts:
- Direct radiation, G_D
- Diffuse radiation, G_d
- Reflected radiation, G_R

Total solar radiation on a horizontal surface, G_{tH}

$G_{tH} = G_{DH} + G_{dH}$, $G_R=0$

Total radiation on a tilted surface

$G_{t\theta} = G_{D\theta} + G_{d\theta} + G_R$

All the three parts are the functions of normal direct radiation, G_{ND}

$$G_{ND} = \frac{A}{\exp(B/\sin\beta)}$$

where A = apparent solar radiation (T_Table 6-1, A_Table 8)
B = atmospheric extinction coefficient (T_Table 6-1, A_Table 8)
β = solar altitude

- **Direct** solar radiation on a surface of arbitrary orientation

$G_{D\theta} = G_{ND} C_N \cos \theta$

where C_N = clearness number (T_Figure 6-7)
θ = incident angle
On a horizontal surface, $\cos \theta = \sin \beta$
$G_{DH} = G_{ND} C_N \sin \beta$

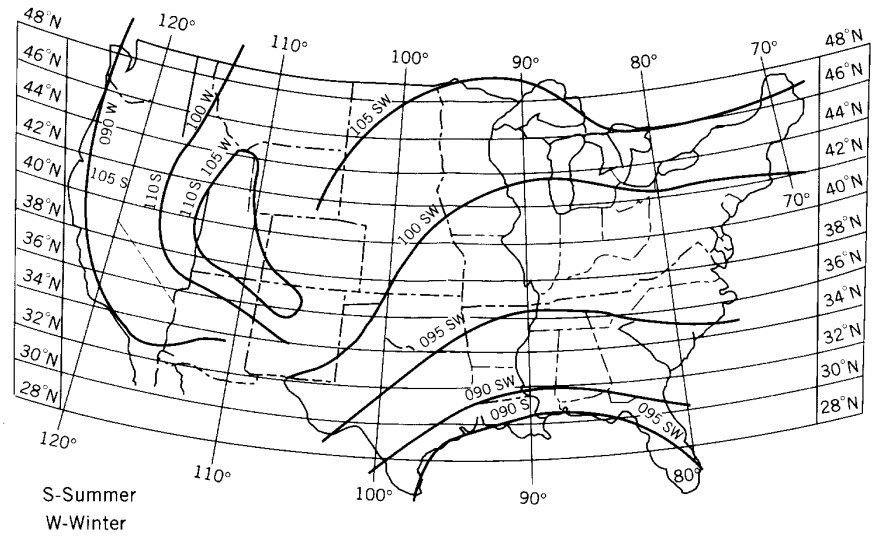


Figure 6-7 Estimated atmospheric clearness numbers C_N in the United States for nonindustrial localities, percent. (Reprinted by permission from *ASHRAE Handbook, Fundamentals Volume*, 1989)

- **Diffuse** radiation

On a horizontal surface:

$G_{dH} = C G_{ND} / C_N^2$

where C = ratio of diffuse on a horizontal surface to direct normal radiation (T_Table 6-1, A_Table 8)

On a tilted surface

$G_{d\theta} = (C G_{ND} / C_N^2) F_{ws}$

F_{ws} = view factor between the surface and the sky

$F_{ws} = (1 + \cos \Sigma)/2$

For a horizontal surface $F_{ws} = 1$ and for a vertical surface $F_{ws} = 0.5$.

- **Reflected** radiation

$G_R = (G_{DH} + G_{dH})\ \rho_g\ F_{wg}$

where ρ_g = reflectance of ground
F_{wg} - view factor between the surface and the ground

$F_{wg} = (1 - \cos \Sigma)/2$

For a horizontal surface $F_{wg} = 0$ and for a vertical surface $F_{wg} = 0.5$.

Example 7-4

Continue from Example 7-3. If the ground reflectivity is 0.3, find the total solar radiation on the vertical window surface.

Solution

From T_Table 6-1 or A_Table 8, A = 1187 W/m^2, B = 0.142, and C = 0.104.
From T_Figure 6-7, $C_N = 1.0$

$$G_{ND} = \frac{A}{\exp(B/\sin\beta)}$$
$$= \frac{1187}{\exp(0.142/\sin 37.85^o)}$$

= 942 W/m^2

$G_{D\theta} = G_{ND}\ C_N \cos\theta$
= 0 No direct solar radiation

$G_{dH} = C\ G_{ND} / C_N^2$
$= 0.104 \times 942 / 1.0^2$
=98 W/m^2

$G_{d\theta} = G_{dH}\ F_{ws}$
$= G_d\ (1 + \cos \Sigma)/2$
$= 98\ (1 + \cos 90^o)/2$
= 49 W/m^2

$G_R = (G_{DH} + G_{dH})\ \rho_g\ F_{wg}$
$= (G_{ND}\ C_N \sin\beta + G_d)\ \rho_g\ (1 - \cos \Sigma)/2$
$= (942 \times 1.0 \times \sin\ 37.85^o + 49) \times 0.3 \times (1 -\ \cos 90^o)/2$
= 94 W/m^2

$G_{t\theta} = G_{D\theta} + G_{d\theta} + G_R$
$= 0 + 49 + 94$
$= 143\ W/m^2$

7.1.5 Solar radiation with a cloudy sky

Total solar radiation on a horizontal surface under a cloudy sky, G_{tc}

$$G_{tHc} = CCF\ (G_{DH} + G_{dH})$$

The CCF is the cloud cover factor determined from

$$CCF = P + Q\ CC + R\ CC^2$$

where P. Q. R = coefficients
CC = cloud cover (a variable between 0 - 10)

7.2 Heat Gain through Fenestrations

7.2.1 Solar optical properties

Any optical material obey the following law under any given wave length:

$\tau + \alpha + \rho = 1$
τ, α, and ρ are function of λ

where τ = transmittance
α = absorptance
ρ = reflectance
λ = wave length (m)

Further,

$\lambda f = c$

where f = frequency (HZ)
c = light speed ($3x10^8$ m/s)

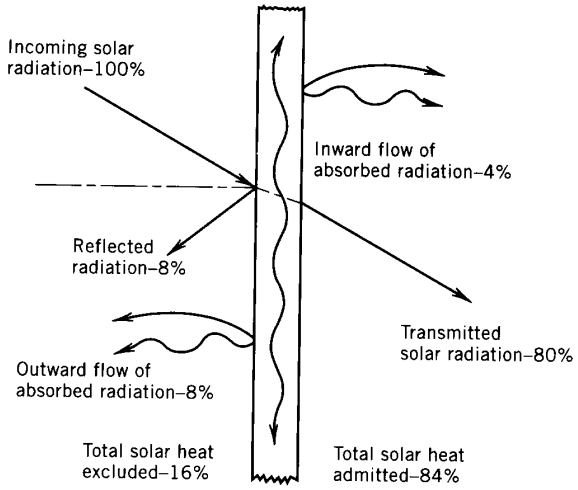


Figure 6-10 Distribution of solar radiation falling on clear plate glass.

Architectural glass:

Opaque to long-wave radiation
Transparent to short-wave radiation (sun light)
This is the well-known greenhouse effect.

7.2.2 Heat gain through fenestrations

Total heat gain = (Solar heat gain)
+ (Conduction heat gain)

Conduction heat gain = $U (T_o - T_i)$

Define:

Solar heat gain factor: SHGF
Transmitted solar heat gain factor: TSHGF
Absorbed solar heat gain factor: ASHGF

$$SHGF = TSHGF + ASHGF$$

- For double-strength sheet glass (DSA):

$$TSHGF = G_D \tau_D + G_d \tau_d$$

with

$$\tau_D = \sum_{j=0}^{5} t_j (\cos\theta)^j$$

and

$$\tau_d = 2\sum_{j=0}^{5} t_j /(j+2)$$

where θ = incident angle
t_j = values in T_Table 6-2 or A_Table 23

$$ASHGF = G_D \alpha_D + G_d \alpha_d$$

with

$$\alpha_D = \sum_{j=0}^{5} a_j (\cos\theta)^j$$

and

$$\alpha_d = 2\sum_{j=0}^{5} a_j /(j+2)$$

where a_j = values in T_Table 6-2 or A_Table 23.

- For non-DSA materials:

Transmitted heat gain, TSHG:

TSHG = (SC) (TSHGF)

and absorbed heat gain, ASHG:

ASHG = (SC) (ASHGF) (N_i)

where SC = shading coefficients in T_Tables 6-3 to 6-6 or A_Tables 24 to 27.
N_i = inward flowing fraction of absorbed heat gain.

$N_i = h_i / (h_i + h_o)$

Example 7-5

Continue from Example 7-4. Assume the window uses a 2.4 mm single-glass with light venetian blinds. Find the solar heat gain under a clear sky with an outside wind speed of 3.35 m/s and an inside convective heat transfer coefficient of 4.0 W/m^2 K.

Solution

From T_Table 6-4 or A_Table, SC = 0.67

$$\tau_d = 2\sum_{j=0}^{5} t_j /(j+2) = 0.8$$

$$\alpha_d = 2\sum_{j=0}^{5} a_j /(j+2) = 0.54$$

$N_i = h_i / (h_i + h_o) = 4.0/(4.0 + 23.0) = 0.148$

Transmitted solar heat gain:

$TSHGF = G_D \tau_D + G_d \tau_d$
$= 0 + (G_{d\theta} + G_{R\theta}) \tau_d$
$= 0 + (49 + 94) \times 0.8$
$= 114$ W/m^2

TSHG = (SC) (TSHGF)
= 0.67 x 114
= 76 W/m^2

Absorbed solar heat gain:

$ASHGF = G_D \alpha_D + G_d \alpha_d$
$= 0 + (G_{d\theta} + G_{R\theta}) \alpha_d$

$= 0 + (49 + 94) \times 0.54$
$= 77\ W/m^2$

$ASHG = (SC)\ (ASHGF)\ (N_i)$
$= 0.67 \times 77 \times 0.148$
$= 8\ W/m^2$

Total solar heat gain:

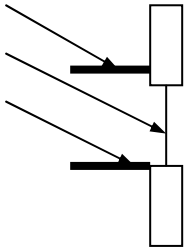
$SHG = TSHG + ASHG$
$= 77 + 8$
$= 85\ W/m^2$

7.2.3 High Performance Glazing

- Multiple glazing

Venetian blinds are treated as a special type of "glazing". Multiple transmission, reflection, and absorption are taken into account.

- External and internal shading



By draperies:

Use T_Table 6-6 or A_Table 29

By external overhangs and side fins:

Calculate the angles and further the shading areas.

- Air curtain windows

Room air is exhausted through window gap.
The air stream removes window load.

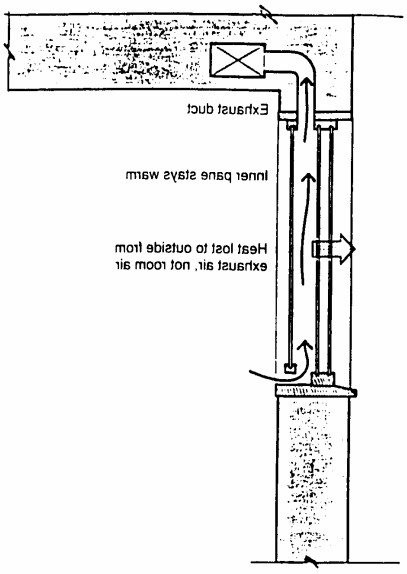


FIGURE 6.24
Air curtain window. (From
Nissen and Dart, 1985.)

Chapter 8 Cooling and Heating Load Calculations

8.1 Residential building cooling load
8.2 Residential building heating load
8.3 Nonresidential cooling and heating load (manual methods)
8.4 Nonresidential cooling and heating load (computer methods)

Readings: Chapters 7 and 8 of the textbook,
ASHRAE 2001 Fundamentals Handbook, Chapters 28 and 29

Residential features:

- 24-hour conditioned
- small internal loads
- single zone
- small capacity
- dehumidification for cooling only
- thermostats control

8.1 Residential building cooling load

Heat gain: The rate at which energy is transferred to or generated within a space
Cooling load: The rate at which energy must be removed from a space to maintain the temperature and humidity at the design values

Heat gain ≠ Cooling load (radiation, thermal mass, thermal lag)

Heat extraction: The rate at which energy is removed from the space by the cooling and dehumidification equipment.

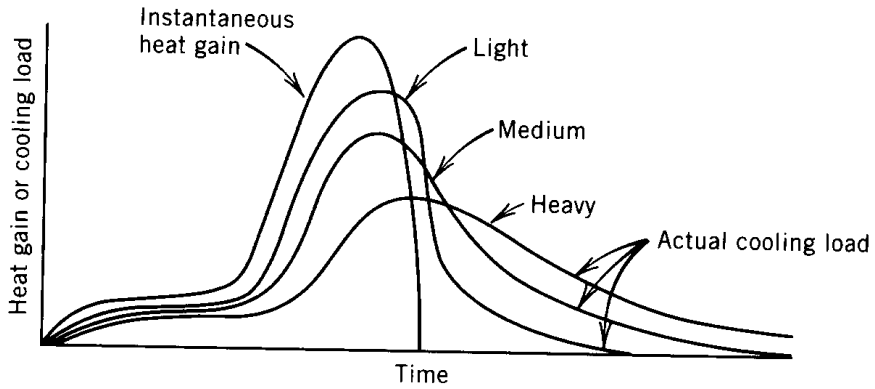


Figure 8-2 Actual cooling load and solar heat gain for light, medium, and heavy construction.

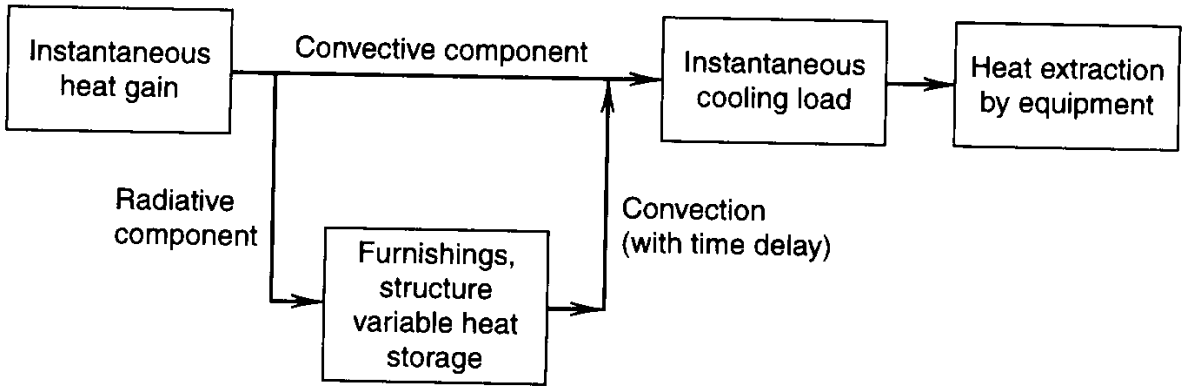


Figure 8-1 Schematic relation of heat gain to cooling load.

Cooling loads:

- through structural components
- through windows
- by infiltration
- due to occupants and appliances

CLTD/GLF Method

The method uses regression data of computer generated transfer function solution.

Calculation procedure:

- Cooling load through structural components:

  Cooling Load Temperature Difference (CLTD) combines the temperature difference between indoor and outdoor and solar radiation and considers thermal capacity of the enclosure.

  Q = U A (CLTD)

  U = overall heat transfer coefficient
  A = area of roof, wall, or glass
  CLTD = cooling load temperature difference; tabulated for flat roofs, walls, glasses (Table A27-1 and A27-22)

- Cooling load through windows:

  Glass Load Factor (GLF) includes effects of both transmission and solar radiation.

  Q = (GLF) A

GLF = glass load factor (Table A27-3 and A27-4)

- Cooling load due to infiltration:

  $Q = (ACH)\ V_{room}\ (T_o - T_i) \times 1200/3600$

- Cooling load due to occupants and appliances:

  People:
  Q = 70 W/person
  Appliances:
  Q = 470 W for both kitchen and laundry for single family
  Q = 350 W for multi-family

For latent cooling load:

- Calculate from individual components or estimate as 30% of the sensible load

8.2 Residential heating load

Heating loads:

- through structural components and windows
- through floors and walls below grade
- by infiltration

Calculation procedure:

- Heating load through structural components and windows

  $Q = U\ A\ (T_o - T_i)$

- Heating loads through floors and walls below grade

  $Q = U\ A\ (T_{earth} - T_i)$

- Heating load by infiltration

  $Q = \rho\ V_{infiltration}\ (T_o - T_i)$

8.3 Nonresidential building cooling and heating load (manual methods)

In nonresidential buildings, it is important to include
(1) the time lag in conductive heat gain through opaque exteriors, and
(2) the thermal storage in converting radiant heat gain to cooling load.

Calculation of a building cooling loads requires detailed information on:
- building characteristics (materials, size, and shape)
- configuration (location, orientation and shading)
- outdoor design conditions
- indoor design conditions
- operating schedules (lighting, occupancy, and equipment)
- date and time
- additional considerations (type of air-conditioning system, fan energy, fan location, duct heat loss and gain, duct leakage, type and position of air return system,…)

8.3.1 Radiant Time Series (RTS) Procedure (spreadsheet method)

ASHRAE Fundamentals 2001 in Chapter 29.27-29.37.

The RTS method is suitable for peak design load calculations, but it should not be used for annual energy simulations due to its simplified assumptions such as steady-periodic conditions.

Conduction time factor (CTS) and radiant time factor (RTS) takes into account (1) and (2).

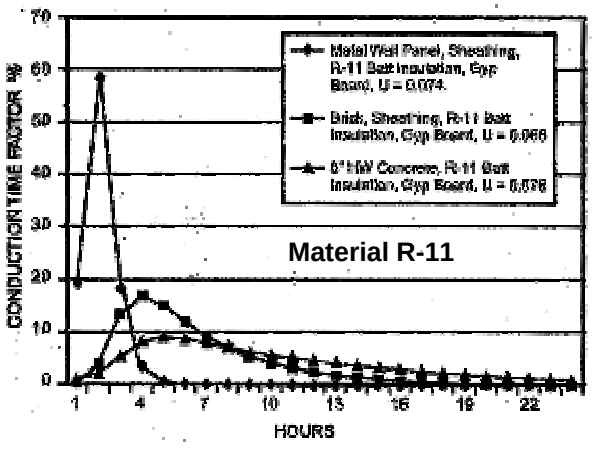


Fig. 9 CTS for Light to Heavy Walls

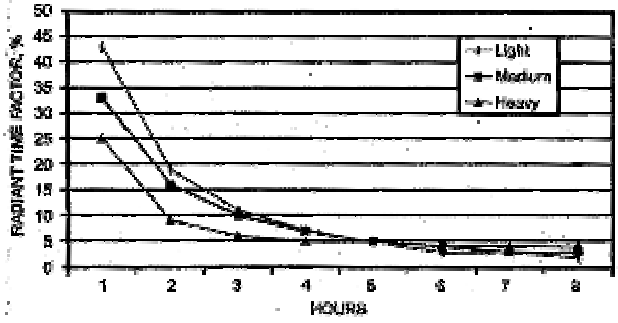


Fig. 11 RTS for Light to Heavy Construction

The RTS calculation procedure as presented on the following figure:
1. calculate 24h profile of heat gain for design day; apply CTS
2. split heat gain into radiant and convective part
3. apply RTS
4. calculate cooling load as a sum of convective heat gain and delayed radiant part of heat gain

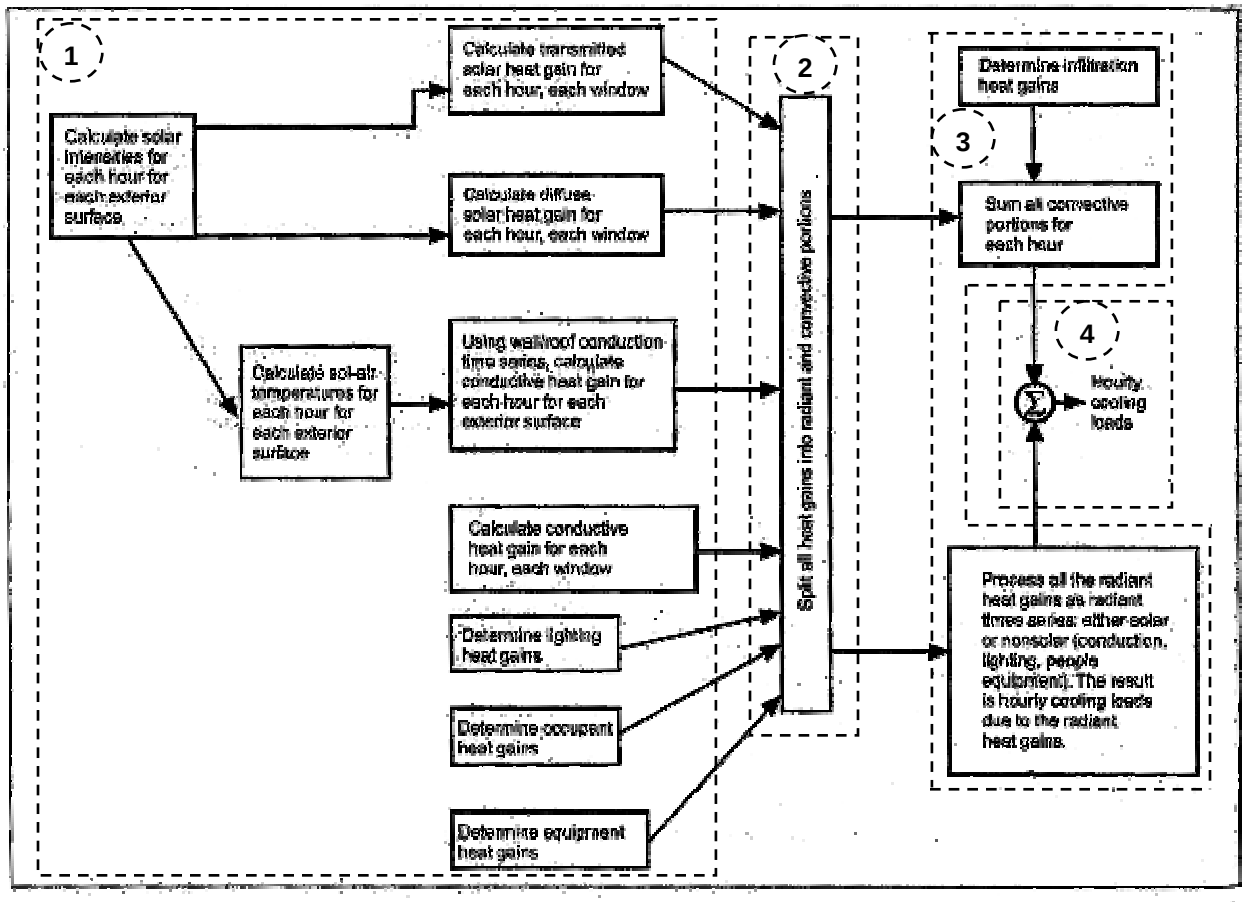


Fig. 8 , Overview of Radiant Time Series Method

The sol-air temperature represents outdoor design air temperature that combines convection to the outdoor air, radiation to the ground and sky, and solar radiation heat transfer effects on the outer surface of a building.

A practical method for cooling load calculations requires computer application. With computer application available simplified methods are not necessary because heat balance equations can be numerically solved in a few seconds on a PC.

The following manual method is suppressed, but not invalidated or suppressed.

8.3.2 CLTD/SCL/CLF Method (manual method)

ASHRAE Fundamentals 1997 in Chapter 28.39-28.56.

- External cooling load

  Roofs, walls, and conduction through glass:

  $Q = U\ A\ (CLTD)$

where U = design heat transfer coefficient (Table A24-4, A29-5)
A = area of roof, wall, or glass
CLTD = cooling load temperature difference (Table A28-32, A28-34)

Solar load through glass:

$Q = A\ (SC)\ (SCL)$

where SC = shading coefficient
SCL = solar cooling load factor (Table A28-36)

Partitions, ceilings, and floors:

$Q = U\ A\ (T_b - T_i)$

where U = design heat transfer coefficient for partition, ceiling, or floor (Table A24-4)
A = area of partition, ceiling, or floor
T_b = adjacent space temperature
T_i = inside air temperature

- Internal cooling load

  People:

  $Q_{sensible} = N$ (sensible heat gain) CLF
  $Q_{latent} = N$ (latent heat gain)

where N = number of people in space (Table A28-3)
CLF = cooling load factor (Table A28-37)

Lights:

$Q_{el} = W\ F_{ul}\ F_{sa}\ (CLF)$

where W = watts input from electrical plans or lighting fixture

F_{ul} = lighting use factor
F_{sa} = special allowance factor
CLF = cooling load factor (a schedule factor, CLF =1 for 24-hour light usage)
(Table A28-38)

Power:

$Q_p = P\ E_F\ CLF$

where P = power rating from electrical plans or manufacturer's data
E_F = efficiency factors and arrangements to suit circumstances
CLF = cooling load factor (a schedule factor) (Table A28-37)

Appliances:

$Q_{sensible} = Q_{is}\ F_{ua}\ F_{ra}\ (CLF)/F_{fl}$
$Q_{latent} = Q_{il}\ F_{ua}$

where Q_{is}, Q_{il} = sensible and latent heat gain from appliances
F_{ua}, F_{ra}, F_{fl} = use factors, radiation factors, flue loss factors
CLF = cooling load factor (a schedule factor) (Tables A28-37, A28-39)
Ventilation and infiltration air:

$Q_{sensible} = \rho\ V_{flow}\ (T_o - T_i)$

$Q_{latent} = 2500\ \rho\ V_{flow}\ (W_o - W_i)$

$Q_{total} = \rho\ V_{flow}\ (H_o - H_{room})$

where ρ = air density
V_{flow} = ventilation or infiltration flow rate (Thermal comfort & IAQ requirements)
T_o, T_i = outside, inside air temperature
W_o, W_i = outside, inside air humidity ratio
H_o, H_i = outside, inside air enthalpy

Example 8-1

A PSU classroom is 6 m long, 6 m wide and 3 m high. There is a 2.5 m x 4 m window in the east wall. Only the east wall/window is exterior. Assume the thermal conditions in adjacent spaces (west, south, north, above and below) are the same as those of the classroom. Determine the cooling load at 9:00 am, 12:00 noon on July 21.

Other known conditions include:
Latitude = 40° N
Ground reflectance = 0.2
Clear sky with a clearness number = 1.0
Overall window heat transmission coefficient = 7.0 W/m^2K
Room dry-bulb temperature = 25.5 ^{o}C
Permissible temperature exceeded = 2.5%
Schedule of occupancy: 20 people enter at 8:00 am and stay for 8 hours
Lighting schedule: 300 W on at 8:00 am for 8 hours
Exterior wall structure:
Outside surface, A0
Face brick (100 mm), A2
Insulation (50 mm), B3
Concrete block (100 mm), C3
Inside surface, E0
Exterior window:
Single glazing, 3 mm
No exterior shading, SC = 1.0

Solution:

(1) Exterior wall:

Layer	Unit resistance (m^2 K/W)
A0	0.059
A2	0.076
B3	1.173
C3	0.125
E0	0.121
Total	1.554

U =1/R = 0.643 W/m^2 K

From Table A28-33A, find wall type 13.

From Table A28-32, $CLTD_{9:00}$ = 9 and $CLTD_{12:00}$ = 14 for east wall.

$CLTD_{corrected} = CLTD + (25.5 - T_i) + (T_m - 29.4)$

T_i = inside temperature
T_m = mean outdoor temperature
T_m = (maximum outdoor temperature) - (daily range)/2

at 9:00 am $CLTD_{cor}$ = 9 + (25.5 - 25.5) + (31 -9/2 - 29.4) = 6.1 K
at 12:00 am $CLTD_{cor}$ = 14 + (25.5 - 25.5) + (31 -9/2 - 29.4) = 11.1 K

Q = U A (CLTD)
= 0.643 (W/m^2 K) x (6 x 3 - 4 x 2.5) m^2 x 6.1 = 34 W (at 9 am)
Q = 0.643 (W/m^2 K) x (6 x 3 - 4 x 2.5) m^2 x 11.1 = 62 W (at 12 noon)

(2) Window conduction:

From Table A28-34, $CLTD_{9:00}$ = 1 and $CLTD_{12:00}$ = 5.

$CLTD_{cor}$ = CLTD + (25.5 - T_i) + (T_m - 29.4)
= 1 + 0 + (31 - 9/2 - 29.4) = -1.9 K (at 9 am)
$CLTD_{cor}$ = 5 + 0 + (31 - 9/2 - 29.4) = 2.1 K (at 12 noon)

Q = U A (CLTD)
= 7.0 W/m^2 K x 4 x 2.5 m^2 x (-1.9 K) = -133 W (at 9 am)
Q = 7.0 W/m^2 K x 4 x 2.5 m^2 x 2.1 K = 147 W (at 12 noon)

(3) Solar load through glass:

From Table A28-35B, find zone type is A
From Table A28-36, find CLF = 576 at 9 am and CLF = 211 at 12 noon

Q = A (SC) (SCL)
= (2.5 x 4 m^2) x 1.0 x 576 = 5760 W (at 9 am)
Q = (2.5 x 4 m^2) x 1.0 x 211 =2110 W (at 12 noon)

(4) Cooling load from partitions, ceiling, floors:

Q = 0

(5) People:

From Table A28-35B, find zone type is B
From Table A28-3, find sensible/latent heat gain = 70 W.
From Table A28-37, find $CLF_{9:00}$ (1) = 0.65 and $CLF_{12:00}$ (3) =0.85

$Q_{sensible}$ = N (sensible heat gain) CLF
= 20 x 70 W x 0.65 = 910 W (at 9 am)
$Q_{sensible}$ = 20 x 70 x 0.85 = 1190 W (at 12 noon)

Q_{latent} = N (latent heat gain)
= 20 x 45 = 900 W (at 9 am and 12 noon)

(6) Lighting:

From Table A28-35B, find zone type is B
From Table A28-38, find $CLF_{9:00}$ (1) = 0.75 and $CLF_{12:00}$ (3) = 0.93

Q_{el} = W F_{ul} F_{sa} (CLF)
= 300 W x 1 x 1 x 0.75 = 225 W (at 9 am)
Q_{el} = 300 x 1 x 1 x 0.93 = 279 W (at 12 noon)

(7) Appliances:

$Q_{sensible}$ = 0
Q_{latent} = 0

(8) Infiltration:

Since room air pressure is positive, we have:

$Q_{sensible} \tilde{=} 0$
Q_{latent} = 0

Total cooling load:

Component	9:00 am	12:00 noon
Wall	34	62
Window conduction	-133	147
Window solar transmission	5,760	2,110
Partitions	0	0
People	910	1,190
Lights	225	279
Appliance	0	0
Infiltration	0	0
Total	6,796 W	3,788 W

8.3.2 TETD/TA (manual method)

TETD is called the total equivalent temperature differential method. It is similar to CLTD method but not the same. More information can be found in ASHRAE Handbook - Fundamentals (ASHRAE Fundamentals 28.56 - 28.64).

Practicum Assignment #1

1. For a winter heating in a PSU classroom, steam enters a radiator at 16 psia and 0.97 quality. The steam flows through the radiator, is condensed, and leaves as liquid water at 200°F. If the heating capacity of the radiator is 5000 Btu/hr, at what rate in lbm/hr must steam be supplied?

2. A solar collector panel, shown in Figure 1, has a surface area of 32 ft^2. The panel receives energy from the sun at a rate of 150 Btu/(hr ft^2-of collector surface). Forty percent of the incoming energy is lost to the surrounding. The reminder is used to warm liquid water from 130°F to 160°F. The water passes through the solar collector with a negligible pressure drop.

(a) (15 points) Neglecting kinetic and potential energy effects, determine at steady state the mass flow rate of water in [lb/min]. Hint: Write the assumptions.

(b) (10 points) How many solar collectors would be needed to provide a total of 40 gal of 160°F water in 30 min? Hint: To obtain water properties assume the atmospheric pressure.

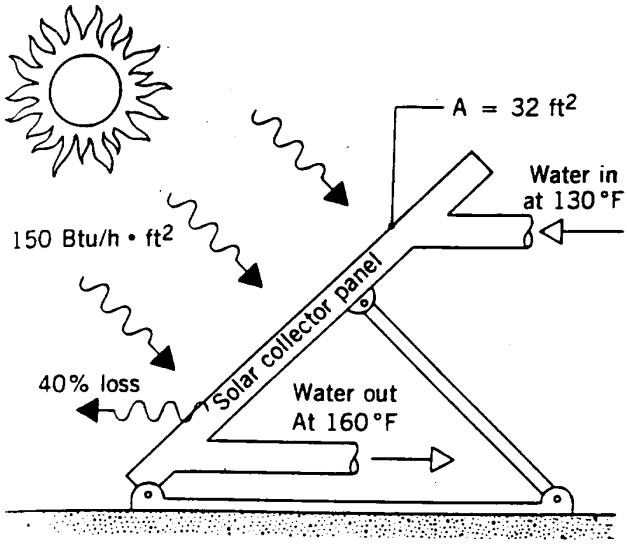


Figure 1. The solar collector for problem 3

3. Steam at 7,000 Pa and 50°C enters a condenser operating at steady state and is condensed to saturated liquid at 7,000 Pa on the inside of tubes through which cooling water flows. The mass flow rate of steam is 25 kg/s. In passing through the tubes, the cooling water increases in temperature by 10°C and experiences no pressure drop. Neglecting kinetic and potential energy effects and ignoring heat transfer from the outside of the condenser, determine:

(a) the mass flow arte of cooling water, in kg/s

(b) the rate of energy transfer, in kW, from the condenser to the cooling water.

Solutions for the Practicum Assignment #1

1. For a winter heating in a PSU classroom, steam enters a radiator at 16 psia and 0.97 quality. The steam flows through the radiator, is condensed, and leaves as liquid water at 200°F. If the heating capacity of the radiator is 5000 Btu/hr, at what rate in lbm/hr must steam be supplied?

$P_1 = 16$ psia
$x_1 = 0.97$
$T_2 = 200^oF$
$x_2 = 0$

Use these values to find h_1 and h_2 with Table A-1a.

$$\dot{m} = \frac{\dot{Q}}{h_1 - h_2} = \frac{5000 Btu/hr}{1123 Btu/lbm - 168.1 Btu/lbm} = 5.236 lbm/hr$$

2. A solar collector panel, shown in Figure 1, has a surface area of 32 ft^2. The panel receives energy from the sun at a rate of 150 Btu/(hr ft^2-of collector surface). Forty percent of the incoming energy is lost to the surrounding. The reminder is used to warm liquid water from 130°F to 160°F. The water passes through the solar collector with a negligible pressure drop.
   (a) Neglecting kinetic and potential energy effects, determine at steady state the mass flow rate of water in [lb/min]. Hint: Write the assumptions.
   (b) How many solar collectors would be needed to provide a total of 40 gal of 160°F water in 30 min? Hint: To obtain water property use thermophysical properties table in your text assuming the atmospheric pressure.

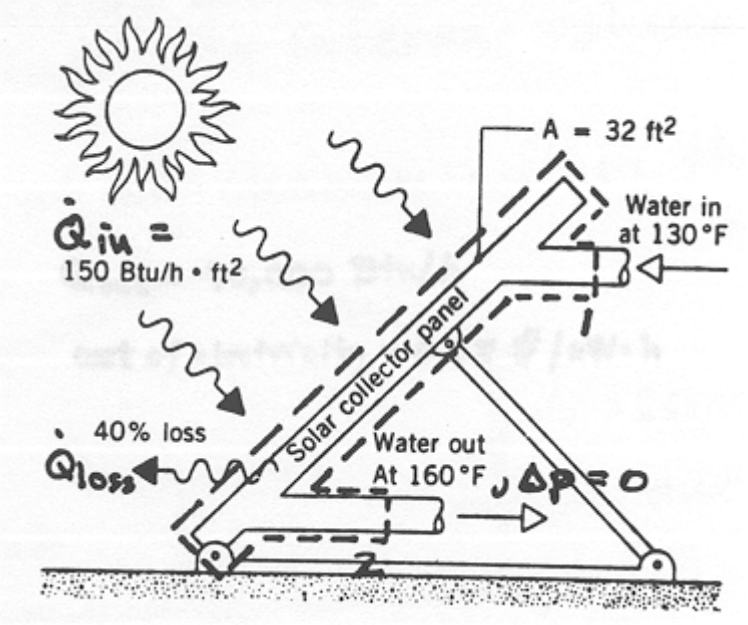


Figure 1.

(a) Assumptions:

1. The control volume is at steady state
2. For the control volume shown, $W_{cv} = 0$
3. Kinetic and potential energy effects are negligible
4. The water is modeled as an incompressible liquid, with constant specific heat $c = 1\ \frac{Btu}{lb \cdot R}$.

Analysis:

The mass flow rate is determined using the steady-state energy balance as follow:

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}[(i_1 - i_2) + (\frac{V_1^2 - V_2^2}{2}) + g\,(Z_1 - Z_2)]$$

Where $\dot{m}_1 = \dot{m}_2 = \dot{m}$ with $\dot{Q}_{cv} = \dot{Q}_{in} = \dot{Q}_{loss}$ and assumption (3)

$$0 = \dot{Q}_{in} - Q_{loss} + \dot{m}(h_1 - h_2)$$

For water as an incompressible liquid, $i_1 - i_2 = c(T_1 - T_2) + v(p_1 - p_2)$

Thus

$$\dot{m} = \frac{\dot{Q}_{in} - \dot{Q}_{loss}}{c\,(T_2 - T_1)}$$

From the given data, $\dot{Q}_{in} = (150\frac{Btu}{hr \cdot ft^2})(32\ ft^3) = 4800\frac{Btu}{hr}$ and $\dot{Q}_{loss} = (0.4)\dot{Q}_{in} = 1920\frac{Btu}{hr}$

Inserting values:

$$\dot{m} = \frac{(4800-1920)\frac{Btu}{h}}{1\frac{Btu}{lb\cdot R}\times(620-590)R}(\frac{1h}{60\,\text{min}}) = 1.6\frac{lb}{\text{min}}$$

(b) The mass of water in 40 gallon is with density $\rho = 60.01\frac{lb}{ft^3}$

$$m_{tot} = \rho V = (60.98\frac{lb}{ft^3})(40gal)(\frac{0.13368\,ft^3}{1gal}) = 326\ lb\ of\ water$$

Each collector provides

$$m = \int_1^t \dot{m}\,dt = (1.6\frac{lb}{\text{min}})(30\,\text{min}) = 48\ \frac{lb}{Collector}$$

Thus

$$\text{No of Collectors} = \frac{326}{48} \approx 7$$

Comments:
Forty gallons is about the capacity of a typical home water heater, the total collector area required is 224 ft^2.

3. Steam at 7,000 Pa and 50°C enters a condenser operating at steady state and is condensed to saturated liquid at 7,000 Pa on the inside of tubes through which cooling water flows. The mass flow rate of steam is 25 kg/s. In passing through the tubes, the cooling water increases in temperature by 10°C and experiences no pressure drop. Neglecting kinetic and potential energy effects and ignoring heat transfer from the outside of the condenser, determine:

(a) the mass flow arte of cooling water, in kg/s
(b) the rate of energy transfer, in kW, from the condenser to the cooling water.

Given: Steam and water pass through separate streams through a condenser.

Assuming: 1.) Control Volume is a steady state.
2.) Heat transfer from outside is negligible.
3.) Potential and Kinetic Energy effects are negligible.
4.) Cooling water is idealized as an incompressible fluid w/ constant specific heat.

Find: a.) Mass flow rate of cooling water, $\dot{m}_{cw}$
b.) Heat transfer rate to the cooling water. $\dot{Q}_{cw}$

Solution:

a.) By conservation of mass: $\dot{m}_1=\dot{m}_2=\dot{m}_{steam}$ and $\dot{m}_a=\dot{m}_b=\dot{m}_{cw}$

-Using steady state energy balance:

$0=Q(dot)_{cv}-W(dot)_{cv}+m(dot)_{st}[(h_1-h_2)+((v_1-v_2)/2)+g(z_1-z_2)]+m(dot)_{cw}[(h_a-h_b)+((v_a-v_b)/2)+g(z_a-z_b)]$

$0=m(dot)_{st}\ (h_1-h_2)+m(dot)_{cw}(h_a-h_b)$

$m(dot)_{cw}=[m(dot)_{st}\ (h_1-h_2)]/Cp\ (T_a-T_b)]$

$m(dot)_{cw}=[(25kg/s)(2571.5kJ/kg-163.0kJ/kg)]/[(4.18\ kJ/(kg*K)(10K)] =$ **1440.5 kg/sec**

b.) Using an energy balance of the cooling water:

$0=Q(dot)_{cv}-W(dot)_{cv}+ m(dot)_{cw}\ [(h_a-h_b)+((v_a-v_b)/2)+g(z_a-z_b)]$

$Q(dot)_{cw}=m(dot)_{cw}(h-h)=m(dot)Cp(T_a-T_b)$

$Q(dot)_{cw}=(1440.5\ kg/sec)(4.18\ kJ/(kg*K)(10K) = 60215.5\ W =$ **60.2kW**

Practicum Assignment #2

1. Running water is heated from 50°F to 120°F by mixing it with saturated steam. Steam gauge pressure may vary from 15 to 25 psi.
   (a) Sketch the system.
   (b) For a heated water flow rate of 14 cfm, calculate and plot (create graph of) steam mass flow rate in lb/s. Assume sea level atmospheric pressure for water flow.

2. By supplying 90,000 Btu/h, a heat pump maintains the temperature of dwelling at 70°F when the outside air is at 32°F.
   (a) Sketch the system.
   (b) What is the minimum work required for this cycle? Define the assumptions.

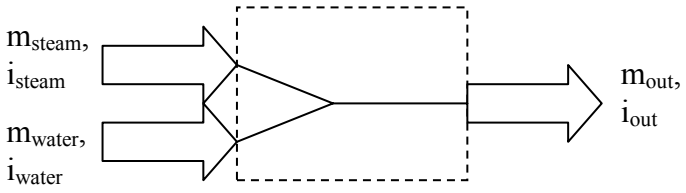
3. A heat pump driven by a 0.4 kW electric motor provides heating for a building on a day when the outside is at -10°C and energy is lost through the walls and roof at rate of 16200 kJ/h. What is the maximum theoretical temperature that can be maintained within building, in ^{o}F?

4. Infiltration of outside air into building through miscellaneous cracks can represent a significant load on the heating or cooling equipment. A particular office building has a total crack length of 440 ft around its doors and windows. On windy day, about 0.4 cfm of air enters per foot of crack. In addition, door openings account for about 100 cfm of outside air infiltration on average. The internal volume of the building is 20,000 ft^3. Assuming ideal gas behavior, estimate n the number of times per hour (ACH – air change rate per hour), at steady state, that the air within building is changed due to infiltration. Hint: n[ACH] = Volume flow rate/Volume.

Solutions for the Practicum Assignment #2

1. Running water is heated from 50°F to 120°F by mixing it with saturated steam. Steam gauge pressure may vary from 15 to 25 psi.
   (a) Sketch the system.
   (b) For a heated water flow rate of 14 cfm, calculate and plot (create graph of) steam mass flow rate in lb/s. Assume sea level atmospheric pressure for water flow.

(a)



(b)

Comparing steam mass flow to steam pressure:

First, lets write a governing energy conservation equation for out process. $E_{in}=E_{out}$

$$\dot{m}_{w1} * i_{w1} + \dot{m}_{steam} * i_{steam} = \dot{m}_{w2} * i_{w2}$$

Note that w1 refers to the cool water and w2 refers to the heated water. Now lets rearrange to create an expression for $m(dot)_{steam.}$

$$\dot{m}_{steam} = \frac{\dot{m}_{w2} * i_{w2} - \dot{m}_{w1} * i_{w1}}{i_{steam}}$$

We are given a volumetric flow rate fom the heated water of 14 cfm. $V(dot)\rho=m(dot)$

So we can calculate m(dot) of the heated water as:

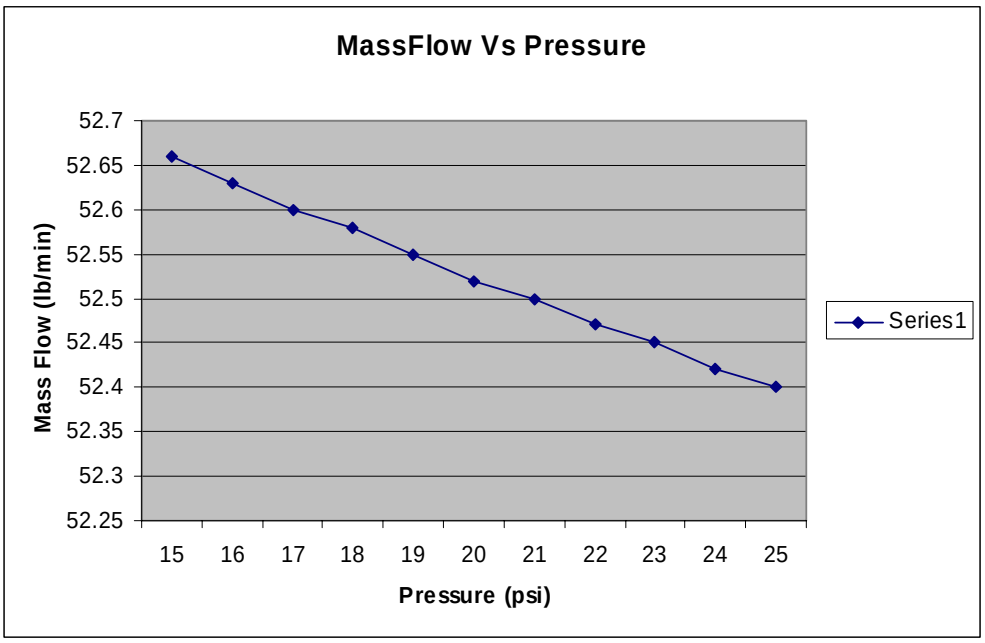
$(14cfm)(60.01\ lb/ft^3)*(1min/60sec)= 14lbs/sec$

Enthalpy (i_{w2}) of heated water at 120 degrees = 88 btu/lb

As for the cool water, the mass flow rate is constant so it will not affect the shape of our graph. The enthalpy for the 50 degree water is 18.1 btu/lb.
Use table A-1a for the above values.

Ok here's the trick, you now go to the steam tables and look up enthalpies for 15psi to 25psi. At each enthalpy you run the above equation and calculate the $m(dot)_{steam}$. This is how you graph steam pressure Vs steam mass flow rate. Actually you could of came up with a graph based off of our initial equation with out any numbers at all, IF you realize the direct relationship between pressure and enthalpy as well as the indirect relationship between enthalpy and mass flow rate.

Here is the Excel Calculation:



Pressure (psi)	Mass Flow Rate Lb/Min
15	52.66
16	52.63
17	52.6
18	52.58
19	52.55
20	52.52
21	52.5
22	52.47
23	52.45
24	52.42
25	52.4

2. By supplying 90,000 Btu/h, a heat pump maintains the temperature of dwelling at 70°F when the outside air is at 32°F.

(a) Sketch the system.

(b) What is the minimum work required for this cycle? Define the assumptions.

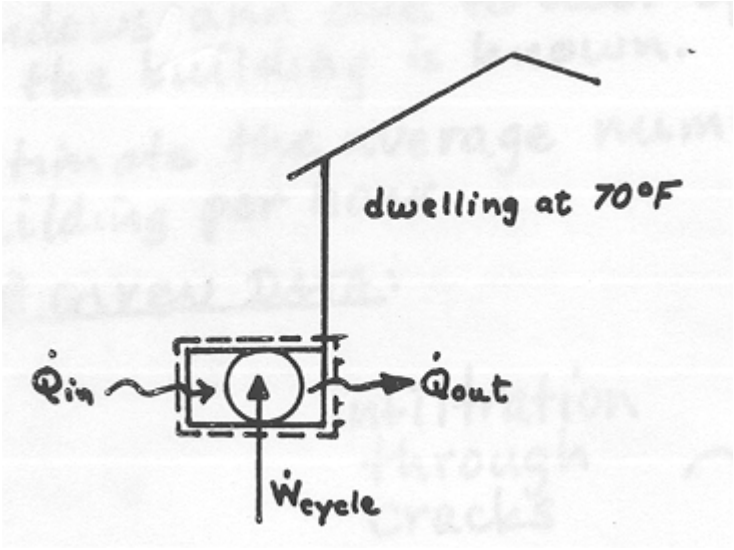


Figure 1.

(a) Assumptions:

1. The system shown on the accompanying figure undergoes a heat pump cycle.
2. The data are for operation at steady state.
3. The dwelling and the surroundings play the roles of hot and cold reservoirs, respectively.

(b) Analysis:

The minimum theoretical cost for any heat pump cycle under the stated conditions is the cost for a reversible cycle operating between reservoirs at T_H=530R (70 °F) and T_C=492R (32°F). The power required by such a cycle can be obtained from:

$$\dot{W}_{cycle} = \frac{\dot{Q}_{out}}{COP_{carnot}}$$

$$COP_{carnot} = \frac{T_H}{T_H - T_L} = \frac{530}{530 - 492} = 13.95$$

$$\dot{W}_{cycle} = \frac{\dot{Q}_{out}}{COP_{carnot}} = \frac{90{,}000 \frac{Btu}{hr}}{13.95} = 6451.6 \frac{Btu}{hr}$$

3. A heat pump driven by a 0.4 kW electric motor provides heating for a building on a day when the outside is at –10°C and energy is lost through the walls and roof at rate of 16200 kJ/h. What is the maximum theoretical temperature that can be maintained within building, in °F?

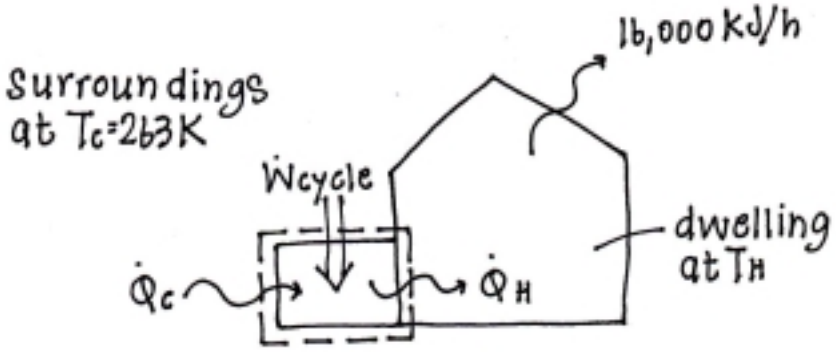


Figure 2.

(a) Assumptions:

1. The system shown in the accompanying figure undergoes a heat pump cycle.
2. The data are for operation at steady state.
3. The dwelling and the surroundings play the roles of hot and cold reservoirs, respectively.

(b) Analysis:

At steady state, the heat pump cycle must provide energy to the dwelling equal to the energy leaking through the walls and roof.

$$\dot{Q}_H = 16{,}200 kJ/h$$

From Section 5.4.2, we know that the coefficient of performance of the heat pump must be less than or equal to the coefficient of performance of a reversible heat pump operating between reservoirs at T_C=263K (-10°C) and T_H. Then with equation 5.10:

$$\frac{\dot{Q}_H}{\dot{W}_{cycle}} \le \frac{T_H}{T_H - T_C}$$

Inserting known values:

$$\frac{[16{,}000 kJ/hr] \times \left[\frac{1hr}{3600s}\right]}{[0.4kW] \times \left[\frac{1kJ/s}{1kW}\right]} \le \frac{T_H}{T_H - 263}$$

$$11.11 \le \frac{1}{1 - 263/T_H}$$

$$T_H \le 289K = 16°C = 61°F$$

4. Infiltration of outside air into building through miscellaneous cracks can represent a significant load on the heating or cooling equipment. A particular office building has a total crack length of 440 ft around its doors and windows. On windy day, about 0.4 cfm of air enters per foot of crack. In addition, door openings account for about 100 cfm of outside air infiltration on average. The internal volume of the building is 20,000 ft^3. Assuming ideal gas behavior, estimate n the number of times per hour (ACH – air change rate per hour), at steady state, that the air within building is changed due to infiltration. Hint: n[ACH] = Volume flow rate/Volume.

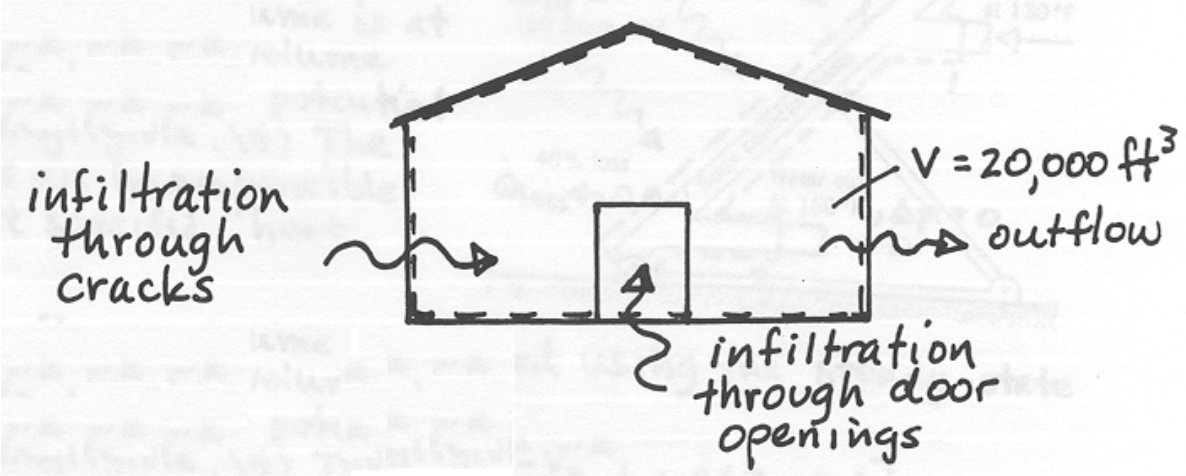


Figure 3.

(a) Assumptions:

1. The control volume shown is at steady state.
2. The air behaves as an ideal gas.
3. The densities of the incoming air and of the air in the building are nearly equal.

(b) Analysis:

At the steady state, the mass balance reduces to:

$$\dot{m}_{outflow} = \dot{m}_{cracks} + \dot{m}_{door\ openings}$$

$$\rho_i (AV)_{outflow} = \rho_o [(AV)_{cracks} + (AV)_{door\ openings}]$$

Where ρ_i and ρ_o are the inside and outside air densities, respectively. Assuming ideal gas behavior:

$$\frac{\rho_i}{\rho_o} = \frac{P_i / RT_i}{P_o / RT_o}$$

If $P_i = P_o$ and $T_i = T_o$, $\rho_i = \rho_o$, Thus

$$(AV)_{outflow} = (AV)_{cracks} + (AV)_{door\ openings}$$

$$= (0.4 \frac{ft^3}{\min \cdot\ ft})(440\ ft) + 100 \frac{ft^3}{\min}$$

$$= 276 \frac{ft^3}{\min} = 16{,}560 \frac{ft^3}{hr}$$

$$\frac{AirChanges}{hr} = \frac{AV_{outflow}}{V} = \frac{16{,}5600 \frac{ft^3}{hr}}{20{,}000 \frac{ft^3}{air\ change}} = 0.828 \frac{air\ changes}{hr}$$

Practicum Assignment #3

Do not use the psychrometric chart.

1. Calculate values of humidity ratio, enthalpy, and specific volume for saturated air at one standard atmosphere using perfect gas relations for temperature of 20 ^{o}C and 0 ^{o}C.

2. The temperature of a certain room is 22 ^{o}C and the relative humidity is 50%. The barometric pressure is 100 kPa. Find (a) the partial pressures of the air and water vapor, (b) the vapor density, and humidity ratio of the mixture.

3. Compute the enthalpy of moist air at 16 ^{o}C and 80% relative humidity for an elevation of (a) sea level and (b) 1500 m.

4. The condition within a room is 20 ^{o}C (dry bulb), 50% relative humidity, and 101325 Pa pressure. The inside surface temperature of the window is 5 ^{o}C. Will moisture condense on the window glass? Explain why.

5. Moist air exists at a relative humidity of 60%, and a pressure of 96.5 kPa. The dewpoint temperature of this moist air is 18 ^{o}C. Determine (a) the humidity ratio and (b) the volume in m^3/kg.

6. A duct has moist air flowing at a rate of 2 m^3/s. What is the mass flow rate of the dry air, where the dry bulb temperature is 16 ^{o}C, the relative humidity is 80% and where the pressure inside the duct corresponds to (a) sea level, and (b) 2000 m?

Solutions for the Practicum Assignment #3

1. Calculate values of humidity ratio, enthalpy, and specific volume for saturated air at one standard atmosphere using perfect gas relations for temperature of 20 ^{o}C and 0 ^{o}C.

2. The temperature of a certain room is 22 ^{o}C and the relative humidity is 50%. The barometric pressure is 100 kPa. Find (a) the partial pressures of the air and water vapor, (b) the vapor density, and humidity ratio of the mixture.

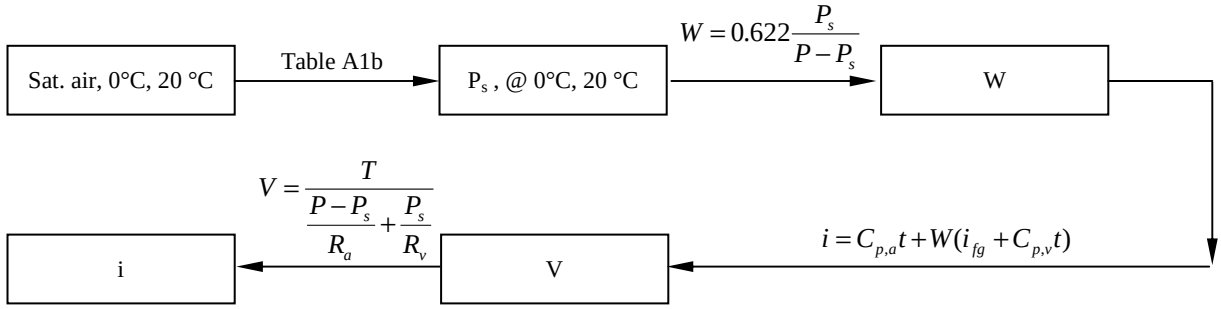
3. Compute the enthalpy of moist air at 16 ^{o}C and 80% relative humidity for an elevation of (a) sea level and (b) 1500 m.

4. The condition within a room is 20 ^{o}C (dry bulb), 50% relative humidity, and 101325 Pa pressure. The inside surface temperature of the window is 5 ^{o}C. Will moisture condense on the window glass? Explain why.

5. Moist air exists at a relative humidity of 60%, and a pressure of 96.5 kPa. The dewpoint temperature of this moist air is 18 ^{o}C. Determine (a) the humidity ratio and (b) the volume in m^3/kg.

6. A duct has moist air flowing at a rate of 2 m^3/s. What is the mass flow rate of the dry air, where the dry bulb temperature is 16 ^{o}C, the relative humidity is 80% and where the pressure inside the duct corresponds to (a) sea level, and (b) 2000 m?

1. Calculate values of humidity ratio, enthalpy, and specific volume for saturated air at one standard atmosphere using perfect gas relations for temperature of 20 °C and 0 °C.



Constants:

P = 101325 Pa, $C_{p,a}$ = 1.01 kJ/kg·K, $i_{f,g}$ = 2501.3 kJ/kg·K, $C_{p,v}$ = 1.86 kJ/kg·K

R_a = 287 J/ kg·K, R_v = 462 J/ kg·K

	P_v=P_s (Pa)	W (kgw/kga)	i (kJ/kg)	V (m^3/kg)
0°	611.3	3.78×10^{-3}	9.45	0.775
20°	2339	1.47×10^{-2}	57.5	0.838

Comments:

Check your answer in the psychrometric chart.

2. The temperature of a certain room is 22 °C and the relative humidity is 50%. The barometric pressure is 100 kPa. Find (a) the partial pressures of the air and water vapor, (b) the vapor density, and humidity ration of the mixture.

From table A-1b $P_{v,s} = 2672\ Pa$

$$\phi = \frac{P_v}{P_{v,s}}, \qquad P_v = \phi P_{v,s} = 1336\,Pa \quad P_a = P - P_v = 98{,}664\,Pa$$

$$\rho_v = \frac{P_v}{R_v T} = \frac{1336}{462 \cdot 295.15} = 9.80 \times 10^{-3}\ (kg/m^3)$$

$$W = 0.622 \frac{P_v}{P - P_v} = 0.622 \frac{1336}{1 \times 10^5 - 1336} = 8.42 \times 10^{-3}\ (kgw/kga)$$

3. Compute the enthalpy of moist air at 16 °C and 80% relative humidity for an elevation of (a) sea level and (b) 1500 m.

(a) At sea level: P = 101325 Pa, From Table A-1b,

$P_{v,s} = 1836\ Pa \quad P_v = \phi P_{v,s} = 1468.8\ Pa$

$$W = 0.622 \frac{P_v}{P - P_v} = 9.15 \times 10^{-3}\ (kgw/kga)$$

$$i = C_{p,a} t + W(2501.3 + 1.86t) = 39.3\ (kJ/kg)$$

(b) At 1500 m level: P = a+bH = 99.436 - 0.01(1500)=84.436 kPa,

$$W = 0.622 \frac{P_v}{P - P_v} = 0.622 \frac{1468.8}{84436 - 1468.8} = 0.0110\ (kgw/kga)$$

$$i = C_{p,a} t + W(2501.3 + 1.86t) = 44.03\ (kJ/kg)$$

4. The condition within a room is 20 °C (dry bulb), 50% relative humidity, and 101325 Pa pressure. The inside surface temperature of the window is 5 °C. Will moisture condense on the window glass? Explain why.

There are a couple of ways to solve this problem.

(1) Calculate the dewpoint temperature T_d,
if $T_d > T_{surface}$, condensation occurs.

(2) Calculate humidity ratio W at 20°C, Φ = 50%, and W at 5°C, 100% RH
if $W|_{20°C,\ 50\%} > W|_{5°,\ 100\%}$, condensation occurs.

(3) Calculate partial pressure of vapor at 20°C, 50% RH and 5C, 100% RH
if $P_v|_{20°C,\ 50\%} > P_{v,s}|_{5°,\ 100\%}$, condensation occurs.

Here I just present (1):

$P_{v,s}|_{20°C}$ = 2339 Pa, $P_v = \phi P_{v,s} = 1169.5\ Pa$

From table A-1b, we can find $T_d \cong 9.25$ °C

$T_d > T_{surface}$, condensation occurs!

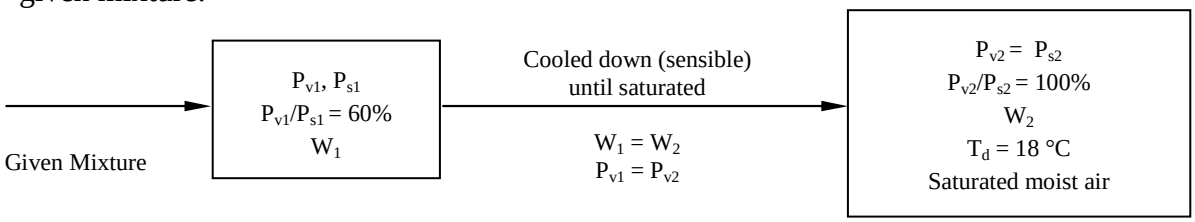
5. Moist air exists at a relative humidity of 60%, and a pressure of 96.5 kPa. The dewpoint temperature of this moist air is 18°C. Determine (a) the humidity ratio and (b) the volume in m^3/kg.

This is the most confusing problem in this problem set. Let's do it step by step.

1) Recall what you heard in weather forecast (if you watch TV sometime). The told you "the temperature in State College is 68°F, dewpoint temperature is 65°F, raining in the morning ....." They are talking about out HVAC concept: The current temperature of air does not necessarily equal to the dewpoint temperature of the air.

2) So, the temperature if the air, in other words, the dry-bulb temperature of the air must be some temperature other than 18°C since Φ = 60%. Let's imagine a psychart (you can use it actually, but at this moment, I just use formula), when you know "2" quantities, either T_{dry}, T_{wet}, or T_{dew}, Φ, you are able to determine a point on the psychart. Now, you have T_d and Φ, so be confident!

3) Recall the definition of the dewpoint temperature from our text: "Dewpoint temperature t_d is the temperature of saturated moist air at the same pressure and humidity ratio as the given mixture."



4) Now we are quite clear what we are going to do: since W and P_v do not change, From Table A-1b, At 18°C, P_s= 2064 Pa , $P_v|_t = P_s|_{18°C}$

$$W = 0.622\frac{P_s}{P - P_s} = 0.622\frac{2064}{96500 - 2064} = 0.0136(kgw/kga)$$

$$P_v = 2064\ Pa \qquad P_{v,s} = \frac{P_v}{\phi} = 3440\ Pa$$

From Table A-1b: T ≅ 26.37 °C=299.5 K

$$V = \frac{T}{\frac{P - P_s}{R_a} + \frac{P_s}{R_v}} = \frac{299.5}{\frac{96500 - 2064_s}{287} + \frac{2064}{462}} = 0.898(m^3/kg)$$

Comments:

Note the keys of this problem are:

- $T_d \neq T_{drybulb}$ when Φ ≠ 100%
- W and P_v do not change during our sensible cooling process
- Using chart or computer program is much easier

6. A duct has moist air flowing at a rate of 2 m^3/s. What is the mass flow rate of the dry air, where the dry bulb temperature is 16°C, the relative humidity is 80% and where the pressure inside the duct corresponds to (a) sea level, and (b) 2000 m?

From Table A-1b, for 16°C, $P_{v,s} = 1818.1\,Pa$ $P_v = \phi P_{v,s} = 1454.5\,Pa$

(a) Sea level: P=101325 Pa $P_a = P - P_v = 99.87 kPa$

$$\rho_a = \frac{P_a}{R_a T} = \frac{99870}{287 \cdot 289.15} = 1.203\ (kg/m^3)$$

$$\dot{m}_a = \rho_a \dot{V} = 2.407\ (kg/\sec)$$

(b) 2000 m elevation: P = a+bH = 99.436 - 0.01(2000)=79.436 kPa,

$$P_a = P - P_v = 79436 - 1454.5 = 77981.5\,Pa$$

$$\rho_a = \frac{P_a}{R_a T} = \frac{77981.5}{287 \cdot 289.15} = 0.94\ (kg/m^3)$$

$$\dot{m}_a = \rho_a \dot{V} = 1.879\ (kg/\sec)$$

Comments:

- Some useful relationships: $\dot{m}_a = \rho_a \dot{V} = \frac{\dot{V}}{v}$
- The mass flow rate changes with elevation but the volumetric flow rate does not.

Practicum Assignment #4

1. Air leaves the cooling coil of an air conditioning system has a relative humidity of 90% and a dewpoint temperature of 55°F at 5000 ft elevation. From the psychrometric chart find:
   (a) Dry bulb air temperature
   (b) Wet bulb temperature
   (c) Air density
   (d) Humidity ratio
   (e) Enthalpy
   Identify your results in the chart and submit the chart with the solutions.

2. To save energy, the environmental conditions in a room are to be regulated so that the dry bulb temperature will be greater than or equal to 78°F (24°C) and the dew point will be less than or equal to 64°F (17°C). Find the maximum relative humidity that can occur for standard barometric pressure.

3. It is desired to heat and humidify 2000 cfm of air from an initial state defined by a temperature of 60°F db and relative humidity of 30% to a final state of 110°F db and 30% relative humidity. The air will first be heated by a hot water coil followed with saturated vapor at 5psig. Assume sea level pressure Using the psychrometric chart:
   (a) Find the heat transfer rate for the heating coil
   (b) Find the mass flow rate of the water vapor
   (c) Sketch the processes on a psychrometric chart.

4. Moist air enters a cooling coil at 28°C dry-bulb temperature and 50% relative humidity and exits the coil at 13°C dry-bulb temperature and 90% relative humidity. The flow rate through the coil is 1.50 kg/s and the process occurs at the sea level pressure.
   (a) Determine the sensible heat factor (SHF) for the process
   (b) Determine the cooling coil capacity (heat transfer-rate)
   (c) Sketch the process in the psychrometric chart denoting sensible and latent heat

5. Saturated steam is sprayed into a stream of moist air. The initial condition of the air is 55°F dry-bulb temperature and 35°F dew-point temperature. The mass airflow rate is 2000 lbm/min. Barometric pressure is 14.696 psi. Determine:
   (a) How much steam must be added in lbm/min to produce a saturated air condition, and
   (b) The resulting dry-bulb and wet-bulb temperature of the saturated air

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1. **Air leaves the cooling coil of an air conditioning system and has a relative humidity of 90% and a dewpoint temperature of 55°F at 5000 ft elevation. From the psychrometric chart find:**
   (a) Dry bulb air temperature
   (b) Wet bulb temperature
   (c) Air density
   (d) Humidity ratio
   (e) Enthalpy
   Identify your results in the chart and submit the chart with the solutions.

   If you follow a constant humidity ratio (w) line from 55 degrees and saturation to where it intersects with the 90% relative humidity line you will have the condition of the air. Read all other properties off of the chart.

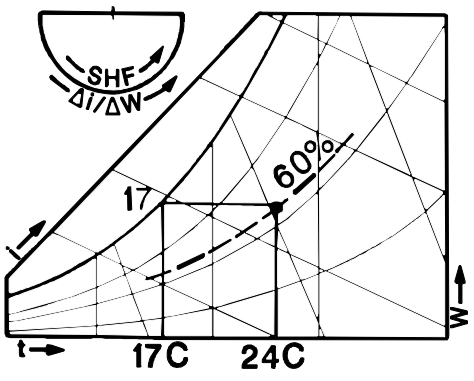
   From psych chart:

   T_{DB} = 57.7°F, T_{WB} = 55.9°F, v = 15.93 ft^3/lb, ρ = 0.0628 lb/ft^3
   W = 0.011 lbv/lba, i = 25.9 Btu/lb

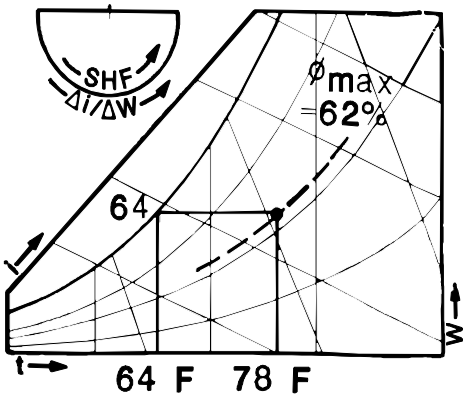
2. **To save energy, the environmental conditions in a room are to be regulated so that the dry bulb temperature will be greater than or equal to 78°F (24°C) and the dew point will be less than or equal to 64°F (17°C). Find the maximum relative humidity that can occur for standard barometric pressure.**

   Follow maximum humidity ratio (w) line (where dew point = 17°C) until it intersects with dry bulb temperature = 24°C line. Read relative humidity lines off of chart as shown below.

   Remember - relative humidity for a constant humidity ratio decreases as dry bulb increases.



From Chart 1b
ϕ_{max} = 60%



From Chart 1a
ϕ_{max} = 62%

3. It is desired to heat and humidify 2000 cfm of air from an initial state defined by a temperature of 60°F db and relative humidity of 30% to a final state of 110°F db and 30% relative humidity. The air will first be heated by a hot water coil followed with saturated vapor at 5 psig. Using the psychrometric chart:
(a) find the heat transfer rate for the heating coil
(b) find the mass flow rate of the water vapor
(c) sketch the processes on a psychrometric chart.
Assume sea level pressure.

The air here undergoes two distinct processes - a sensible only (constant humidity ratio) heating through the coil and the addition of humidity as saturated steam. In this case the steam is saturated and therefore that process follows a constant dry bulb line.

From the psych chart:
Air at initial state 1:
$T_{DB,1}$ = 60°F, Φ_1 = 30%, i_1= 17.1 Btu/lb,
v_1 = 13.18 ft^3/lb, W_1= 0.00335 lbv/lba

Air at state 2:
$T_{DB,2}$ = 110°F, i_2 = 30.2 Btu/lb, W_2 = 0.00335 lbv/lba

Air at final state 3:
$T_{DB,3}$ = 110°F, Φ_3 = 30%, i_3 = 45 Btu/lb,
v_3 = 14.75 ft^3/lb, W_3= 0.01679 lbv/lba

Coil load is only the part of the load that increases the dry bulb temperature since the coil only adds heat and no moisture.

Coil load:

$$\dot{Q}_{coil} = \dot{m}_a (i_2 - i_1) = \frac{\dot{\forall}}{v_1}(i_2 - i_1) = \frac{2000}{13.18}(30.2 - 17.1) = 1988 \ \ (Btu/\min) = 119280\,(Btu/hr)$$

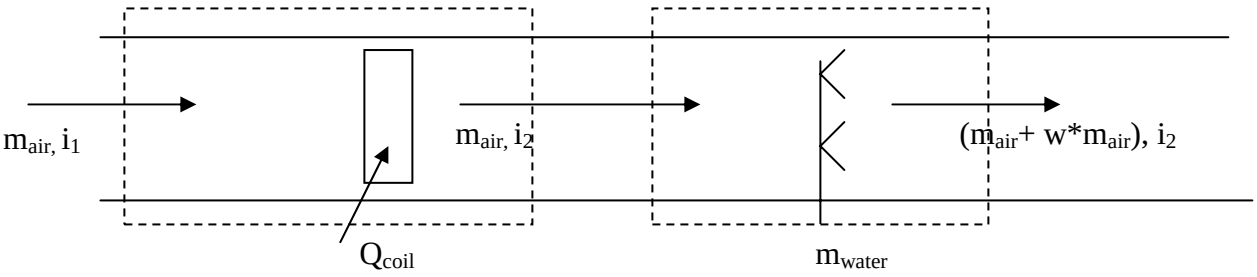
Vapor mass flow rate:

$$\dot{m}_v = \dot{m}_a (W_3 - W_2) = \frac{\dot{\forall}}{v_1}(W_3 - W_2) = \frac{2000}{13.18}(0.01679 - 0.00335) = 2.04\ (lb/min) = 122.4$$

(*lb/min*)

ALTERNATE SOLUTION
An energy balance can also be used to solve this problem keeping in mind that the properties given on the psych. table are given per mass of DRY air. When solving for the mass flow rate of dry air, however, you can neglect the contribution of the water in the volume flow rate given.

Diagram:



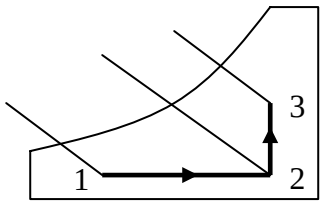
1st Law Energy Balance:

$$\mathbf{m_a * i_2 + m_w * i_s - m_a * i_3 = 0}$$

Where i_s = 1130 Btu/lb - enthalpy of saturated steam at 5 psig (interpolate from table A-1b in text)

$$m_w = m_a * (i_3 - i_2)/i_s = \frac{2000}{13.18}\frac{(45-30.2)}{1131} = 1.99\ (lb/\min)$$

On the Psych. chart the process will look like:



4. Moist air enters a cooling coil at 28°C dry-bulb temperature and 50% relative humidity and exits the coil at 13°C dry-bulb temperature and 90% relative humidity. The flow rate through the coil is 1.50 kg/s and the process occurs at the sea level pressure.
(a) Determine the sensible heat factor (SHF) for the process
(b) Determine the cooling coil capacity (heat transfer-rate)
(c) Sketch the process in the psychrometric chart denoting sensible and latent heat

(a) From the chart: SHF = 0.635

(b) From the chart: i_1=58.6 kJ/kg, i_2=34.2 kJ/kg

Neglecting condensate,

$$\dot{Q}_{CC} = \dot{m}_a \times (i_1 - i_2) = (1.50 kg/s) \times (58.6 kJ/kg - 34.2 kj/kg) = 36.6 kW$$

(c) See psychrometric chart

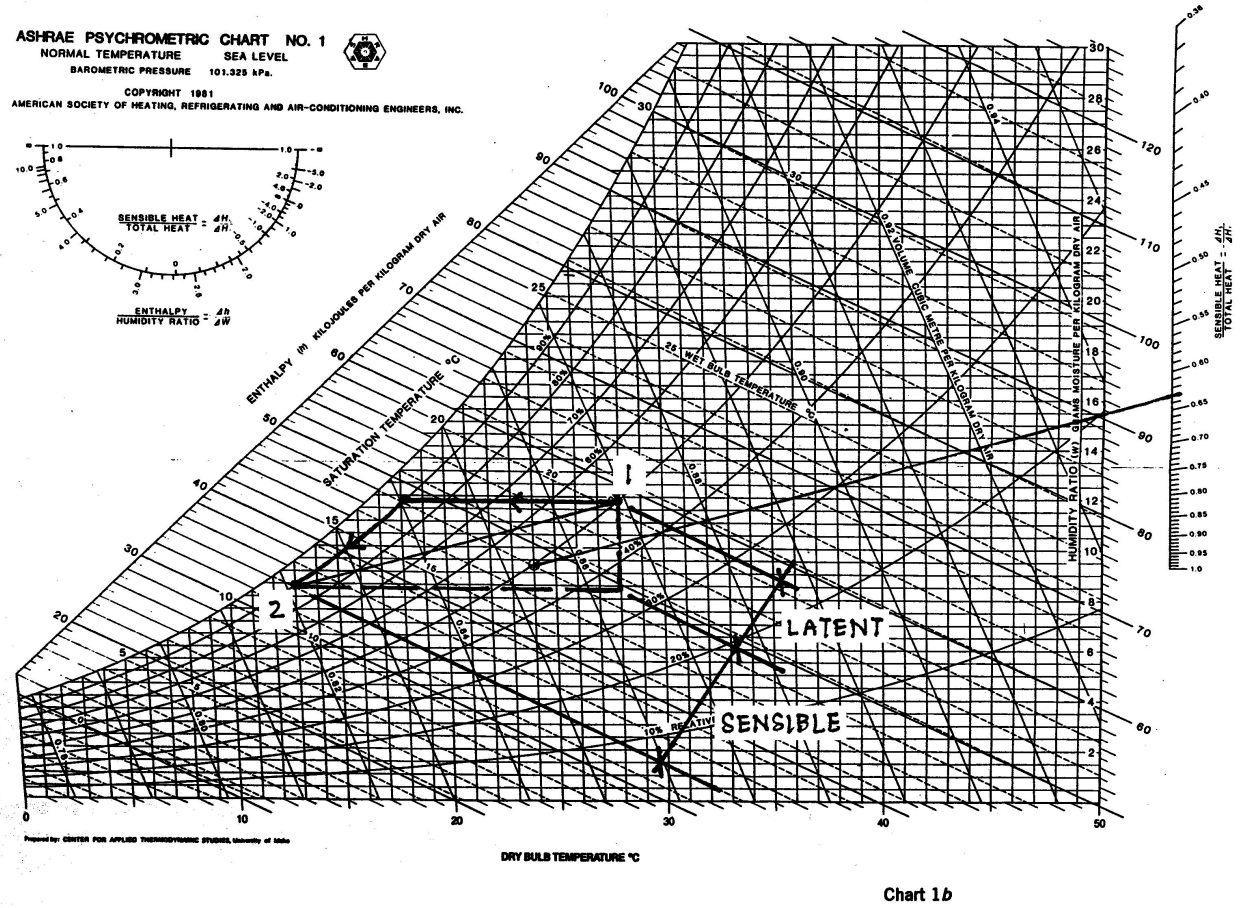


Chart 1*b*

5. Saturated steam is sprayed into a stream of moist air. The initial condition of the air is 55°F dry-bulb temperature and 35°F dew-point temperature. The mass airflow rate is 2000 lbm/min. Barometric pressure is 14.696 psi. Determine:

(a) How much steam must be added in lbm/min to produce a saturated air condition, and

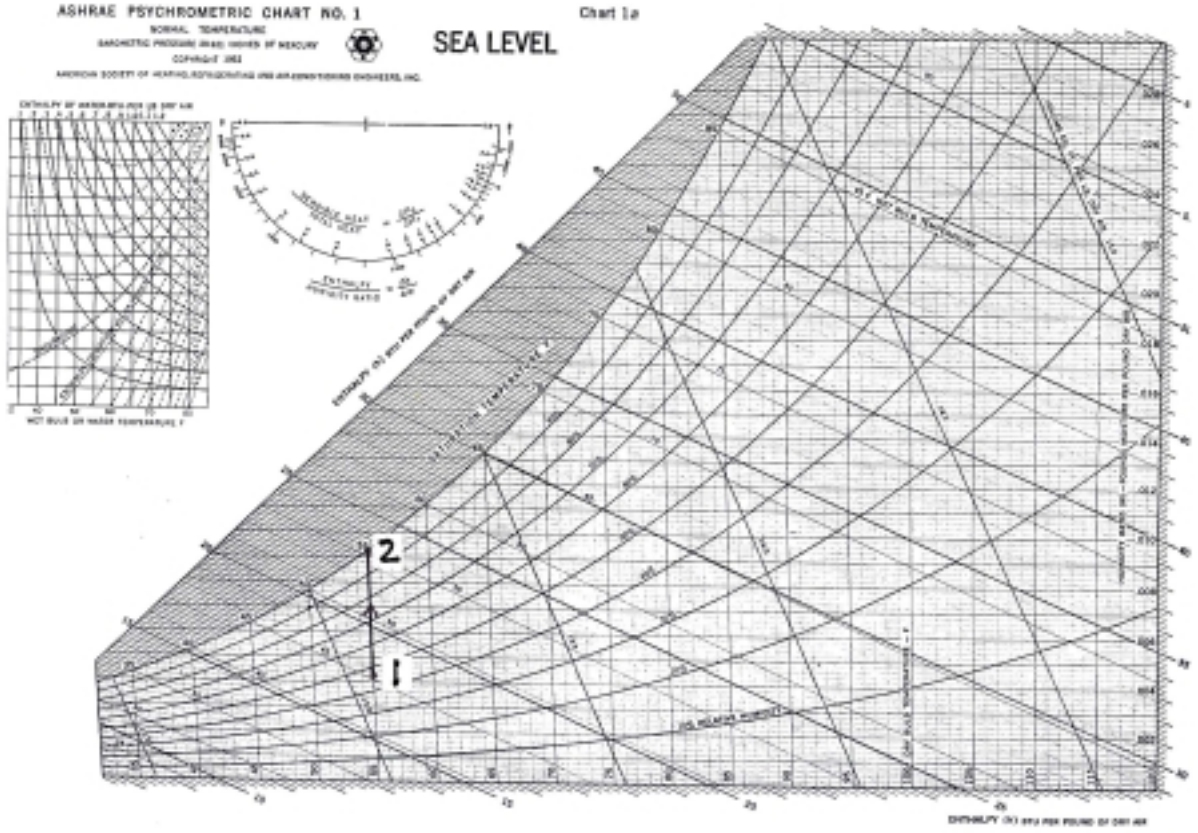
(b) The resulting dry-bulb and wet-bulb temperature of the saturated air

(a) End state (state 2) is saturated. See construction on chart.

From the chart: $w_1 = 0.0043 lb_m / lb_a$, $w_2 = 0.010 lb_m / lb_a$

$$\dot{m}_v = \dot{m}_a \times (w_2 - w_1) = (2000 lb_m / \min) \times (0.010 - 0.0043 lb_m / lb_a) = 11.4 lb_m / \min$$

(b) From the chart: T_{DB2}=55°F, T_{WB2}=55°F


ASHRAE PSYCHROMETRIC CHART NO. 1
Chart 1a
SEA LEVEL
2
1

Practicum Assignment #5

1. A space is to be maintained at 72°F and 30% relative humidity during the winter months. The sensible heat loss from the space is 500,000 Btu/hr and the latent heat loss due to infiltration is 50,000 Btu/hr. Construct the condition line in the psychrometric chart.

2. Air at 10°C db and 5°C wb is mixed with air at 25°C db and 18°C wb in a steady-flow process at standard atmospheric pressure. The volume flow rates are 10 m^{3}/s and 6m^{3}/s, respectively.
   (a) Compute the mixture conditions (enthalpy and humidity ratio)
   (b) Find the mixture conditions using the psychrometric chart.

3. A building has a total load of 200,000 Btu/hr. The sensible heat factor for this space is 0.8. The space is to be maintained at 72°F and 40% relative humidity. Outdoor air at 40°F and 20% relative humidity in the amount of 1000 cfm is required. Air is supplied to the space at 120°F. Find:
   (a) the conditions and amount of air supplied to the space,
   (b) the temperature rise of the air through the furnace,
   (c) the amount of water at 50°F required by the humidifier, and
   (d) the capacity of the furnace.
   Assume sea pressure level.

4. A large warehouse located in Denver, Colorado (elevation = 5000 ft) is to be conditioned using an evaporative cooling system. Assume that the space is to be maintained at 80°F and 50% relative humidity by a 100% outdoor air system. Outdoor design conditions are 91°F dry bulb and 59°F wet bulb. The cooling load is estimated to be 100 tons with a sensible heat factor of 0.9. The supply air fan is located just downstream of the spray chamber and is estimated to require 20 hp. Determine the volume flow rate of air to the space, and sketch the process on a psychrometric chart.

Solutions for the Practicum Assignment #5

1. A space is to be maintained at 72°F and 30% relative humidity during the winter months. The sensible heat loss from the space is 500,000 Btu/hr and the latent heat loss due to infiltration is 50,000 Btu/hr. Construct the condition line in the psychrometric chart.

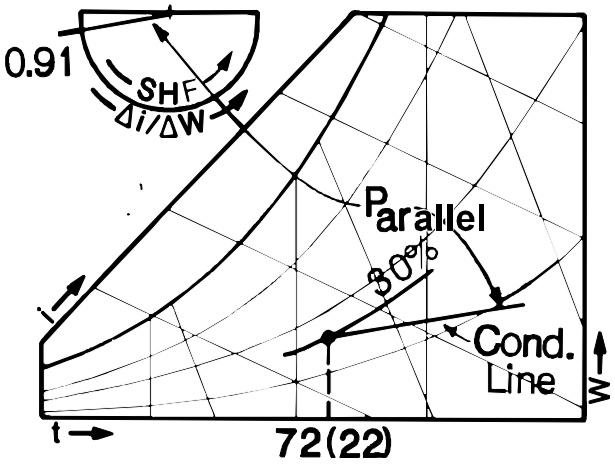
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1. A space is to be maintained at 72°F and 30% relative humidity during the winter months. The sensible heat loss from the space is 500,000 Btu/hr and the latent heat loss due to infiltration is 50,000 Btu/hr. Construct the condition line in the psychrometric chart.

$$SHF = \frac{500{,}000}{550{,}000} = 0.91$$



2. Air at 10°C db and 5°C wb is mixed with air at 25°C db and 18°C wb in a steady-flow process at standard atmospheric pressure. The volume flow rates are 10 m^3/s and 6 m^3/s, respectively.
(a) Compute the mixture conditions (enthalpy and humidity ratio)
(b) Find the mixture conditions using the psychrometric chart.

Use mass weighted averages in order to find the mixed condition of the air.

(a) $\dot{m}_{a1} = \dot{Q}_1 / v_1 = 10 / 0.81 = 12.346 \text{ kg / s}$

$$\dot{m}_{a2} = 6 / 0.86 = 6.977 \text{ kg /s}$$

$$i_3 = \frac{(12.436 \text{ x } 18.6) + (6.977 \text{ x } 50.9)}{(12.346 + 6.977)} = 30.26 \text{ kJ / kg}$$

$$W_3 = \frac{(12.436 \text{ x } 0.0034) + (6.977 \text{ x } 0.0102)}{(12.346 + 6.977)}$$

$$W_3 = 0.0059 \text{ kgv /kga}$$

(b) The ratio between the mass flow rates of the air streams locates where the mixed condition is on a line drawn between the two incoming states on the psych. chart.

$$\frac{\dot{m}_{a1}}{\dot{m}_{a3}} = \frac{12.346}{(12.346 + 6.977)} = 0.64 = \frac{\overline{32}}{\overline{\overline{12}}}$$

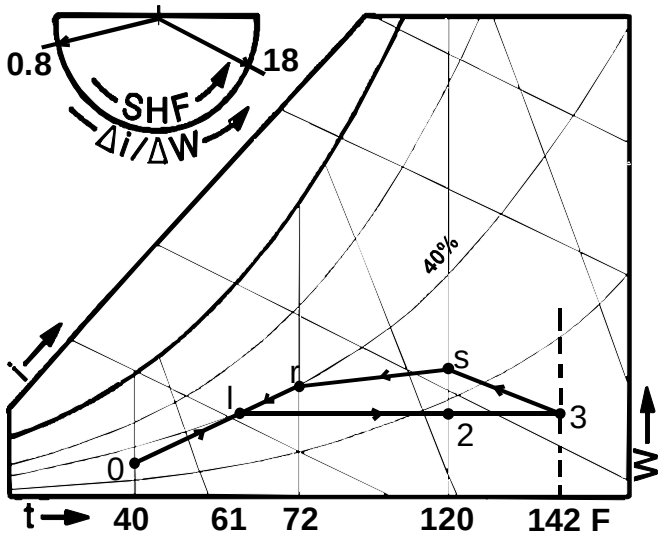
This indicates the mixed condition is 64% of the way on the line towards air stream 1

Read: $i_3 = 30.0$ Btu / lba; $W_3 = 0.0058$ kgv / kga

3. A building has a total load of 200,000 Btu/hr. The sensible heat factor for this space is 0.8. The space is to be maintained at 72°F and 40% relative humidity. Outdoor air at 40°F and 20% relative humidity in the amount of 1000 cfm is required. Air is supplied to the space at 120°F. Find:
   a. the conditions and amount of air supplied to the space,
   b. the temperature rise of the air through the furnace,
   c. the amount of water at 50°F required by the humidifier, and
   d. the capacity of the furnace.

Assume sea pressure level.

$\dot{V}_0 = 1000$ cfm



(a) From Chart 1a

$t_s = 120 / 76$ F

$$\dot{m}_s = \frac{\dot{q}}{(i_s - i_r)} = \frac{200{,}000}{(39.4 - 24.4)}$$

= 13,333 lb/hr = $\dot{m}_1$

$\dot{Q}_s = \dot{m}_s v_s = \dot{m}_s (14.8)/60 = 3280$ ft³ / min

(b) $\dot{m}_o = \dot{Q}_o / v_o = 1000$ x 60 / 12.6 = 4760 lb/hr

$$\frac{\dot{m}_v}{\dot{m}_1} = \frac{13{,}333 - 4762}{13{,}333} = 0.642; \quad \textit{From} \text{ Chart } \quad t_1 = 61/49 \text{ F}$$

$t_3 - t_1 =$ (*See* (d) below)

(c) $\dot{m}_w = \dot{m}_s (W_s - W_1) = 13{,}333\,(0.0095 - 0.0046) = 65.3$ lbm/hr

(d) $\dot{q}_f = \dot{m}_1 (i_3 - i_1) = 13{,}333\,(39.4 - 19.7) = 262{,}660$ Btu/hr

Where $i_3 = i_s$ from chart

Then $\dot{q}_f = 262{,}660 = 13{,}333\,(0.24)(t_3 - 61)$; *and* $t_3 = 143\,\text{F}$

$\Delta t_{fur} = t_3 - t_1 \;\; = 143 - 61 = 82\,\text{F}$

4. A large warehouse located in Denver, Colorado (elevation = 5000 ft) is to be conditioned using an evaporative cooling system. Assume that the space is to be maintained at 80°F and 50% relative humidity by a 100% outdoor air system. Outdoor design conditions are 91°F dry bulb and 59°F wet bulb. The cooling load is estimated to be 100 tons with a sensible heat factor of 0.9. The supply air fan is located just downstream of the spray chamber and is estimated to require 20 hp. Determine the volume flow rate of air to the space, and sketch the process on a psychrometric chart.

$\dot{q}_r = \dot{m}_s(i_r - i_s)$; $\dot{W}_{\text{fan}} = \dot{m}_s(i_s - i_c)$

(a) $i_c = 28\,\text{Btu/lbm}$; $i_r = 30.6\,\text{Btu/lbm}$

Using Psychrometric Chart:

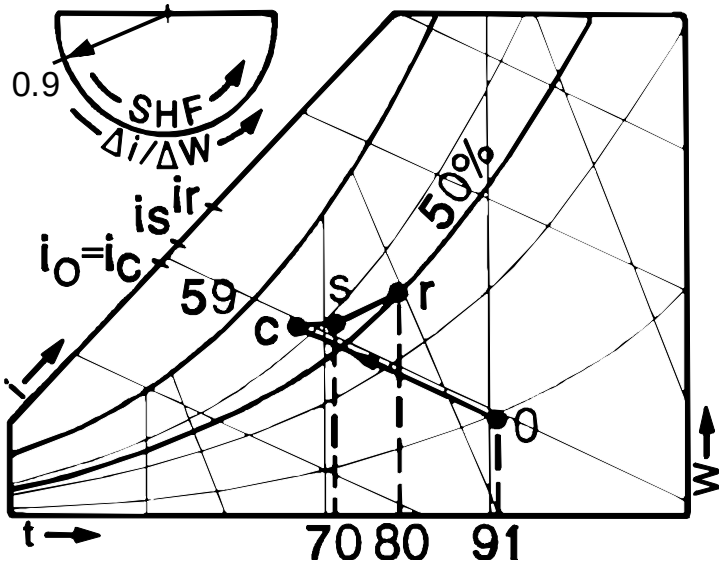
$\dot{q}_r = 1{,}200{,}000\,\text{Btu/hr}$

$\dot{W}_{fan} = 50{,}900\,\text{Btu/hr}$

$i_3 = 28.1\,\text{Btu/lbm}$; $\dot{m}_s = 480{,}000\,\text{lba/hr}$

$\dot{Q}_s = 480{,}000 \text{ x } 16.3/60 = 130{,}400\,\text{cfm}$

(b) $\dot{Q}_s = 61.54\,\text{m}^3/s$; Using Psychrometric Chart



Practicum Assignment #6

1. A condition exists where it is necessary to cool and dehumidify air from 80°F db and 67°F wb to 60°F db and 54°F wb.
   (a) Discuss the feasibility of doing this in one process with a cooling coil. (Hint: Determine the apparatus dewpoint temperature for the process.)
   (b) Describe a practical method of achieving the required process, and sketch it on a psychrometric chart.

2. Continue to design the air-conditioning system of the classroom at PSU for winter heating. The example in the lecture notes (pg. 36 of Chapter2) specifies design conditions.

   (a) Design a heating system with a heating coil and a saturated-steam humidifier and identify the corresponding air-handling processes in a psychrometric chart. The system should include the components used for summer cooling.
   (b) Size the equipment of the air-conditioning system proposed in (a). Use the data for cooling if necessary.

   Use the psych chart to perform this design assignment.

3. A space is to be maintained at 78°F db and 68°F wb. The cooling system is a variable-air-volume (VAV) type where the quantity of air supplied and the supply temperature are controlled. Under design conditions, the total cooling load is 150 tons with a sensible heat factor of 0.6, and the supply temperature is 60°F db. At minimum load, about 18 tons with SHF of 0.8, the air quantity may be reduced no more than 80% by volume of the full load design value.

   (a) Determine the supply air conditions for minimum load.
   (b) Show all the conditions on a psychrometric chart.

Solutions for the Practicum Assignment #6

1. A condition exists where it is necessary to cool and dehumidify air from 80°F db and 67°F wb to 60°F db and 54°F wb.
   (a) Discuss the feasibility of doing this in one process with a cooling coil. (Hint: Determine the apparatus dewpoint temperature for the process.)
   (b) Describe a practical method of achieving the required process, and sketch it on a psychrometric chart.

2. Continue to design the air-conditioning system of the classroom at PSU for winter heating. The example in the lecture notes (pg. 36 of Chapter 2) specifies design conditions.

   (a) Design a heating system with a heating coil and a saturated-steam humidifier and identify the corresponding air-handling processes in a psychrometric chart. The system should include the components used for summer cooling.
   (b) Size the equipment of the air-conditioning system proposed in (a). Use the data for cooling if necessary.

   Use the psych chart to perform this design assignment.

3. A space is to be maintained at 78°F db and 68°F wb. The cooling system is a variable-air-volume (VAV) type where the quantity of air supplied and the supply temperature are controlled. Under design conditions, the total cooling load is 150 tons with a sensible heat factor of 0.6, and the supply temperature is 60°F db. At minimum load, about 18 tons with SHF of 0.8, the air quantity may be reduced no more than 80% by volume of the full load design value.

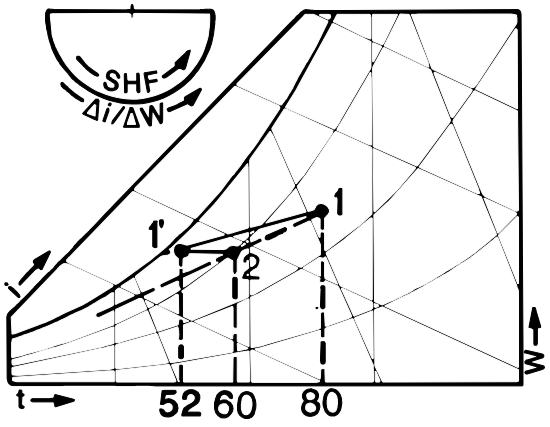
   (a) Determine the supply air conditions for minimum load.
   (b) Show all the conditions on a psychrometric chart.

1. A condition exists where it is necessary to cool and dehumidify air from 80°F db and 67°F wb to 60°F db and 54°F wb.
   (a) Discuss the feasibility of doing this in one process with a cooling coil. (Hint: Determine the apparatus dewpoint temperature for the process.)
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(a) It is probably impossible to cool the air from 1 to 2 in one process. The extension of line 1-2 does not intersect the saturation curve. Therefore, there is no feasible apparatus dew point.

(b) Cool the air to state 1' and then sensibly heat to state 2.

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2. Continue to design the air-conditioning system of the classroom at PSU for winter heating. The example in the lecture notes (pg. 36 of Chapter 2) specifies design conditions.

   (a) Design a heating system with a heating coil and a saturated-steam humidifier and identify the corresponding air-handling processes in a psychrometric chart. The system should include the components used for summer cooling.
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Given: Winter-heating condition of a PSU classroom
Required: HVAC system Design.
Solution:

This problem is a continuous design of the example in chapter 3.5, which is given as: (only winter conditions are listed below):

Sensible	Heating (W)
Walls	2000
Windows	2000

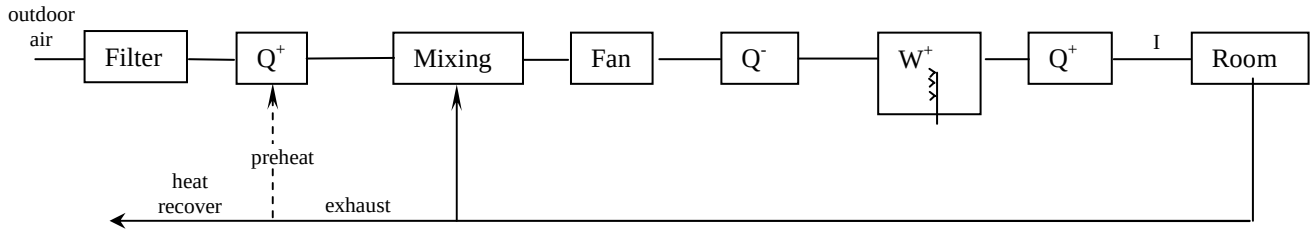
Latent	Heating (W)
/	/

Outdoor Design Conditions	
T	-14°C
T_{wet}	/

Indoor Design Conditions	
T	22°C
ϕ	50%

Minimum Outdoor Air: 80 l/s
Air Supply Temperature: 60°C (max)

(a) The heating system is designed as follow:



Please Note:

- Preheat is needed in winter heating since condensation (freezing) may occur if out air mixes with return air directly.
- Heat recover system is often used by taking heat out of exhaust air
- Cooling coil will be turned off in winter heating
- **Heating coil only supplies sensible heat**

Determine important points: (Find T, W, i, ...)

1) Outdoor (O):
   $T_o = T_{dry} = -14$ °C, ϕ_o=60% (design rule of thumb for winter RH)
   From low T psych chart (attached at end of problem): W_o=0.00068 (kgw/kga), i_o= -12.5 (kJ/kg)

2) Indoor (R):
T_R = 22 °C, ϕ_R=50%
From psych chart: W_R = 0.008225 (kgw/kga), i_R = 43.13 (kJ/kg)

3) Supply Point (I):
Make a choice of supply air temperature T_I < 60 °C , so pick up T_I = 45 °C.
Since no latent heat to remove,
$W_I = W_R$ = 0.008225 (kgw/kga)
From psych chart: i_I = 66.71 kJ/kg
Heating Coil only adds sensible heat - humidification adds the rest of the heat in the form of latent energy - see Psych chart below.
i_I' = 55.6 kJ/kg - from psychrometric chart

4) Preheated point (P):
Suppose we use heat recovery system to preheat the outdoor air, T_p =12.8°C [55°F] is a design rule of thumb.
T_p = 12.8 °C
$W_p = W_o$ = 0.00068 kgw/kga
From psych chart: i_p = 14.6 (kJ/kg)

5) Mixing Point (M):

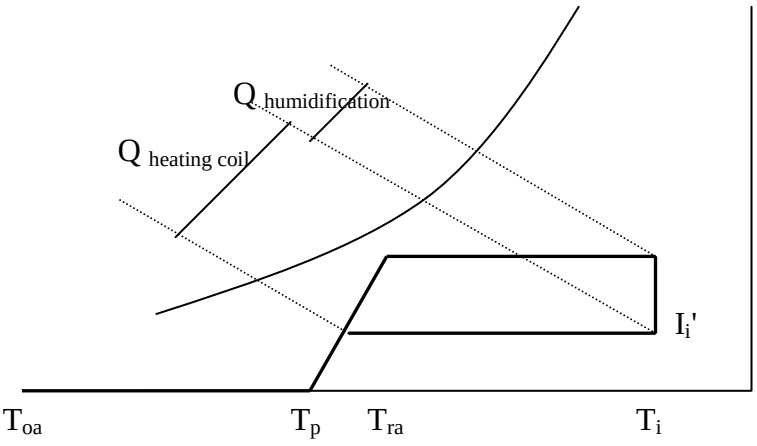
$$m_a = \frac{W_{sen}}{C_p \Delta T} = \frac{4000}{1.01\times10^3 \times (45-22)} = 0.1722 \text{ (kg/s)}$$

m_o = 80 l/s = 80 × 10^{-3} m^3/s × 1.2 kg/m^3 = 0.096 (kg/s)

$$i_M = \frac{m_o i_p + m_R i_R}{m_o + m_R} = \frac{m_o i_p + (m_a - m_o) i_R}{m_a} = \frac{0.096(14.6) + (0.1722-0.096)43.13}{0.1722} = 27.22 \text{ (kJ/kg)}$$

$$W_M = \frac{m_o W_a + m_R W_R}{m_a} = \frac{0.096(0.00068) + (0.1722-0.096)0.008225}{0.1722} = 0.00402 \text{ (kgw/kga)}$$

(b) Now, it is our choice to size the equipments- heating coil, humidifier and fan. Suppose we heat the air from mixing point to same temperature and use adiabatic humidification (maybe not the best way), then



For heating coil $Q_{heating} = m_a (i_I' - i_M) = 0.1722 (55.6 – 27.22) = 4.89$ (kW)

For humidifier: $\Delta W = W_I - W_M = 0.008225 - 0.00402 = 0.004205$ (kgw/kga)

For fan: $\dot{V} = m_a / \rho = m_a \times v = 0.1722 \times 0.828 = 0.1426$ m^3/s = 513 (m^3/hr)

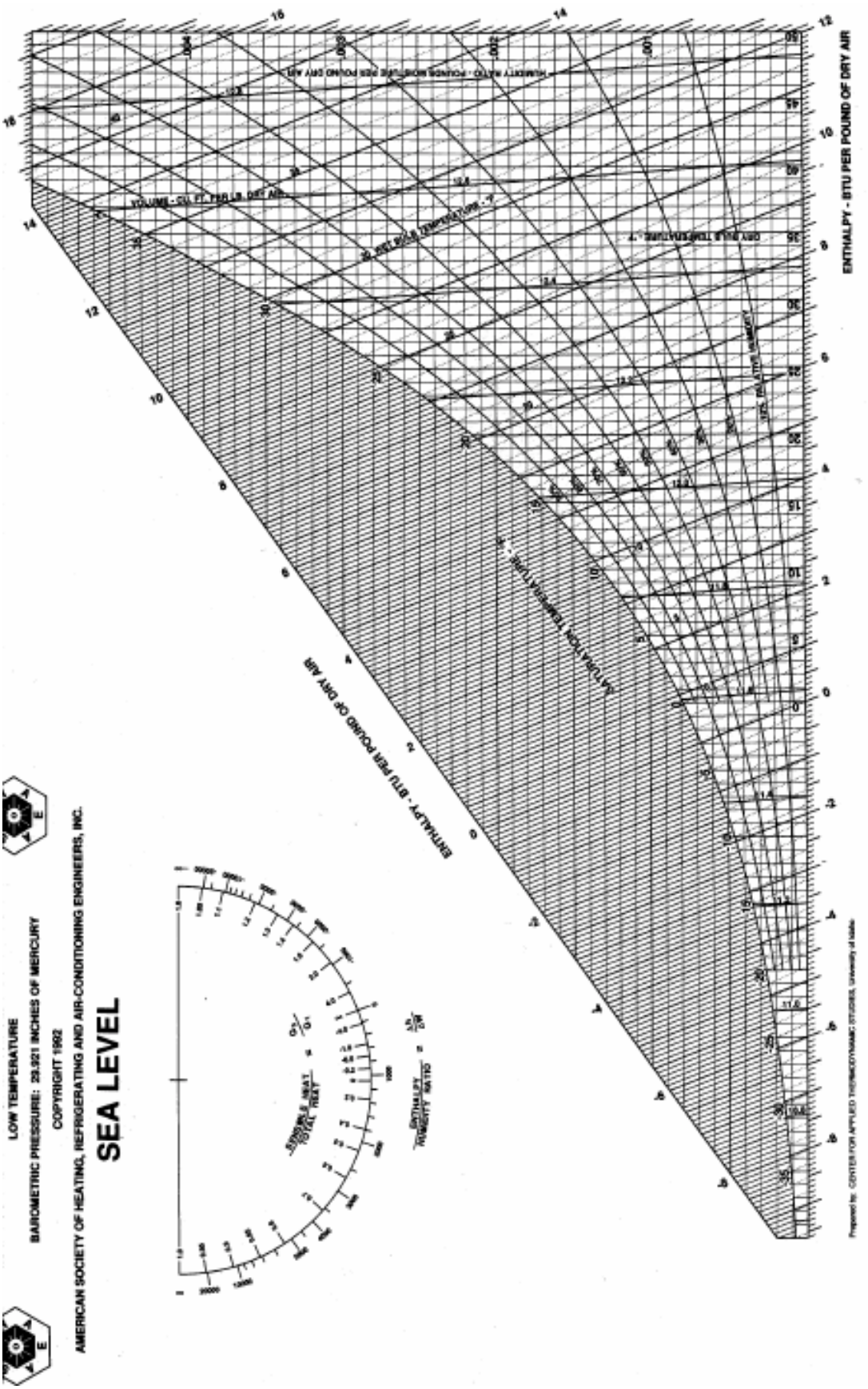
For preheater: $Q_{preh} = m_o (i_P - i_o) = 0.096 (14.6 - (-12.5)) = 2.60$ (kW)

This equipment should be use in both heating (winter) and summer conditions. So we should pick up the great value to size them.

From example:	Fan	1200 m^3/hr	Our calc:	Fan	513 m^3/hr
	$Q_{cooling}$	7.156 kW		$Q_{cooling}$	—
	$Q_{heating}$	0.4 kW		$Q_{heating}$	4.89 kW

So our final choice will be:

Fan:	1200 m^3/hr
Cooling Coil:	7.156 kW
Heating Coil:	4.89 kW
Humidifier:	0.004205 (kgw/kga)
Pre-heater:	2.60 kW


LOW TEMPERATURE
BAROMETRIC PRESSURE: 29.921 INCHES OF MERCURY
COPYRIGHT 1992
AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS, INC.
SEA LEVEL
ENTHALPY - BTU PER POUND OF DRY AIR
ENTHALPY - BTU PER POUND OF DRY AIR

3. A space is to be maintained at 78°F db and 68°F wb. The cooling system is a variable-air-volume (VAV) type where the quantity of air supplied and the supply temperature are controlled. Under design conditions, the total cooling load is 150 tons with a sensible heat factor of 0.6, and the supply temperature is 60°F db. At minimum load, about 18 tons with SHF of 0.8, the air quantity may be reduced no more than 80% by volume of the full load design value.

(a) Determine the supply air conditions for minimum load.
(b) Show all the conditions on a psychrometric chart.

(a) $\dot{q}_d = \dot{m}_d (i_r - i_s)$

Using psych chart 1a, i_r = 32.4 Btu/hr, i_s = 25.0Btu/hr

$$\dot{m}_d = \frac{\dot{q}_d}{(i_r - i_s)} = \frac{(150 tons) \times \left[\frac{12{,}000 Btu/hr}{1 ton}\right]}{32.4 - 25.0 Btu/lb} = 243{,}243 lbm/hr$$

$$\dot{V}_d = \dot{m}_d \upsilon_s = \frac{(243{,}243 lbm/hr) \times (13.3\, ft^3/lb)}{60 \,\text{min}/hr} = 53{,}920 cfm$$

$$\dot{V}_m = 0.2\dot{V}_d = 0.2 \times 53{,}920 cfm = 10{,}784 cfm$$

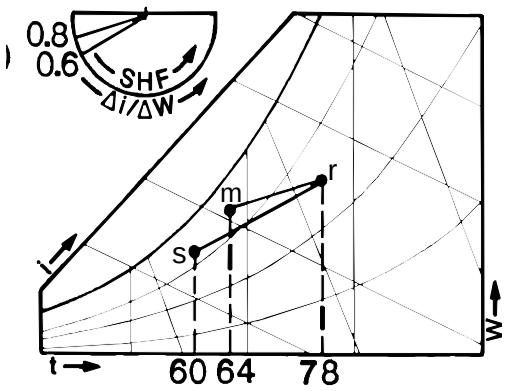
$$\dot{m}_m = \frac{\dot{V}_m}{\upsilon_m} = \frac{(10{,}784 cfm) \times (60 \,\text{min}/hr)}{13.5\, ft^3/lb} = 47{,}928 lbm/hr \text{ where } \upsilon_m \text{ is assumed}$$

$$\dot{q}_m = \dot{m}_m (i_r - i_m)$$

$$i_m = i_r - \frac{\dot{q}_m}{\dot{m}_m} = 32.4 Btu/lb - \frac{(18 tons) \times \left[\frac{12{,}000 Btu/hr}{1 ton}\right]}{47{,}928 lbm/hr} = 27.9 Btu/lb$$

(b) $T_{DB,m} = 64^\circ$ F

$T_{WB,m} = 62^\circ$ F



Practicum Assignment #7

1. During the winter months it is possible to cool and dehumidify a space using outdoor air. Suppose an interior zone of a large building is designed to have a supply airflow rate of 5000 cfm, which can be all outdoor air. The cooling load is constant at 10 tons with a SHF of 0.8 the year round. Indoor conditions are 78°F db and 67°F wb.

   (a) What is the maximum outdoor air dry bulb temperature and humidity ratio that would satisfy the load condition?
   (b) Consider a different time when the outdoor air has a temperature of 40°F db and 20% relative humidity. Return air and outdoor air may be mixed to cool the space, but humidification will be required. Assume that saturated water vapor at 14.7 psia is used to humidify the mixed air, and compute the amounts of outdoor and return air in cfm.
   (c) At another time, outdoor air is at 70°F db with a relative humidity of 90%. The cooling coil is estimated to have a minimum apparatus dew point of 50°F. What amount of outdoor air and return air should be mixed before entering the coil to satisfy the given load condition?
   (d) What is the refrigeration load for the coil of part (c) above?

2. An economizer mixes outdoor air with room return air to reduce the refrigeration load on the cooling coil.

   (a) For a space condition of 25°C db and 20°C wb, describe the maximum wet bulb and dry bulb temperatures that will reduce the coil load.
   (b) Suppose a system is designed to supply 5m^{3}/s abd 18°C db and 17°C wb to a space maintained at the conditions given in part (a) above. What amount outdoor air at 20°C db and 90% relative humidity can be mixed with the return air if the coil SHF is 0.6?
   (c) What are the apparatus dew point and the bypass factor in part (b) above?
   (d) Compare the coil refrigeration load in part (b) above with the outdoor air to that without outdoor air.

Practicum Assignment #8

1. A classroom in a school is designed for 100 people.
   (a) What is the minimum amount of fresh outdoor air required?
   (b) The floor area is 1500 ft^2. What is the outdoor air ventilation requirement on the bases of floor area?

2. What level, in ppm, will the CO_2 concentration be in a space in steady state, if CO_2 is being released into the space at the rate of 0.25 cfm and outdoor air with a CO_2 concentration of 200 ppm is being supplied to the space at the rate of 1000 cfm? Assume complete mixing.

3. The same space as in problem 2 uses a mixing with the recirculation rate of 0.4 in order to conserve energy. If the CO_2 release rate and fresh air supply is the same as in problem 2, calculate the CO_2 concentration (ppm) in the space in steady state. Note: Recirculation rate R=Recirculated Air / Total Supply Air